

Techniques in Combinatorics

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Introduction

To many beginners, combinatorics is the easiest topic because one does not need to know deep knowledge to work on combinatorial problems. Usually, the arguments are intuitively obvious, and some are indeed common sense. However, due to the same reason, combinatorics is also the hardest topic to many people. This is because learning deep knowledge seems to be useless.

It is also hard to judge the difficulty of a combinatorial problem. If one looks into the IMO problems, there are occasionally some C3 or C4 problems for which only a small proportion of the contestants can get complete solutions. One reason is that a combinatorial problem usually only requires a key observation or a key idea. This makes the solution short. But sometimes the idea may be hard to think of.

In this note, we emphasize more on introducing various techniques in combinatorics. This allows one to learn more tools to work on combinatorial problems. Although these techniques will be introduced under separated topics, one should bear in mind that every method is useful in every topic. Sometimes having a limited mindset is not a good thing.

In section 1, we introduce different counting methods, including bijection and generating function. In section 2, we will talk about some standard methods in combinatorics, like induction, pigeonhole principle, colouring, etc. In section 3, we use algorithm and invariance to solve some problems involving games or operations. Section 4 is devoted to graph theory, while section 5 is devoted to combinatorial geometry. All techniques introduced in previous sections are useful for both topics.

1 Counting

Counting is the most elementary topic in combinatorics. There are various counting methods, each of which has its own advantage. We will go through different counting methods in this section.

1.1 Basic Counting Tools

First of all, we review the basic counting tools.

Definition 1.1. The *method of exhaustion*(窮舉法/枚舉法) refers to the counting method which lists out all possible cases.

This is the most fundamental way of counting. A clear disadvantage of listing everything out is that it is easy to make mistakes. It is very likely that some cases are missed out during the counting. However, sometimes the exhaustion method cannot be avoided in part of the steps. In order to count more accurately, one should follow a certain order when listing out the cases. In particular, the *lexicographic order* (or *dictionary order*(字典排列法)) is useful. The name is self-explanatory. For example, if we use an ordered tuple $(a_1, a_2, \dots, a_n)$ of numbers to identify a specific case, then we can list out all the cases in ascending order of their first entries a_1 . If some of them have the same first entry, then we order them using their second entries and so on.

Proposition 1.1. (*Addition principle/Sum rule*(加法原理)) For any pairwise disjoint finite sets $S_1, S_2, \dots, S_n$, we have

$$|S_1 \cup S_2 \cup \dots \cup S_n| = |S_1| + |S_2| + \dots + |S_n|.$$

Recall that for any set S , the notation $|S|$ denotes the number of elements in S . Therefore, proposition 1.1 states that the number of elements in at least one of the sets $S_1, S_2, \dots, S_n$ is equal to the sum of number of elements of these sets, if they are pairwise disjoint. This is rather obvious. For application on counting, it simply means that we can count by partitioning all possibilities into a few cases. Each case has its own feature which allows us to count quickly by using some other methods. Note that we should ensure that each situation falls into exactly one case in order to apply the **addition principle**.

Proposition 1.2. (*Multiplication principle/Product rule*(乘法原理)) For any finite sets $S_1, S_2, \dots, S_n$, we have

$$|S_1 \times S_2 \times \dots \times S_n| = |S_1| \times |S_2| \times \dots \times |S_n|.$$

The Cartesian product $S_1 \times S_2 \times \cdots \times S_n$ is the set of all elements $(a_1, a_2, \dots, a_n)$ such that $a_j \in S_j$ for each $j = 1, 2, \dots, n$. Suppose we would like to make some choices in a few steps so that the number of choices in each step is independent of the other steps. Then we can simply take the product of the number of choices in all steps to obtain the total. This saves a lot of time in many cases.

Problem 1.1. Let n be a positive integer. How many n -digit numbers are there such that all digits of n are nonzero and have the same parity?

Analysis. We demonstrate the simplest use of the **addition principle** and the **multiplication principle**. For this simple problem, it is obvious that we should divide into two cases, according to the parity of the digits.

Solution. If all digits are odd, then there are 5 choices for each of the n digits, namely, 1, 3, 5, 7, 9. Hence, there are 5^n such numbers.

Similarly, if all digits are even, then each digit can be one of 2, 4, 6, 8. There are 4^n such numbers.

In total, we have $5^n + 4^n$ numbers satisfying the condition.

Problem 1.2. Let n be a positive integer. How many n -digit numbers are there whose sums of digits are multiples of 5?

Analysis. Suppose we choose the digits one by one. Since the condition involves the sum of digits, it seems that the choice for each digit depends on the other digits. It is true that the choices are dependent, but the number of choices is still a fixed number. Therefore, we can apply the **multiplication principle** as usual.

Solution. When $n = 1$, there are two possibilities, 0 and 5. When $n > 1$, we choose the first $n - 1$ digits arbitrarily. The first digit is nonzero, and hence there are 9 choices. For each of the remaining $n - 2$ digits, there are 10 choices.

After the first $n - 1$ digits are chosen, let S be the sum of these digits. Then we have to choose a digit k for the unit digit such that $S + k$ is a multiple of 5. This means k is one of 0, 1, 2, $\dots$, 9 with the constraint $k \equiv -S \pmod{5}$. Note that there are exactly 2 choices for k no matter what S is.

Therefore, there are $9 \times 10^{n-2} \times 2 = 18 \times 10^{n-2}$ numbers satisfying the condition for $n > 1$.

Definition 1.2. (*Permutation*(排列)) Let n and r be nonnegative integers satisfying $0 \leq r \leq n$. Then the number of ways to choose r objects from n distinct objects in order is denoted by P_r^n .

Definition 1.3. (*Combination*(組合)) Let n and r be nonnegative integers satisfying $0 \leq r \leq n$. Then the number of ways to choose r objects from n distinct objects without specifying the order is denoted by $\binom{n}{r}$ or C_r^n .

It is customary in China to use the notations P_n^r (or A_n^r) for permutation and C_n^r for combination (with the n and r being swapped). To avoid misunderstanding, we will use the notation $\binom{n}{r}$ in this note. This symbol is also known as the *binomial function*(二項式函數) or *binomial coefficient*(二項式係數), due to the following theorem.

Theorem 1.1. (Binomial theorem(二項式定理)) Let n be a nonnegative integer. Then

$$(x + y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots + \binom{n}{n}y^n.$$

Proof. When we expand the left-hand side, each term corresponds to the product of n factors, each of which is either x or y . Therefore, each term must be of the form $x^{n-k}y^k$ for some $k = 0, 1, \dots, n$.

On the other hand, to obtain a term $x^{n-k}y^k$, we need to choose exactly k copies of y from n brackets (and the remaining $n - k$ copies of x from the other brackets). There are $\binom{n}{k}$ ways to do so by definition. (Although the brackets look the same, they are treated as different objects when doing the expansion.) This means the coefficient of $x^{n-k}y^k$ is $\binom{n}{k}$. Summing over k from 0 to n gives the desired identity. $\square$

Many counting problems can be solved using P_r^n and $\binom{n}{r}$ (indeed, the latter is more useful). The following two propositions give the useful formulae related to these numbers.

Proposition 1.3. Let n and r be nonnegative integers satisfying $0 \leq r \leq n$. Then

$$(a) \quad P_r^n = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!},$$

$$(b) \quad P_n^n = n!,$$

$$(c) \quad P_1^n = n,$$

$$(d) \quad P_0^n = 1 \text{ for } n \geq 1.$$

Proof. To choose r objects from n objects, we can choose them one by one. There are n choices for the first one, $n - 1$ choices for the second one, $\dots$, and $n - r + 1$ choices for the last one (the r th one). By the **multiplication principle**, we have $P_r^n = n(n-1)\cdots(n-r+1)$. The second part of (a) follows from the definition of factorial. The remaining parts can be deduced by substituting $r = n, 1, 0$ respectively in this formula. Note that the product $n(n-1)\cdots(n-r+1)$ is empty when we put

$r = 0$. By convention, an empty product equals 1. Similarly, we define $0! = 1$ by convention. $\square$

Proposition 1.4. Let n, r, k be nonnegative integers satisfying $0 \leq r \leq n$. Then

$$(a) \binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!} = \frac{n!}{r!(n-r)!},$$

$$(b) \binom{n}{r} = \binom{n}{n-r},$$

$$(c) \binom{n}{n} = \binom{n}{0} = 1,$$

$$(d) \binom{n}{1} = n,$$

$$(e) \text{ (Pascal's identity (帕斯卡恒等式)) } \binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1} \text{ for } r \geq 1,$$

$$(f) \binom{n}{n} + \binom{n+1}{n} + \cdots + \binom{n+k}{n} = \binom{n+k+1}{n+1},$$

$$(g) \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Proof. Suppose we want to choose r objects from n distinct objects in order. This can be done in 2 steps. Firstly, we choose r objects without specifying the order. There are $\binom{n}{r}$ ways to do so. Then we rearrange these r chosen objects to give the order, where there are P_r^r ways to do so. Hence, the total number of ways is $\binom{n}{r} \times P_r^r$. Note that this is equal to P_r^n by definition. This implies

$$\binom{n}{r} = \frac{P_r^n}{P_r^r} = \frac{n(n-1)\cdots(n-r+1)}{r!}$$

by proposition 1.3. This proves (a). In particular, we have the corollary of part (c) and part (d).

For (b), the simplest way to see is as follows. If we want to choose r objects from n distinct objects, then we simply need to choose $n-r$ objects from these n objects and remove them. The remaining r objects will become our choice. This easily establishes the formula.

Next, **Pascal's identity** can be proved in a combinatorial way. Note that we have to choose r objects from $n+1$ objects. We first label one of these $n+1$ objects as special. If we do not choose the special object, then we have to choose r objects from the remaining n objects. There are $\binom{n}{r}$ ways to do so. If we instead choose the

special object, then we only need to choose $r-1$ objects from the remaining n objects. There are $\binom{n}{r-1}$ ways to do so. Thus, the total number is the right-hand side of the identity. This establishes **Pascal's identity**. Applying this identity repeatedly, we obtain

$$\begin{aligned}
\binom{n+k+1}{n+1} &= \binom{n+k}{n} + \binom{n+k}{n+1} \\
&= \binom{n+k}{n} + \binom{n+k-1}{n} + \binom{n+k-1}{n+1} \\
&= \binom{n+k}{n} + \binom{n+k-1}{n} + \binom{n+k-2}{n} + \binom{n+k-2}{n+1} \\
&= \dots \\
&= \binom{n+k}{n} + \binom{n+k-1}{n} + \binom{n+k-2}{n} + \dots + \binom{n+1}{n} + \binom{n+1}{n+1} \\
&= \binom{n+k}{n} + \binom{n+k-1}{n} + \binom{n+k-2}{n} + \dots + \binom{n+1}{n} + \binom{n}{n}
\end{aligned}$$

since $\binom{n+1}{n+1} = 1 = \binom{n}{n}$. This proves (f).

Finally, (g) follows from the **binomial theorem** by substituting $x = y = 1$. $\square$

Problem 1.3. (HKIMO Prelim 2012) Given a, b, c, d (where $a < b < c < d$) are positive integers not exceeding 2012, and they form an arithmetic sequence in this order. How many different sets of possible values of (a, b, c, d) are there?

Analysis. Although we need to choose 4 numbers from the first 2012 positive integers, there is nothing to do with $\binom{2012}{4}$. Note that the arithmetic sequence condition imposes constraints on all but 2 variables. In other words, if 2 of a, b, c, d are fixed, then the remaining numbers are uniquely determined (but they need not be integers and need not be smaller than or equal to 2012). To deal with the upper bound and the implicit lower bound $a > 0$, one way is to fix a and d .

Solution. Suppose a, b, c, d form an arithmetic sequence in this order. Then we can write $b = a + s$, $c = a + 2s$ and $d = a + 3s$ for some positive integer s . Therefore, we must have $d \equiv a \pmod{3}$.

Conversely, for any fixed a and d satisfying $d \equiv a \pmod{3}$, we can define $b = \frac{2a+d}{3}$ and $c = \frac{a+2d}{3}$ so that both of them are integers and a, b, c, d form an arithmetic sequence. Therefore, the problem is the same as finding positive integers a and d not exceeding 2012 such that they leave the same remainder when divided by 3.

If $a \equiv d \equiv 0 \pmod{3}$, they can be one of $3, 6, 9, \dots, 2010$. There are $\binom{670}{2}$ choices.

Similarly, if $a \equiv d \equiv 1 \pmod{3}$, there are $\binom{671}{2}$ choices. If $a \equiv d \equiv 2 \pmod{3}$, there are $\binom{671}{2}$ choices.

Thus, the total number of possible tuples is

$$\binom{670}{2} + \binom{671}{2} + \binom{671}{2} = \frac{670 \times (669 + 671 + 671)}{2} = 673685.$$

Problem 1.4. (HKIMO TST 2013 Test 2 Problem 2) Let $S = \{1, 2, 3, \dots, 2n\}$, where n is a positive integer greater than or equal to 1. For any subset T of S , T is called a good subset if in T the number of even elements is greater than the number of odd elements.

- (i) Find the total number of good subsets of S .
- (ii) Find the sum of all the elements in all the good subsets of S .

Analysis. We are going to present a more algebraic solution to this problem. This means we first write down a formula for the answer using the definition of binomial coefficients and simplify this formula using algebraic tricks and the properties of binomial coefficients. To count in this algebraic way, one should know how to deal with the summation sign. In particular, dealing with double (multiple) sum is not an easy task.

Solution.

- (i) There are n odd numbers and n even numbers in S . Suppose we choose i odd elements and j even elements. The requirement for a subset to be good is $0 \leq i < j \leq n$. Therefore, we have to evaluate

$$N = \sum_{i=0}^{n-1} \sum_{j=i+1}^n \binom{n}{i} \binom{n}{j}.$$

There is no neat formula for the product of two binomial coefficients. However, sometimes a sum over such summands can be simplified. One well-known example is

$$\sum_{i=0}^k \binom{n}{i} \binom{n}{k-i} = \binom{2n}{k}. \quad (1)$$

Indeed, to choose k objects from $2n$ objects, we can partition the $2n$ objects into two equal groups, choose i objects from the first group and another $k-i$ objects from the second group. Here, i runs from 0 to k , hence suggesting the formula (1).

In view of this, we try to rewrite the summand of N as something similar to that in (1). This can be done via substitution. Note that

$$N = \sum_{i=0}^{n-1} \sum_{j=i+1}^n \binom{n}{i} \binom{n}{j} = \sum_{i=0}^{n-1} \sum_{j=i+1}^n \binom{n}{i} \binom{n}{n-j}.$$

Let $k = n - j + i$ so that the summand becomes $\binom{n}{i}\binom{n}{k-i}$. Then the condition on k (which comes from the bounds on j) is $i \leq k \leq n - 1$. This means

$$N = \sum_{i=0}^{n-1} \sum_{k=i}^{n-1} \binom{n}{i} \binom{n}{k-i}.$$

We would like the inner sum to be a sum over i so that we can apply (1). This can be done by swapping the order of the double sum. Note that k is bounded below by 0, and i is bounded above by k . Hence, we have

$$N = \sum_{k=0}^{n-1} \sum_{i=0}^k \binom{n}{i} \binom{n}{k-i}.$$

Using (1), this becomes $\sum_{k=0}^{n-1} \binom{2n}{k}$. This can be further simplified using proposition 1.4(g). Since

$$\begin{aligned} 2^{2n} &= \binom{2n}{0} + \cdots + \binom{2n}{n-1} + \binom{2n}{n} + \binom{2n}{n+1} + \cdots + \binom{2n}{2n} \\ &= \binom{2n}{0} + \cdots + \binom{2n}{n-1} + \binom{2n}{n} + \binom{2n}{n-1} + \cdots + \binom{2n}{0} \\ &= N + \binom{2n}{n} + N, \end{aligned}$$

we deduce $N = 2^{2n-1} - \frac{1}{2} \binom{2n}{n}$. This is the simplest form that we can give.

- (ii) Due to symmetry, it suffices to count the number of times the integers 1 and 2 occur in the good subsets respectively.

For the integer 1, we count the number of good subsets containing 1. Suppose we choose i other odd numbers and j even numbers. This time we need $j \geq i + 2$. Hence, the desired number is

$$S_1 = \sum_{i=0}^{n-2} \sum_{j=i+2}^n \binom{n-1}{i} \binom{n}{j}.$$

Using the same trick as in part (i) (with the substitution $k = n - j + i$), we find that

$$\begin{aligned} S_1 &= \sum_{i=0}^{n-2} \sum_{j=i+2}^n \binom{n-1}{i} \binom{n}{n-j} = \sum_{i=0}^{n-2} \sum_{k=i}^{n-2} \binom{n-1}{i} \binom{n}{k-i} \\ &= \sum_{k=0}^{n-2} \sum_{i=0}^k \binom{n-1}{i} \binom{n}{k-i} = \sum_{k=0}^{n-2} \binom{2n-1}{k}. \end{aligned}$$

As

$$\begin{aligned}
2^{2n-1} &= \binom{2n-1}{0} + \cdots + \binom{2n-1}{n-2} + \binom{2n-1}{n-1} + \binom{2n-1}{n} + \binom{2n-1}{n+1} + \cdots + \binom{2n-1}{2n-1} \\
&= \binom{2n-1}{0} + \cdots + \binom{2n-1}{n-2} + \binom{2n-1}{n-1} + \binom{2n-1}{n-1} + \binom{2n-1}{n-2} + \cdots + \binom{2n-1}{0} \\
&= S_1 + 2\binom{2n-1}{n-1} + S_1,
\end{aligned}$$

we obtain $S_1 = 2^{2n-2} - \binom{2n-1}{n-1}$.

Similarly, for the integer 2, the desired number is

$$S_2 = \sum_{i=0}^{n-1} \sum_{j=i}^{n-1} \binom{n}{i} \binom{n-1}{j}.$$

We have (here $k = n - 1 - j + i$)

$$\begin{aligned}
S_2 &= \sum_{i=0}^{n-1} \sum_{j=i}^{n-1} \binom{n}{i} \binom{n-1}{n-1-j} = \sum_{i=0}^{n-1} \sum_{k=i}^{n-1} \binom{n}{i} \binom{n-1}{k-i} \\
&= \sum_{k=0}^{n-1} \sum_{i=0}^k \binom{n}{i} \binom{n-1}{k-i} = \sum_{k=0}^{n-1} \binom{2n-1}{k} \\
&= 2^{2n-2}.
\end{aligned}$$

Therefore, the total sum is

$$\begin{aligned}
&S_1 \times (1 + 3 + \cdots + (2n-1)) + S_2 \times (2 + 4 + \cdots + 2n) \\
&= 2^{2n-2} \times n^2 - \binom{2n-1}{n-1} \times n^2 + 2^{2n-2} \times n(n+1) \\
&= 2^{2n-2} n(2n+1) - n^2 \binom{2n-1}{n-1}.
\end{aligned}$$

Remark. To obtain this kind of solution, we always use the properties of binomial coefficients. The more useful properties include $\binom{n}{r} = \binom{n}{n-r}$ and the **binomial theorem**.

Equation (1) is an example of Vandermonde's identity(范德蒙恒等式). The general form of this identity is

$$\sum_{i=0}^k \binom{n}{i} \binom{m}{k-i} = \binom{n+m}{k}$$

where $n, m \geq k$.

Sometimes when we try to count by dividing into a few scenarios, some cases may fall into more than one scenario. Then the **addition principle** cannot be applied. Indeed, we need to modify the formula by subtracting the overlapping cases. This is the spirit of the following theorem.

Theorem 1.2. (Inclusion-exclusion principle(容斥原理)) Let $S_1, S_2, \dots, S_n$ be any finite sets. Then

$$|S_1 \cup S_2 \cup \dots \cup S_n| = \sum_{i=1}^n |S_i| - \sum_{i < j} |S_i \cap S_j| + \sum_{i < j < k} |S_i \cap S_j \cap S_k| - \dots + (-1)^{n+1} |S_1 \cap S_2 \cap \dots \cap S_n|.$$

Proof. It suffices to show that every element in $S_1 \cup S_2 \cup \dots \cup S_n$ contributes exactly 1 to the right-hand side. Indeed, suppose an element a belongs to exactly m of the sets $S_1, S_2, \dots, S_n$. Then its contribution to the first sum is m , its contribution to the second sum is $\binom{m}{2}$ since we have to choose 2 of those m sets and take the intersection.

Similarly, its contribution to the i th sum is $\binom{m}{i}$ for $i = 0, 1, \dots, m$. Hence, the total contribution is

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \dots + (-1)^{m+1} \binom{m}{m}.$$

Using the **binomial theorem** to expand $0 = (1 - 1)^m$, we find that the above quantity is equal to $\binom{m}{0} = 1$. $\square$

Problem 1.5. (HKIMO TST 2014 Test 1 Problem 4) A number is good if the sum of its digits is 18 (for example 130518 is a good number). Find the number of good 6-digit numbers.

Analysis. This is equivalent to solving the equation $x_1 + x_2 + \dots + x_6 = 18$ where each variable is an integer with a certain bound. The solution using binomial coefficients is standard.

Solution. We first prove the following well-known fact.

Claim. Let n and k be positive integers. Then the number of nonnegative integer solutions to $x_1 + x_2 + \dots + x_k = n$ is $\binom{n+k-1}{n} = \binom{n+k-1}{k-1}$.

To show this, suppose there are n identical balls and $k - 1$ identical sticks. We rearrange these $n + k - 1$ objects. For each arrangement, the sticks partition the balls into k regions. For $i = 1, 2, \dots, k$, the number of balls in the i th region corresponds to the value of x_i . This corresponds to a solution to the Diophantine equation, and such a correspondence is a bijection (see subsection 1.4 for more discussions on bijection). From this combinatorial point of view, it is easy to count, since we only need to decide the places for which the n identical balls are. There are $\binom{n+k-1}{n}$ possibilities.

Back to the original problem. We are going to solve $x_1 + x_2 + \dots + x_6 = 18$ in integers where $1 \leq x_1 \leq 9$ (since the leading digit cannot be 0) and $0 \leq x_i \leq 9$ for $i = 2, 3, 4, 5, 6$. We first use the substitution $y = x_1 - 1$ so that $0 \leq y \leq 8$. The equation becomes $y + x_2 + x_3 + \dots + x_6 = 17$. The advantage is that every variable can be 0.

From the claim, there are $\binom{17+6-1}{6-1}$ solutions for which $y, x_2, \dots, x_6$ are nonnegative integers. However, those solutions for which some variables exceed their upper bounds should be rejected. For example, we eliminate those solutions satisfying $y \geq 9$ and $x_2, \dots, x_6 \geq 0$. Rewriting the equation in the form $(y-9) + x_2 + \dots + x_6 = 8$, we know that there are $\binom{8+6-1}{6-1}$ such cases. Similarly, the number of solutions for a particular x_i ($2 \leq i \leq 6$) satisfying $x_i \geq 10$ is $\binom{7+6-1}{6-1}$.

By the **inclusion-exclusion principle**, we still need to see whether it is possible for 2 of the inequalities $y \geq 9$, $x_2 \geq 10$, $\dots$, $x_6 \geq 10$ to hold simultaneously. If this could happen, then those cases should be added up again. Indeed, such cases cannot happen due to the fact that $y + x_2 + x_3 + \dots + x_6 = 17$. Thus, the final answer is

$$\binom{22}{5} - \binom{13}{5} - 5 \times \binom{12}{5} = 26334 - 1287 - 5 \times 792 = 21087.$$

1.2 Double Counting

Double counting is not a counting method. Indeed, it is a useful tool in solving combinatorial problems with the help of counting. The method is to count a certain quantity in two or more different ways to obtain a nontrivial equality. More generally, we can obtain bounds for that quantity to get an inequality that is useful for our purpose. Many identities involving the binomial coefficients can be proved in this way. We have seen such an example when we prove **Pascal's identity**.

Problem 1.6. Let n be a positive integer. Prove that $\sum_{\substack{d=1 \\ d|n}}^n \varphi(d) = n$.

Analysis. Recall that $\varphi(d)$ is Euler's totient function. It counts the number of positive integers not exceeding d which are relatively prime to d . The sum on the left-hand side runs through all positive integers d from 1 to n which divide n .

The idea of double counting is simple. Since we want to show that the sum equals n , we may try to partition the first n positive integers into certain groups so that the size of each group is $\varphi(d)$ for some d dividing n . The combinatorial meaning of $\varphi(d)$ will be useful to set up the groups.

Solution. For each positive integer m from 1 to n , we put it into the set S_d where $d = \gcd(m, n)$. This defines a partition on the set of integers from 1 to n so that we must have

$$|S_1| + |S_2| + \dots + |S_n| = n.$$

Since the subscript is the greatest common divisor of m and n , it must be a divisor of n . Therefore, only those S_d with $d \mid n$ are nonempty. We have to count the number of elements in S_d for $d \mid n$. As $\gcd(m, n) = d$ if and only if (iff) $\gcd\left(\frac{m}{d}, \frac{n}{d}\right) = 1$, we have

$$S_d = \left\{ m : 1 \leq m \leq n, \gcd(m, n) = d \right\} = \left\{ m : 1 \leq \frac{m}{d} \leq \frac{n}{d}, \gcd\left(\frac{m}{d}, \frac{n}{d}\right) = 1 \right\}.$$

By the definition of Euler's totient function, this set contains exactly $\varphi\left(\frac{n}{d}\right)$ elements. Hence,

$$n = \sum_{\substack{d=1 \\ d \mid n}}^n |S_d| = \sum_{\substack{d=1 \\ d \mid n}}^n \varphi\left(\frac{n}{d}\right) = \sum_{\substack{d=1 \\ d \mid n}}^n \varphi(d).$$

In this last step, we have replaced d by $\frac{n}{d}$. This is allowed since $d \mid n$ iff $\frac{n}{d} \mid n$. This completes the proof.

Problem 1.7. Let n and m be positive integers satisfying $n \geq m$. Prove that

$$\sum_{i=0}^n \binom{n}{i} \binom{n-i}{m-i} = 2^m \binom{n}{m}.$$

Analysis. Note that $\binom{n}{m}$ counts the number of ways to choose m objects from n objects. Also, $\binom{n}{i}$ and $\binom{n-i}{m-i}$ have the combinatorial meaning of choosing i and $m-i$ objects respectively, which sum up to m objects. This gives a linkage between the two sides. We only need to see how to deal with the extra factor 2^m . For example, we may assign colours to each of the m chosen objects.

Solution. Suppose we want to choose m objects from n objects and use one of the 2 colours, black and white, to colour each chosen object. There are obviously $\binom{n}{m} \times 2^m$ ways to do so.

Another way to count is as follows. We first choose i objects from the n objects and colour them black. Then we choose $m-i$ objects from the remaining $n-i$ objects and colour them white. There are $\binom{n}{i} \binom{n-i}{m-i}$ ways to do so for each fixed $i = 0, 1, \dots, n$. Summing over these i 's, we obtain the left-hand side of the identity. This establishes the identity.

Problem 1.8. (CHKMO 2007 Problem 3) In a school there are 2007 male and 2007 female students. Each student joins not more than 100 clubs in the school. It is known that any two students of opposite genders have joined at least one common club. Show that there is a club with at least 11 male and 11 female members.

Analysis. It is natural to prove by contradiction. From the key condition, we may try to count something related to a club joined by two students of opposite genders. The condition gives a lower bound for such a quantity while the other numerical values provide upper bounds. It is not hard to obtain a contradiction.

Solution. Suppose on the contrary that no club contains at least 11 male and 11 female members. We shall count the number of triples (M, F, C) where M is a male student, F is a female student and C is a club joined by these two students.

Firstly, for each of the 2007 choices of M and each of the 2007 choices of F , the condition shows that there is at least one club C for which (M, F, C) should be counted. This gives at least 2007^2 triples.

Secondly, we can count according to the clubs. We label the clubs as $C_1, C_2, \dots, C_k$. For each $i = 1, 2, \dots, k$, let m_i and f_i be the number of male students and female students in the club C_i respectively. Then the number of triples (M, F, C) is

$$\sum_{i=1}^k m_i f_i.$$

From the assumption, we can divide these clubs into two groups. The first group consists of those clubs with at most 10 male members (say $C_1, C_2, \dots, C_\ell$). The second group consists of those clubs with at least 11 male members, and hence at most 10 female members (say $C_{\ell+1}, C_{\ell+2}, \dots, C_k$). Then

$$\begin{aligned} \sum_{i=1}^k m_i f_i &= \sum_{i=1}^{\ell} m_i f_i + \sum_{i=\ell+1}^k m_i f_i \\ &\leq 10 \sum_{i=1}^{\ell} f_i + 10 \sum_{i=\ell+1}^k m_i \\ &\leq 10 \sum_{i=1}^k f_i + 10 \sum_{i=1}^k m_i. \end{aligned}$$

Note that the sum over f_i counts the total number of female members in all clubs. By another simple double counting, this is equal to the total number of clubs joined by all the female students. Therefore, we have

$$\sum_{i=1}^k f_i \leq 100 \times 2007.$$

The same upper bound holds for the sum over m_i . Combining all these, we obtain

$$2007^2 \leq 10 \times 100 \times 2007 + 10 \times 100 \times 2007 = 2000 \times 2007.$$

This is a contradiction. Hence, there must be a club with at least 11 male and 11 female members.

Problem 1.9. (IMO shortlist 2001 C8/Problem 3) Twenty-one girls and twenty-one boys took part in a mathematical contest.

- Each contestant solved at most six problems.
- For each girl and each boy, at least one problem was solved by both of them.

Prove that there was a problem that was solved by at least three girls and at least three boys.

Analysis. The setting is exactly the same as the previous problem. However, the same solution does not apply directly since we would obtain the inequality

$$21^2 \leq 2 \times 6 \times 21 + 2 \times 6 \times 21 = 24 \times 21$$

if we follow the same steps. Therefore, we need to modify the arguments to strengthen the bounds.

Solution. Suppose on the contrary that no problem was solved by at least 3 girls and 3 boys. We count the number of triples (P, G, B) where P was a problem solved by a girl G and a boy B .

Firstly, for each of the 21 choices of G and each of the 21 choices of B , there exists a P such that (P, G, B) should be counted. This gives at least 21^2 triples.

Secondly, we divide the problems into two groups. Group I consists of those problems which were solved by at most 2 girls. Group II consists of those problems which were solved by at least 3 girls, and hence at most 2 boys. We first consider those triples (P, G, B) for which each problem belongs to group I. We call these special triples.

Intuitively, since these problems were solved by a small number of girls, each such girl was involved in many triples. This in turn means that each such problem had to be solved by many boys, which may contradict the given conditions. This is the idea to proceed.

For each boy B , we claim that B solved at most 5 group I problems. Indeed, if B solved 6 group I problems, then he cannot solve any more problem due to the constraint. For each of these 6 problems, there are at most $6 \times 2 = 12$ triples to be counted, since there are at most 2 choices of G for each P . This contradicts the fact that there are at least 21 triples containing B . This proves the claim. Therefore, we have at most $5 \times 2 = 10$ special triples containing B . Since this holds for each boy, there can be at most $10 \times 21 = 210$ special triples.

Similarly, there can be at most 210 triples consisting of problems which belong to group II. This yields the contradiction

$$441 = 21^2 \leq 210 + 210 = 420.$$

Hence, there was a problem that was solved by at least 3 girls and at least 3 boys.

1.3 Recurrence Relation

In some cases, when we count step by step, we cannot simply count by using the **multiplication principle** as the steps are not independent. However, usually we can still obtain some relationships between consecutive steps. A useful tool is to use induction to apply these relationships. Alternatively, we can make use of recurrence relation to find the general term when more and more steps are involved. This means we can set up an algebraic relation connecting the k th step and the $(k + 1)$ st step and use algebraic method to solve the problem.

Before working on combinatorial problems, we first have a review on finding the general terms of some recurrence relations.

Proposition 1.5.

- (a) Let $\{a_n\}_{n \geq 0}$ be a sequence defined by $a_n = ca_{n-1} + f(n)$ for some constant c and some function f . Then

$$a_n = c^n a_0 + f(n) + cf(n-1) + c^2 f(n-2) + \cdots + c^{n-1} f(1).$$

- (b) Let $\{a_n\}_{n \geq 0}$ be a sequence defined by $a_n = ta_{n-1} + ua_{n-2}$ for some nonzero constants t and u . Let α, β be the roots of the characteristic equation $\lambda^2 - t\lambda - u = 0$.

- If $\alpha \neq \beta$, then there exist constants A, B such that

$$a_n = A\alpha^n + B\beta^n. \quad (1)$$

- If $\alpha = \beta$, then there exist constants A, B such that

$$a_n = A\alpha^n + Bn\alpha^n. \quad (2)$$

Proof.

- (a) Adding the equations

$$\begin{aligned} a_n &= ca_{n-1} + f(n), \\ ca_{n-1} &= c^2 a_{n-2} + cf(n-1), \\ c^2 a_{n-2} &= c^3 a_{n-3} + c^2 f(n-2), \\ &\dots \\ c^{n-1} a_1 &= c^n a_0 + c^{n-1} f(1), \end{aligned}$$

we obtain the desired formula.

- (b) Firstly, suppose $\alpha \neq \beta$. Then the equation

$$\begin{cases} A\alpha + B\beta = a_1, \\ A + B = a_0 \end{cases}$$

has a unique solution

$$A = \frac{a_1 - a_0\beta}{\alpha - \beta}, \quad B = \frac{a_1 - a_0\alpha}{\beta - \alpha}.$$

Suppose (1) holds for $n = k$ and $n = k + 1$. Then

$$\begin{aligned} a_{k+2} &= ta_{k+1} + ua_k \\ &= t(A\alpha^{k+1} + B\beta^{k+1}) + u(A\alpha^k + B\beta^k) \\ &= (t\alpha + u)A\alpha^k + (t\beta + u)B\beta^k \\ &= \alpha^2 \cdot A\alpha^k + \beta^2 \cdot B\beta^k \\ &= A\alpha^{k+2} + B\beta^{k+2}. \end{aligned}$$

Hence, (1) is true for all integers $n \geq 0$ by induction.

Similarly, if $\alpha = \beta$, in which case we must have $\alpha = \frac{t}{2}$, we consider the equation

$$\begin{cases} A\alpha + B\alpha = a_1, \\ A = a_0. \end{cases}$$

There exists a unique solution

$$A = a_0, \quad B = \frac{a_1}{\alpha} - a_0.$$

Suppose (2) holds for $n = k$ and $n = k + 1$. Then

$$\begin{aligned} a_{k+2} &= ta_{k+1} + ua_k \\ &= t(A\alpha^{k+1} + B(k+1)\alpha^{k+1}) + u(A\alpha^k + Bk\alpha^k) \\ &= (t\alpha + u)A\alpha^k + (t\alpha + u)Bk\alpha^k + tB\alpha^{k+1} \\ &= \alpha^2 \cdot A\alpha^k + \alpha^2 \cdot Bk\alpha^k + 2\alpha \cdot B\alpha^{k+1} \\ &= A\alpha^{k+2} + B(k+2)\alpha^{k+2}. \end{aligned}$$

Hence, (2) is true for all integers $n \geq 0$ by induction. □

Note that the constants A, B can be determined by the initial conditions, i.e. the values of a_0, a_1 .

Problem 1.10. (HKIMO Prelim 2016) 10 boys are seated in a row, and each of them puts on a red, green or blue hat randomly. What is the probability that everyone finds at least one neighbour (another boy sitting on his left or right) who has put on a hat of the same colour as his own?

Analysis. We can generalize the same problem for n boys instead of 10. The condition does not allow us to count by assigning hats to the boys one by one. Indeed, we have to consider some neighbours as a whole and deduce some recurrence relations.

Solution. Let a_n be the number of ways to assign a hat (in one of the 3 colours) to n boys in a row such that each boy finds at least one neighbour whose hat has the same colour as his. Clearly, we have $a_1 = 0$ and $a_2 = 3$.

Consider a row of n boys with $n \geq 3$. Note that the hat of the n th boy must have the same colour as that of the $(n-1)$ st boy. If this colour is not the same as that of the $(n-2)$ nd boy, then there are a_{n-2} ways to assign hats to the first $n-2$ boys by definition. Afterwards, there are 2 ways to assign hats to the last 2 boys so that the colour is distinct from that of the $(n-2)$ nd boy.

If the colour of hats of the $(n-2)$ nd boy is the same as that of the $(n-1)$ st boy, then there are a_{n-1} ways to assign hats to the first $n-1$ boys. Then the colour of hat of the last boy is uniquely determined.

Therefore, we have established a recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 3$. Solving the characteristic equation $\lambda^2 - \lambda - 2 = 0$, we obtain two distinct roots 2 and -1 . This shows

$$a_n = A \cdot 2^n + B \cdot (-1)^n$$

for some constants A, B . Solving

$$\begin{cases} 4A + B = 3, \\ 2A - B = 0, \end{cases}$$

we get $A = \frac{1}{2}$ and $B = 1$. This means $a_n = 2^{n-1} + (-1)^n$ for all integers $n \geq 1$. The probability is thus $\frac{2^{n-1} + (-1)^n}{3^n}$. In particular, when $n = 10$, we obtain

$$a_{10} = \frac{513}{3^{10}} = \frac{19}{2187}.$$

Problem 1.11. (CHKMO 2003 Problem 2) Let $ABCDEF$ be a regular hexagon of side length 1, and O be the centre of the hexagon. In addition to the sides of the hexagon, line segments are drawn from O to each vertex, making a total of twelve unit line segments. Find the number of paths of length 2003 along these line segments that start at O and terminate at O .

Analysis. This is another standard example which can be solved using recurrence relation. Note that it is more convenient to introduce 2 sequences corresponding to different terminating points, since there may not be any intermediate step which stays at point O .

Solution. For each integer $n \geq 0$, let a_n be the number of paths of length n starting at O and terminating at O . Also, let b_n be the number of paths of length n starting at O and terminating at A . By symmetry, b_n is also the number of those paths terminating at any of B, C, D, E, F .

To count a_n for $n \geq 1$, the $(n-1)$ st step must terminate at one of A, B, C, D, E, F . For each of these 6 points, there are b_{n-1} paths terminating at that point. This gives a total of $6b_{n-1}$ paths. In other words, we have

$$a_n = 6b_{n-1}. \quad (1)$$

To count b_n for $n \geq 1$, suppose the n th step terminates at A . Then the $(n-1)$ st step may terminate at B, F or O . If it is B or F , then there are b_{n-1} paths. If it is O , then there are a_{n-1} paths. Therefore, we have

$$b_n = 2b_{n-1} + a_{n-1}. \quad (2)$$

We now combine (1) and (2) to obtain a recurrence relation for the sequence $\{a_n\}$. This can be done by replacing every b_n by a_{n+1} in view of (1). We have

$$\frac{1}{6}a_{n+1} = b_n = 2b_{n-1} + a_{n-1} = \frac{1}{3}a_n + a_{n-1}.$$

This can be rewritten as $a_n = 2a_{n-1} + 6a_{n-2}$ for $n \geq 2$. The roots of the characteristic equation $\lambda^2 - 2\lambda - 6 = 0$ are $1 \pm \sqrt{7}$. Suppose $a_n = A(1 + \sqrt{7})^n + B(1 - \sqrt{7})^n$. Using the initial conditions $a_0 = 1$ and $a_1 = 0$, we solve

$$\begin{cases} A(1 + \sqrt{7}) + B(1 - \sqrt{7}) = 0, \\ A + B = 1. \end{cases}$$

This yields $A = \frac{7 - \sqrt{7}}{14}$ and $B = \frac{7 + \sqrt{7}}{14}$. Therefore, the answer is

$$a_{2003} = \frac{7 - \sqrt{7}}{14}(1 + \sqrt{7})^{2003} + \frac{7 + \sqrt{7}}{14}(1 - \sqrt{7})^{2003}.$$

Remark. Note that the same recurrence relation holds for the sequence $\{b_n\}$, i.e. $b_n = 2b_{n-1} + 6b_{n-2}$. This is a common phenomenon of this kind of problem.

Problem 1.12. (IMO shortlist 2008 C4/Problem 5) Let n and k be positive integers with $k \geq n$ and $k - n$ an even number. Let $2n$ lamps labelled $1, 2, \dots, 2n$ be given, each of which can be either on or off. Initially all the lamps are off. We consider sequences of steps: at each step one of the lamps is switched (from on to off or from off to on).

Let N be the number of such sequences consisting of k steps and resulting in the state where lamps 1 through n are all on, and lamps $n + 1$ through $2n$ are all off.

Let M be the number of such sequences consisting of k steps, resulting in the state where lamps 1 through n are all on, and lamps $n + 1$ through $2n$ are all off, but where none of the lamps $n + 1$ through $2n$ is ever switched on.

Determine the ratio $\frac{N}{M}$.

Analysis. By working on the small values, we can observe that the ratio depends on n and k . Thus, we shall write $N(n, k)$ and $M(n, k)$ instead of N and M for convenience.

We can find that

$$\begin{aligned}\frac{N(1, 1)}{M(1, 1)} &= \frac{1}{1} = 1, \quad \frac{N(1, 3)}{M(1, 3)} = \frac{4}{1} = 4, \quad \frac{N(1, 5)}{M(1, 5)} = \frac{16}{1} = 16, \\ \frac{N(2, 2)}{M(2, 2)} &= \frac{2}{2} = 1, \quad \frac{N(2, 4)}{M(2, 4)} = \frac{32}{8} = 4, \quad \frac{N(2, 6)}{M(2, 6)} = \frac{512}{32} = 16,\end{aligned}$$

etc. It is not hard to guess the answer from these values. Indeed, we claim that $\frac{N(n, k)}{M(n, k)} = 4^{\frac{k-n}{2}} = 2^{k-n}$. We can use recurrence relation or induction to prove this formula.

Solution. For the case $n = 1$, there are only 2 lamps. For $N(1, k)$, lamp 2 should be switched for an even number of times. Therefore,

$$N(1, k) = \sum_{i=0}^{\frac{k-1}{2}} \binom{k}{2i}.$$

Note that

$$\begin{aligned}2^k &= \binom{k}{0} + \binom{k}{2} + \cdots + \binom{k}{k-1} + \binom{k}{k} + \binom{k}{k-2} + \cdots + \binom{k}{1} \\ &= \binom{k}{0} + \binom{k}{2} + \cdots + \binom{k}{k-1} + \binom{k}{0} + \binom{k}{2} + \cdots + \binom{k}{k-1} \\ &= 2N(1, k).\end{aligned}$$

This gives $N(1, k) = 2^{k-1}$. It is obvious that $M(1, k) = 1$. Hence, $\frac{N(1, k)}{M(1, k)} = 2^{k-1}$.

For the case $k = n$, we have $N(k, k) = M(k, k) = k!$. This implies $\frac{N(k, k)}{M(k, k)} = 1$.

We claim that $\frac{N(n, k)}{M(n, k)} = 2^{k-n}$ for any n, k satisfying the conditions. Suppose this holds for all such pairs with $n \leq n_0$, $k \leq k_0$ and $(n, k) \neq (n_0, k_0)$. We are going to show that the same holds for the case (n_0, k_0) .

Firstly, for $M(n_0, k_0)$, suppose lamp n_0 is switched i times where i is an odd integer ranging from 1 to $k_0 - n_0 + 1$. Then there are $\binom{k_0}{i}$ ways to choose these steps, and the remaining $k_0 - i$ steps are used for the first $n_0 - 1$ lamps only. This corresponds to the case $(n_0 - 1, k_0 - i)$. Therefore, we obtain

$$M(n_0, k_0) = \sum_{\substack{i=1 \\ i \text{ odd}}}^{k_0 - n_0 + 1} \binom{k_0}{i} M(n_0 - 1, k_0 - i). \quad (1)$$

Secondly, for $N(n_0, k_0)$, suppose lamp n_0 is switched i times (where $1 \leq i \leq k_0 - n_0 + 1$ is odd) and lamp $2n_0$ is switched j times (where $0 \leq j \leq k_0 - n_0 - i + 1$ is even). Similar as above, we obtain

$$N(n_0, k_0) = \sum_{\substack{i=1 \\ i \text{ odd}}}^{k_0-n_0+1} \sum_{\substack{j=0 \\ j \text{ even}}}^{k_0-n_0-i+1} \binom{k_0}{i} \binom{k_0-i}{j} N(n_0-1, k_0-i-j). \quad (2)$$

From the inductive hypothesis, we have

$$N(n_0-1, k_0-i-j) = 2^{k_0-n_0-i-j+1} M(n_0-1, k_0-i-j).$$

By comparing (1) and (2), we should replace $i+j$ in the double sum by a single variable. Let $m = i+j$. Then m is odd and $1 \leq m \leq k_0 - n_0 + 1$. This gives

$$N(n_0, k_0) = \sum_{\substack{m=1 \\ m \text{ odd}}}^{k_0-n_0+1} \sum_{\substack{i=1 \\ i \text{ odd}}}^m \binom{k_0}{i} \binom{k_0-i}{m-i} 2^{k_0-n_0-m+1} M(n_0-1, k_0-m).$$

We should now focus on the term $\sum_{\substack{i=1 \\ i \text{ odd}}}^m \binom{k_0}{i} \binom{k_0-i}{m-i}$. Analogous to problem 1.7, we can prove that this sum is equal to $\binom{k_0}{m} 2^{m-1}$. Hence,

$$\begin{aligned} N(n_0, k_0) &= \sum_{\substack{m=1 \\ m \text{ odd}}}^{k_0-n_0+1} \binom{k_0}{m} 2^{m-1} \cdot 2^{k_0-n_0-m+1} M(n_0-1, k_0-m) \\ &= 2^{k_0-n_0} \sum_{\substack{m=1 \\ m \text{ odd}}}^{k_0-n_0+1} \binom{k_0}{m} M(n_0-1, k_0-m) = 2^{k_0-n_0} \cdot M(n_0, k_0). \end{aligned}$$

This completes the inductive step and hence $\frac{N(n, k)}{M(n, k)} = 2^{k-n}$ for all possible pairs (n, k) as desired.

Problem 1.13. (APMO 2017 Problem 3) Let $A(n)$ denote the number of sequences $a_1 \geq a_2 \geq \dots \geq a_k$ of positive integers for which $a_1 + \dots + a_k = n$ and each $a_i + 1$ is a power of two ($i = 1, 2, \dots, k$). Let $B(n)$ denote the number of sequences $b_1 \geq b_2 \geq \dots \geq b_m$ of positive integers for which $b_1 + \dots + b_m = n$ and each inequality $b_j \geq 2b_{j+1}$ holds ($j = 1, 2, \dots, m-1$).

Prove that $A(n) = B(n)$ for every positive integer n .

Analysis. A natural strategy is to establish a certain recurrence relation for each of the sequences $\{A(n)\}$ and $\{B(n)\}$, and complete the proof by induction. Therefore, we first consider some small cases to look for those relationships.

When $n = 1$, the only representation is $1 = 1$, and hence $A(1) = B(1) = 1$.

When $n = 2$, we have $A(2) = 1$, which corresponds to $1 + 1 = 2$, and we have $B(2) = 1$, which corresponds to $2 = 2$.

When $n = 3$, we have $A(3) = 2$, which corresponds to $1 + 1 + 1 = 3$ and $3 = 3$, and we have $B(3) = 2$, which corresponds to $2 + 1 = 3$ and $3 = 3$.

When $n = 4$, we have $A(4) = 2$, which corresponds to $1 + 1 + 1 + 1 = 4$ and $3 + 1 = 4$, and we have $B(4) = 2$, which corresponds to $3 + 1 = 4$ and $4 = 4$. Similarly, we find that $A(5) = B(5) = 2$.

When $n = 6$, we have $A(6) = 3$, which corresponds to $1 + 1 + 1 + 1 + 1 + 1 = 6$, $3 + 1 + 1 + 1 = 6$ and $3 + 3 = 6$, and we have $B(6) = 3$, which corresponds to $5 + 1 = 6$, $4 + 2 = 6$ and $6 = 6$.

When $n = 7$, we have $A(7) = 4$, which corresponds to $1 + 1 + 1 + 1 + 1 + 1 + 1 = 7$, $3 + 1 + 1 + 1 + 1 = 7$, $3 + 3 + 1 = 7$ and $7 = 7$, and we have $B(7) = 4$, which corresponds to $4 + 2 + 1 = 7$, $6 + 1 = 7$, $5 + 2 = 7$ and $7 = 7$.

We should now have some feeling on these sequences. Firstly, the solutions corresponding to $A(n)$ and $B(n)$ are not the same (although we need to show that the number of solutions are the same). Secondly, solutions corresponding to $A(n)$ with smaller number of variables is somehow related to solutions corresponding to $B(n)$ with larger number of variables. For example, when $n = 2^s - 1$ (e.g. $n = 7$), the special solution $n = n$ contributes 1 to $A(n)$, while the special solution (in the sense that each variable is twice of the next variable) $2^{s-1} + 2^{s-2} + \dots + 1 = n$ contributes 1 to $B(n)$. These contributions only arise when n changes from $2^s - 2$ to $2^s - 1$. From this observation, we may try to connect the sequences $\{A(n)\}$ and $\{B(n)\}$ via the number of variables or the size of the variables.

Solution. For any positive integers n, s , let $A(n, s)$ be the number of solutions to $a_1 + \dots + a_k = n$ counted by $A(n)$ with the restriction that the largest variable a_1 is equal to $2^s - 1$. Let $B(n, s)$ be the number of solutions to $b_1 + \dots + b_m = n$ counted by $B(n)$ with the restriction that the number of variables m is equal to s . Based on the observation, we prove the following.

Claim. For any positive integers n, s , we have $A(n, s) = B(n, s)$.

If this claim holds, then we easily obtain

$$A(n) = A(n, 1) + A(n, 2) + \dots = B(n, 1) + B(n, 2) + \dots = B(n).$$

We shall prove the claim by induction on n . When $n = 1$, we have $A(1, 1) = 1 = B(1, 1)$ and $A(1, s) = 0 = B(1, s)$ for $s \geq 2$. These can be verified directly.

Now, we proceed to the inductive step and work on the case n . Note that both $A(n, s)$ and $B(n, s)$ are zero if $n < 2^s - 1$. For the special case $n = 2^s - 1$, we have $A(n, s) = 1$ as the only solution is $2^s - 1 = 2^s - 1$, and $B(n, s) = 1$ as the only solution is $2^{s-1} + 2^{s-2} + \dots + 1 = 2^s - 1$. In the following, we consider the general case where $n > 2^s - 1$.

For $A(n, s)$, we first remove the term $a_1 = 2^s - 1$ on the left-hand side of $a_1 + \dots + a_k = n$. Then we have to solve $a_2 + \dots + a_k = n - (2^s - 1)$ under the given constraints and $a_2 \leq 2^s - 1$. This corresponds to the cases $A(n - 2^s + 1, t)$ with $t = 1, 2, \dots, s$. This gives the recurrence relation

$$A(n, s) = A(n - 2^s + 1, 1) + A(n - 2^s + 1, 2) + \dots + A(n - 2^s + 1, s). \quad (1)$$

For $B(n, s)$, in view of (1), we should reduce the equation $b_1 + \dots + b_s = n$ to a sum equal to $n - 2^s + 1$. Since the condition $b_j \geq 2b_{j+1}$ should be kept (and motivated by the special solution as discussed), we define $c_j = b_j - 2^{s-j}$ for $j = 1, 2, \dots, s$ so that the equation becomes $c_1 + \dots + c_s = n - 2^s + 1$. Note that $b_j \geq 2b_{j+1}$ iff $c_j \geq 2c_{j+1}$ for each $j = 1, 2, \dots, s - 1$. This shows we are reducing to cases counted in $B(n - 2^s + 1)$. Although we write $c_1 + \dots + c_s$ here, it does not mean that the equation consists of s variables. This is because each c_j here could be 0. If we remove the zero terms, the equation consists of at most s variables. This gives

$$B(n, s) = B(n - 2^s + 1, 1) + B(n - 2^s + 1, 2) + \dots + B(n - 2^s + 1, s). \quad (2)$$

Comparing (1) and (2), we find that $A(n, s) = B(n, s)$ by induction. Hence, the claim is established and the proof is complete.

1.4 Bijection

A bijection is a function for which every element in the codomain has a unique preimage in the domain. This means there is a correspondence between the elements in the domain and the elements in the codomain. When the domain and the codomain are finite sets, a simple corollary is that they have the same size.

For the application in counting, we can try to rephrase the things that we need to count to some other things through a bijection. Sometimes the counting job becomes easier using different formulations. Indeed, we have seen some examples in previous subsections. For example, when we solve a Diophantine equation of the form $x_1 + x_2 + \dots + x_k = n$ in problem 1.5, we use balls and sticks to transform the problem. This reduces the problem to a counting problem which can be easily done using binomial coefficients. Some other examples can be found when we prove a certain identity using combinatorial meanings.

Problem 1.14. Let $n \geq 7$ be an integer. How many ways are there to choose 4 positive integers from $1, 2, \dots, n$ so that the difference of any 2 chosen integers is at least 2?

Analysis. We need to choose 4 integers from n integers. But the answer is certainly not $\binom{n}{4}$ since some chosen integers may be consecutive in some cases. The **inclusion-exclusion principle** may be used, but such a method is not efficient if 4 is replaced by a larger integer.

Indeed, we would form a bijection so that the condition disappears. Then we simply need to count the number of ways to choose r objects from m objects, which is a standard problem.

Solution. Let the chosen positive integers be x_1, x_2, x_3, x_4 . Without loss of generality (WLOG) assume $x_1 < x_2 < x_3 < x_4$. The condition implies x_j and x_{j+1} are not consecutive for $j = 1, 2, 3$.

Consider the mapping $(x_1, x_2, x_3, x_4) \mapsto (y_1, y_2, y_3, y_4)$ where

$$\begin{aligned} y_1 &= x_1, \\ y_2 &= x_2 - 1, \\ y_3 &= x_3 - 2, \\ y_4 &= x_4 - 3. \end{aligned}$$

Firstly, as x_1 and x_2 are not consecutive and $x_1 < x_2$, we have $x_1 < x_2 - 1$. This means $y_1 < y_2$. Similarly, we find that $y_1 < y_2 < y_3 < y_4$. Secondly, y_1, y_2, y_3, y_4 are positive integers. Also, we have $y_1 \geq 1$ and $y_4 = x_4 - 3 \leq n - 3$. Therefore, we need to choose 4 distinct positive integers from $1, 2, \dots, n - 3$. From the definition, it is easy to see that the mapping from (x_1, x_2, x_3, x_4) to (y_1, y_2, y_3, y_4) is bijective. Indeed, it suffices to show that any set of 4 distinct positive integers from 1 to $n - 3$ corresponds to exactly one set of 4 positive integers satisfying the given condition in the problem.

In this formulation, there is no further restriction on y_1, y_2, y_3, y_4 (indeed, the given condition is used implicitly when we prove the bijectivity). Therefore, it is clear that the answer is $\binom{n-3}{4}$.

Problem 1.15. (CMO 2001 Problem 5) The circumference of a circle with perimeter 24 is divided into 24 equal parts. We are going to choose 8 points from these 24 points of division so that the arc length formed between any two points is not equal to 3 or 8. How many different ways are there to choose the 8 points satisfying the constraint?

Analysis. A natural way is to use numbers to restate the problem. This means we need to choose 8 numbers from $0, 1, \dots, 23$ so that no 2 chosen numbers differ by 3 or 8 (mod 24). Since a difference of 3 or 8 is important, we may further rearrange the numbers so that neighbouring numbers differ by 3 or 8. This reduces to a simple counting problem.

Solution. Label the points from 0 to 23 in this order. We rearrange these 24 numbers as follows:

0	3	6	9	12	15	18	21
8	11	14	17	20	23	2	5
16	19	22	1	4	7	10	13

The key feature of this table is that any 2 numbers from 0 to 23 differ by 3 or 8 (mod 24) iff they are neighbours in this table. Here, 0, 21 are neighbours, 0, 16

are neighbours, etc. Hence, we only need to count the number of ways to choose 8 integers from this table so that no 2 chosen numbers are neighbours.

First of all, we can only choose 1 number from each column because any 2 numbers in the same column are neighbours. As there are 8 columns and we have to choose 8 numbers, we need to choose exactly 1 number from each column. The problem can now be solved easily using recurrence relation.

For any positive integer n , consider a $3 \times n$ table. Let a_n be the number of ways to choose one cell from each column so that no 2 chosen cells are neighbours of each other (recall our convention of neighbours). We have to find a_8 .

Suppose $n \geq 3$. If the chosen cell in column $n - 1$ lies in a different row from the chosen cell in column 1, then there are a_{n-1} ways to choose the cells in the first $n - 1$ columns. In order that the condition is satisfied, the chosen cell in column n is uniquely determined.

If the chosen cell in column $n - 1$ lies in the same row as the chosen cell in column 1, then the chosen cell in column $n - 2$ lies in a different row from the chosen cell in column 1. There are a_{n-2} ways to choose the cells in the first $n - 2$ columns (and hence $n - 1$ columns). Also, there are 2 ways to choose the cell in column n .

We obtain the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 3$. The initial conditions are $a_1 = 0$ (since the first column is neighbour of the last column which is itself) and $a_2 = 6$. Since we are only interested in a_8 , there is no need to find the general term. We have

$$\{a_n\}_{n \geq 1} = 0, 6, 6, 18, 30, 66, 126, 258, \dots$$

The answer is $a_8 = 258$.

Problem 1.16. (IMO shortlist 2002 C3) Let n be a positive integer. A sequence of n positive integers (not necessarily distinct) is called full if it satisfies the following condition: for each positive integer $k \geq 2$, if the number k appears in the sequence then so does the number $k - 1$, and moreover the first occurrence of $k - 1$ comes before the last occurrence of k . For each n , how many full sequences are there?

Analysis. It is not hard to guess there are $n!$ full sequences by considering small cases. Since $n!$ counts the number of permutations of $1, 2, \dots, n$, we may try to think of a bijection between the two sequences. In other words, we have to label the terms in a full sequence from 1 to n so that two full sequences are labelled differently and every label corresponds to a full sequence. From the condition, such a labelling should be related to the size and order of the integers in the sequence.

Solution. Let S be the set of all full sequences and let T be the set of all permutations of $1, 2, \dots, n$. For any full sequence $(a_1, a_2, \dots, a_n)$, we consider a function f defined by $f(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$, where $b_1, b_2, \dots, b_n$ is obtained by labelling the integers in $a_1, a_2, \dots, a_n$ using 1 to n in ascending order of size, and from right to

left if the integers are equal. For example, the full sequence $(2, 1, 1, 2, 3)$ becomes $(4, 2, 1, 3, 5)$, while the full sequence $(2, 1, 2, 3, 1)$ becomes $(4, 2, 3, 5, 1)$. This clearly gives a mapping $f: S \rightarrow T$. We shall show that f is a bijection, so that $|S| = |T| = n!$.

It suffices to find an inverse for f . For any $(b_1, b_2, \dots, b_n) \in T$, we define another function g as follows. We first partition $1, 2, \dots, n$ into several (possibly only one) groups

$$(1, 2, \dots, d_1), (d_1 + 1, d_1 + 2, \dots, d_2), \dots, (d_{s-1} + 1, d_{s-1} + 2, \dots, d_s)$$

so that for any two numbers in the same group, the larger number appears on the left of the smaller number in the sequence $b_1, b_2, \dots, b_n$, and the number $d_i + 1$ appears on the right of d_i for $i = 1, 2, \dots, s - 1$. Then for each $j = 1, 2, \dots, n$, if b_j belongs to the m th group, we define $a_j = m$. Then we define $g(b_1, b_2, \dots, b_n) = (a_1, a_2, \dots, a_n)$. From the construction, it is obvious that the first occurrence of $k - 1$ comes before the last occurrence of k in the sequence $(a_1, a_2, \dots, a_n)$. Therefore, $(a_1, a_2, \dots, a_n) \in S$. We now show that g is the inverse of f .

Suppose $f(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ and $g(b_1, b_2, \dots, b_n) = (c_1, c_2, \dots, c_n)$. Suppose $a_{i_1} = a_{i_2} = \dots = a_{i_v} = m$ with $i_1 < i_2 < \dots < i_v$ and all other terms are distinct from them. Then $b_{i_1}, b_{i_2}, \dots, b_{i_v}$ are consecutive integers in descending order. Therefore, they belong to the same group in the definition of $g(b_1, b_2, \dots, b_n)$. Also, $b_{i_1} + 1$ appears on the right of b_{i_1} , while $b_{i_v} - 1$ (if exists) appears on the left of b_{i_v} . This shows $b_{i_1}, b_{i_2}, \dots, b_{i_v}$ are the only numbers in this group. Therefore, the terms for $m = 1$ correspond to group 1, the terms for $m = 2$ correspond to group 2, etc. This yields $c_{i_1} = c_{i_2} = \dots = c_{i_v} = m$. As this is true for any m , this gives $(a_1, a_2, \dots, a_n) = (c_1, c_2, \dots, c_n)$ and hence $g \circ f$ is the identity on S .

Next, suppose $g(b_1, b_2, \dots, b_n) = (a_1, a_2, \dots, a_n)$ and $f(a_1, a_2, \dots, a_n) = (e_1, e_2, \dots, e_n)$. Consider any term b_j which is the ℓ th term in the group $(d_{i-1} + 1, d_{i-1} + 2, \dots, d_i)$ (suppose $d_0 = 0$). Then $a_j = i$ by definition. Also, there are d_{i-1} terms in $(a_1, a_2, \dots, a_n)$ smaller than i , and ℓ terms in $(a_1, a_2, \dots, a_n)$ which are equal to i with indices larger than or equal to j . By definition of f , we have $e_j = d_{i-1} + \ell = b_j$. Hence, $f \circ g$ is the identity on T .

These yield $g = f^{-1}$ so that $|S| = |T| = n!$.

Remark. The rigorous proof that the function is bijective is a bit clumsy, mainly due to the notations. However, it is rather obvious if one really understands the labelling.

Problem 1.17. (IMO shortlist 2013 C7/Problem 6) Let $n \geq 3$ be an integer, and consider a circle with $n + 1$ equally spaced points marked on it. Consider all labellings of these points with the numbers $0, 1, \dots, n$ such that each label is used exactly once; two such labellings are considered to be the same if one can be obtained from the other by a rotation of the circle. A labelling is called beautiful if, for any four labels

$a < b < c < d$ with $a + d = b + c$, the chord joining the points labelled a and d does not intersect the chord joining the points labelled b and c .

Let M be the number of beautiful labellings, and let N be the number of ordered pairs (x, y) of positive integers such that $x + y \leq n$ and $\gcd(x, y) = 1$. Prove that

$$M = N + 1.$$

Analysis. The main difficulty is to find M . We can look for patterns from small cases. For example, when $n = 7$, we have labellings such as $(0, 3, 6, 2, 5, 1, 4, 7)$ and $(0, 4, 3, 7, 2, 6, 1, 5)$. These look like arithmetic progressions (the former has common difference 3, while the latter has common difference 4). However, these arithmetic progressions are taken modulo m for some m ($m = 7$ for the former case, and $m = 9$ for the latter case) and terms $n + 1, n + 2, \dots, m - 1$ are missing.

From this observation, it is easy to work out some examples of beautiful labellings. The harder thing is to show that all beautiful labellings are more or less obtained in this way. Note that such a labelling is completely determined by the term after 0 and the modulus m , or determined by the two terms next to 0.

Solution. We first find N . Let $x + y = s$ so that $2 \leq s \leq n$. Then $(x, y) = 1$ iff $(x, s) = 1$. Thus, there are exactly $\varphi(s)$ solutions in x , and hence $\varphi(s)$ choices of x, y . This yields

$$N = \varphi(2) + \varphi(3) + \dots + \varphi(n). \quad (1)$$

Next, we proceed to find M . We use a row to represent the labels in clockwise direction. Since a rotation does not change the labelling, we may assume the first term is 0. Suppose the second term is a and the last term is b . We have the following claim.

Claim. Let a and b be distinct positive integers from 1 to n .

- If $a + b > n$ and $(a, b) = 1$, then there is exactly one beautiful labelling of the form $(0, a, \dots, b)$.
- If $a + b \leq n$ or $(a, b) > 1$, then there is no beautiful labelling of the form $(0, a, \dots, b)$.

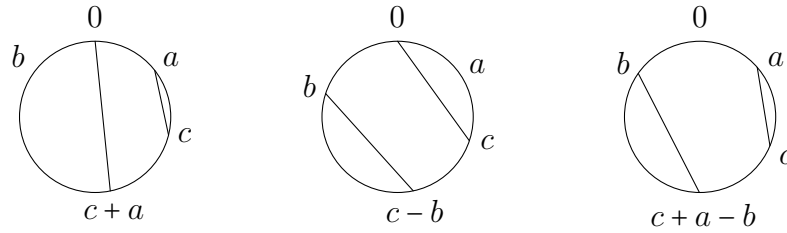
This claim is motivated by the small cases if one tries to find out all beautiful labellings for small n . We first suppose the claim is true. Then it suffices to count the number of solutions to $a + b > n$ with $(a, b) = 1$ and $1 \leq a, b \leq n$. For each $b = 1, 2, \dots, n$, we count the number of a relatively prime to b with $n - b < a \leq n$. Note that a can be one of the b consecutive integers. Hence, exactly $\varphi(b)$ of these are relatively prime to b . Therefore, the number of solutions is

$$\varphi(1) + \varphi(2) + \dots + \varphi(n).$$

Assuming the claim is true, this is the value of M . Comparing with (1), we obtain $M = N + 1$. It remains to prove the claim.

Firstly, suppose $a + b > n$ and $(a, b) = 1$. Consider any beautiful labelling (if exists). We would like to show that every two consecutive terms more or less follow the pattern $+a$ or $-b$. Let c be any number on the arc from a to b (we always refer to the clockwise direction in the following). As $a + b > n$, at most one of $c + a$ and $c - b$ lies between 0 and n .

- If $0 \leq c + a \leq n$, from $(0) + (c + a) = (a) + (c)$, the number $c + a$ must lie on the arc from c to b (if $c + a$ is not one of $0, a, b$, the same for the following).
- If $0 \leq c - b \leq n$, from $(c - b) + (b) = (0) + (c)$, the number $c - b$ must lie on the arc from c to b .



- If both $c + a$ and $c - b$ do not lie in the range $[0, n]$, then we must have $c + a > n$ and $c - b < 0$. In that case, the number $c + a - b$ must satisfy

$$0 \leq n - b < c + a - b < a \leq n.$$

From $(c + a - b) + (b) = (a) + (c)$, the number $c + a - b$ must lie on the arc from c to b .

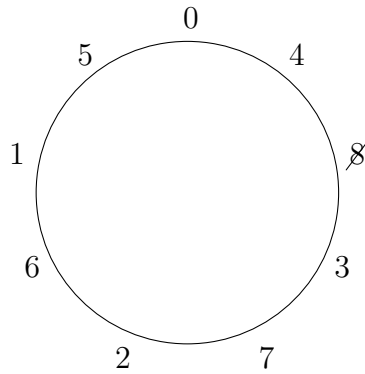
In any case, for any c on the arc from a to b , we can find a corresponding term lying on the arc from c to b . Indeed, if either $c + a$ or $c - b$ lies in $[0, n]$, then it is the corresponding term. If not, then $c + a - b$ is the corresponding term. Therefore, starting from the term after a , we can put more and more terms in the clockwise direction. Since there are only finitely many choices, the process must end, which means the corresponding term of a certain number is one of $0, a, b$. Suppose the process terminates at the number c .

- If $c + a$ is one of $0, a, b$, then since $c + a > a > 0$, we can only have $c + a = b$.
- If $c - b$ is one of $0, a, b$, then since $c \neq b$, we cannot have $c - b = 0$. The case $c - b = a$ is also rejected since this implies $c = a + b > n$. Hence, we can only have $c - b = b$.
- If $c + a - b$ is one of $0, a, b$, then since $c \neq b$, we cannot have $c + a - b = a$. The case $c + a - b = 0$ is also rejected since this implies $c + a = b$ and hence we should consider $c + a$ instead of $c + a - b$ from the process. Hence, we can only have $c + a - b = b$.

In any case, the corresponding term of c must be b . As the same argument works in the anticlockwise direction, we know that the third number (the number after a) in the sequence $(0, a, \dots, b)$ can only be $2a$, $a - b$ or $2a - b$ (which only occurs when both $2a$ and $a - b$ lie outside $[0, n]$). Therefore, we can actually carry out the process starting from the number a . This gives a unique sequence determined by a and b . The order of the numbers in this sequence cannot change for any beautiful labelling with consecutive terms $b, 0, a$. However, it is not guaranteed that such a sequence includes every integer from 0 to n . We are going to prove that this is indeed the case, and such a labelling is beautiful.

Other than following the process directly, we use a modification of that as suggested from our discussion. Consider the arithmetic progression $0, a, 2a, \dots$. Every term is taken modulo $m = a + b$. From $(a, b) = 1$, we have $(a, m) = 1$. Since the common difference and the modulus are relatively prime, the arithmetic progression must form a complete set of residues modulo m . In other words, the terms $0, a, 2a, \dots, (m-1)a$ taken modulo m in the range $[0, m-1]$ is a permutation of $0, 1, \dots, m-1$. The difference between any two consecutive numbers can only be $+a$ or $-b$.

Since $m = a + b > n$, we remove the numbers $n+1, n+2, \dots, m-1$ in this arithmetic progression. Note that no two of these numbers are next to each other before the removal (if $c > n$ for some c , then $c + a > n + a \geq m$ so that the term after c must be $c - b$, which is less than $m - b = a \leq n$). Then we get a permutation of $0, 1, \dots, n$, such that the difference between any two consecutive numbers can only be $+a$, $-b$ or $+a - b$. In addition, it is easy to see that the sequence left is identical to the sequence we obtain by following the above process. This shows the sequence formed from the process does include every integer from 0 to n .



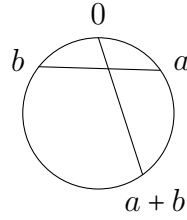
To show that the beautiful condition is satisfied, it suffices to show that the arithmetic progression configuration satisfies the beautiful condition (if this holds, then every subset also holds). Indeed, if $ia + ja \equiv ka + \ell a \pmod{m}$ for some distinct indices i, j, k, ℓ in the range $[0, m-1]$, then $i + j \equiv k + \ell \pmod{m}$ since $(a, m) = 1$. From the range of the indices, we can only have $i + j = k + \ell$ or $i + j = k + \ell \pm m$.

- If $i + j = k + \ell$, we may assume $i < j, k, \ell$. Then we must have $j > k, \ell$. Therefore, both ka and ℓa lie on the arc from ia to ja . The beautiful condition is satisfied.

- If $i + j = k + \ell + m$, then $i + m > i + j = k + \ell + m$ so that $i > k, \ell$. Similarly, $j > k, \ell$. The beautiful condition is satisfied. The same holds if $i + j = k + \ell - m$.

Therefore, we have proven that if $a + b > n$ and $(a, b) = 1$, then there is exactly one beautiful labelling of the form $(0, a, \dots, b)$.

Next, if $a + b \leq n$, then the number $a + b$ (which is larger than $0, a, b$) must lie on the arc from a to b . However, the relation $(0) + (a + b) = (a) + (b)$ contradicts the beautiful condition. Thus, there is no beautiful labelling of the form $(0, a, \dots, b)$.



Lastly, if $a + b > n$ but $(a, b) > 1$, then $a, b \neq 1$ and the number 1 lies on the arc from a to b . We can use the same argument as in the proof of the first claim to put more and more numbers on the arc from 1 to b starting from 1 (note that the deduction of the process does not depend on (a, b)). Let $d = (a, b)$. Since the difference of any two consecutive numbers can only be $+a, -b$ or $+a - b$, this difference must be a multiple of d . Therefore, every number we add in the process must be congruent to 1 modulo d . Recall that the process must terminate at b , which is a multiple of d . This is clearly a contradiction. In other words, the beautiful condition cannot be satisfied, and there is no beautiful labelling of the form $(0, a, \dots, b)$.

This completes the proof of the claim and hence the problem.

Remark. The whole proof mainly depends on the observation that the numbers in a beautiful labelling follow a certain arithmetic progression order. It is not hard to make this observation if one has tried to work out all beautiful labellings for $n = 1, 2, \dots, 7$ (or more). Note that it is not a difficult task as there are only at most $\varphi(1) + \varphi(2) + \dots + \varphi(7) = 18$ cases, and some are symmetric to each other. Also, each beautiful labelling of case n is generated by a beautiful labelling of case $n - 1$.

1.5 Generating Function

Generating function can be used to solve a certain kind of counting problem. Suppose the number of ways to do a task corresponding to a certain quantity n (where n is a nonnegative integer) is a_n . Then we can try to form a function using the numbers $a_0, a_1, \dots$ and deduce some useful properties of the function formed. This sometimes allows us to find the values of a_n in an algebraic way.

The most common type of generating function is

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots.$$

This means we use $a_0, a_1, \dots$ as the coefficients to form a formal power series. We call it a formal power series but not a polynomial because it has infinitely many terms in most cases. For a formal power series, it means we do not need to worry whether the infinite sum is convergent or not. Indeed, as long as x remains as a variable, this does not cause any trouble. Yet we can still do the arithmetic manipulation as usual. For example, we have

$$(a_0 + a_1x + a_2x^2 + \cdots)(b_0 + b_1x + b_2x^2 + \cdots) = c_0 + c_1x + c_2x^2 + \cdots$$

where $c_n = a_0b_n + a_1b_{n-1} + \cdots + a_nb_0$ for each $n = 0, 1, 2, \dots$

One major fact which makes generating function useful is the following **generalized binomial theorem**. To state the theorem, we need to use the generalized binomial coefficients. For any real number a and any positive integer r , we define

$$\binom{a}{r} = \frac{a(a-1)\cdots(a-r+1)}{r!}.$$

We also define $\binom{a}{0} = 1$. These definitions coincide with the definition for integer $a \geq r$. Also, it can be proved easily that $\binom{a}{r} = 0$ if a, r are positive integers with $a < r$. For positive integers n and r , we find that

$$\begin{aligned} \binom{-n}{r} &= \frac{(-n)(-n-1)\cdots(-n-r+1)}{r!} \\ &= (-1)^r \frac{(n)(n+1)\cdots(n+r-1)}{r!} \\ &= (-1)^r \binom{n+r-1}{r}. \end{aligned}$$

We have the following theorem.

Theorem 1.3. (Generalized binomial theorem) Let a be a rational number. Then

$$(x+y)^a = \sum_{r=0}^{\infty} \binom{a}{r} x^{a-r} y^r.$$

The proof of this generalized version is omitted. This is indeed the same as the ordinary **binomial theorem**. The main difference is that the expansion may be an infinite series, since for non-integers a , we do not have $\binom{a}{r} = 0$ for large positive integers r .

For example, we often come across a geometric sequence such as

$$(1 + x + x^2 + \cdots + x^k)^m.$$

This is equal to $\left(\frac{x^{k+1}-1}{x-1}\right)^m = (x^{k+1}-1)^m(x-1)^{-m}$. Then we can use the **generalized binomial theorem** to expand the terms $(x^{k+1}-1)^m$ and $(x-1)^{-m}$.

We now demonstrate the use of generating functions with a few problems.

Problem 1.18. (HKIMO TST 2014 Test 1 Problem 4) A number is good if the sum of its digits is 18 (for example 130518 is a good number). Find the number of good 6-digit numbers.

Analysis. We give an alternative solution to this problem using generating function. It demonstrates the fundamental use of generating functions.

Solution. For each positive integer n , let a_n be the number of 6-digit numbers whose sums of digits are n . Consider the generating function $f(x) = a_1x + a_2x^2 + a_3x^3 + \dots$. Note that the exponent corresponds to the sum of digits. If we only consider the leading digit, then the corresponding generating function should be $x + x^2 + \dots + x^9$. This is because the leading digit can be one of $1, 2, \dots, 9$, and there is one possibility for each digit. Similarly, each of the remaining digits corresponds to $1 + x + \dots + x^9$. To form a 6-digit number, we can choose the 6 digits one by one. The choice for each digit is independent of each other. Therefore, using the **multiplication principle**, we have

$$f(x) = (x + x^2 + \dots + x^9)(1 + x + \dots + x^9)^5.$$

Each of the geometric sequences can be expanded in the way as discussed. We have

$$f(x) = \frac{x^{10}-x}{x-1} \cdot \left(\frac{x^{10}-1}{x-1}\right)^5 = (x^{10}-x)(x^{10}-1)^5(x-1)^{-6}.$$

Using the **generalized binomial theorem**, we find that

$$f(x) = (-x + x^{10}) \left(\sum_{r=0}^{\infty} \binom{5}{r} (-1)^{5-r} x^{10r} \right) \left(\sum_{s=0}^{\infty} \binom{-6}{s} (-1)^{-6-s} x^s \right).$$

We need to find the coefficient of x^{18} . Thus, we can restrict the sum over r to $r=0$ and $r=1$ (as all subsequent terms exceed x^{18}). Then we obtain

$$f(x) = (x - x^{10} - 5x^{11} + \dots) \left(\sum_{s=0}^{\infty} \binom{-6}{s} (-1)^{-6-s} x^s \right)$$

where the part not shown consists of terms x^k with $k > 18$. Hence, the answer is

$$\begin{aligned} a_{18} &= \binom{-6}{17} (-1)^{-6-17} - \binom{-6}{8} (-1)^{-6-8} - 5 \binom{-6}{7} (-1)^{-6-7} \\ &= -\binom{22}{17} - \binom{13}{8} + 5 \binom{12}{7} = 21087. \end{aligned}$$

Remark. The last formula $-\binom{22}{17} - \binom{13}{8} + 5 \binom{12}{7}$ is actually the same as the one we found using the **inclusion-exclusion principle** in problem 1.5. One may see the linkage between the two methods.

Problem 1.19. The Fibonacci sequence $\{F_n\}_{n \geq 0}$ is defined by $F_0 = 0$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Find the general term of F_n .

Analysis. We use generating function to find the general term. Solving recurrence relations is one application of generating functions.

Solution. Define $g(x) = F_0 + F_1x + F_2x^2 + \dots$. Using the recurrence relation, we have

$$\begin{aligned} g(x) &= x + F_2x^2 + F_3x^3 + F_4x^4 + \dots \\ &= x + (F_0 + F_1)x^2 + (F_1 + F_2)x^3 + (F_2 + F_3)x^4 + \dots \\ &= x + x^2(F_0 + F_1x + F_2x^2 + \dots) + x(F_1x + F_2x^2 + F_3x^3 + \dots) \\ &= x + x^2g(x) + xg(x). \end{aligned}$$

Solving for $g(x)$, we obtain $g(x) = -\frac{x}{x^2 + x - 1}$. To express this using an infinite series, since we only know how to do so for a linear denominator, we shall use partial fraction. As $x^2 + x - 1 = (x - \alpha)(x - \beta)$ where $\alpha = \frac{-1 + \sqrt{5}}{2}$ and $\beta = \frac{-1 - \sqrt{5}}{2}$, we equate $g(x) = \frac{A}{x - \alpha} + \frac{B}{x - \beta}$ to obtain

$$\begin{aligned} g(x) &= \frac{1}{\alpha - \beta} \left(-\frac{\alpha}{x - \alpha} + \frac{\beta}{x - \beta} \right) \\ &= \frac{1}{\alpha - \beta} \left(-\alpha \sum_{r=0}^{\infty} \binom{-1}{r} (-\alpha)^{-1-r} x^r + \beta \sum_{r=0}^{\infty} \binom{-1}{r} (-\beta)^{-1-r} x^r \right) \\ &= \frac{1}{\alpha - \beta} \left(\sum_{r=0}^{\infty} \alpha^{-r} x^r - \sum_{r=0}^{\infty} \beta^{-r} x^r \right). \end{aligned}$$

Note that $\alpha^{-1} = -\beta$ and $\beta^{-1} = -\alpha$. Hence, this becomes

$$g(x) = \frac{1}{\alpha - \beta} \sum_{r=0}^{\infty} ((-\beta)^r - (-\alpha)^r) x^r.$$

From this, it is obvious that $F_n = \frac{(-\beta)^n - (-\alpha)^n}{\alpha - \beta}$.

Remark. One can obtain the same formula using recurrence relation.

Problem 1.20. Let n be a positive integer. An object is moved from $(0, 0)$ to (n, n) in the coordinate plane in the following way. In each step, the object can be moved to one step rightward or one step upward. Suppose the object never goes to a point (x, y) for which $y > x$. Find the number of ways to move the object.

Analysis. Let C_n be the number of ways to move the object. With some trials, one can obtain $C_1 = 1$, $C_2 = 2$, $C_3 = 5$ and $C_4 = 14$, etc. There seems to be no obvious pattern. We shall try to find a recurrence relation for the sequence $\{C_n\}$ and use generating function to solve for the general term. In this way, there is no need to guess the answer.

Solution. Let C_n be the number of ways to move the object. We further define $C_0 = 1$ for convenience. For $n \geq 1$, suppose the first time when the object reaches the line $y = x$ (other than $(0,0)$) is at point (r, r) where $r = 1, 2, \dots, n$.

For the path from $(0,0)$ to (r, r) , the first step must go to $(1,0)$ and the step before (r, r) must go to $(r, r-1)$. This means we count the number of paths from $(1,0)$ to $(r, r-1)$ such that the object never goes to a point (x, y) with $y+1 > x$ (we cannot have $y = x$ since this contradicts the smallest choice of r). By translating the configuration one step to the left, this number is exactly C_{r-1} .

For the path from (r, r) to (n, n) , the object is allowed to reach the line $y = x$ in the intermediate steps. Therefore, it corresponds to the original problem moving from $(0,0)$ to $(n-r, n-r)$. There are C_{n-r} ways to do so. Note that the same holds when $r = n$ as $C_0 = 1$.

By the **multiplication principle** and summing over all possible r , we obtain the recurrence relation

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + \dots + C_{n-1} C_0.$$

We use generating function to solve for C_n . Note that the right-hand side of the recurrence relation looks like the coefficient of the square of a power series. Indeed, if we define

$$f(x) = C_0 + C_1 x + C_2 x^2 + \dots$$

as usual, we have

$$\begin{aligned} f(x)^2 &= (C_0 + C_1 x + C_2 x^2 + \dots)^2 \\ &= C_0 C_0 + (C_0 C_1 + C_1 C_0)x + (C_0 C_2 + C_1 C_1 + C_2 C_0)x^2 + \dots \\ &= C_1 + C_2 x + C_3 x^2 + \dots \\ &= \frac{f(x) - 1}{x}. \end{aligned}$$

We can solve the quadratic equation $xf(x)^2 - f(x) + 1 = 0$ as if $f(x)$ is a real number. This gives $f(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$. The square root can be expanded using the **generalized binomial theorem** with rational exponent. We have

$$f(x) = \frac{1}{2x} (1 \pm (1-4x)^{\frac{1}{2}}) = \frac{1}{2x} \left(1 \pm \sum_{r=0}^{\infty} \binom{\frac{1}{2}}{r} (-4x)^r \right).$$

Comparing with $f(x) = C_0 + C_1 x + C_2 x^2 + \dots$, we easily see that the negative sign should be chosen (otherwise there is a term $\frac{1}{x}$). Therefore, we find that

$$f(x) = -\frac{1}{2} \sum_{r=1}^{\infty} \binom{\frac{1}{2}}{r} (-4)^r x^{r-1}.$$

Hence,

$$C_n = -\frac{1}{2} \binom{\frac{1}{2}}{n+1} (-4)^{n+1} = \binom{\frac{1}{2}}{n+1} (-1)^n 2^{2n+1}.$$

This can be further simplified. Indeed, using the definition, we have

$$\begin{aligned}
\binom{\frac{1}{2}}{n+1} &= \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2n-1}{2}\right)}{(n+1)!} \\
&= (-1)^n 2^{-n-1} \frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{(n+1)!} \\
&= (-1)^n 2^{-n-1} \frac{(2n)!}{(2 \times 4 \times 6 \times \cdots \times (2n))(n+1)!} \\
&= (-1)^n 2^{-n-1} \frac{(2n)!}{n!(n+1)!} \\
&= (-1)^n 2^{-n-1} \frac{1}{n+1} \binom{2n}{n}.
\end{aligned}$$

This implies $C_n = \frac{1}{n+1} \binom{2n}{n}$.

Remark. The terms in the sequence $\{C_n\}$ are called *Catalan numbers*. There are various combinatorial meanings of this sequence.

Before proceeding to the next problem, we recall some facts about complex numbers. A complex number can be represented in the form $z = a + bi$ (where $a, b \in \mathbb{R}$ and i is the number satisfying $i^2 = -1$), $z = r(\cos \theta + i \sin \theta)$ (where $r \geq 0$ and $0 \leq \theta < 2\pi$) or $z = re^{i\theta}$. For each positive integer n , the solutions to the equation $z^n = 1$ are

$$z = 1, e^{i\frac{2\pi}{n}}, e^{i\frac{4\pi}{n}}, \dots, e^{i\frac{2(n-1)\pi}{n}}.$$

Let $\omega = e^{i\frac{2\pi}{n}}$. Then the solutions can be represented as $1, \omega, \omega^2, \dots, \omega^{n-1}$. The solutions other than 1 are called roots of unity. If z is such a root of unity, then since

$$(z-1)(z^{n-1} + z^{n-2} + \cdots + 1) = z^n - 1 = 0$$

and $z \neq 1$, we know that $z^{n-1} + z^{n-2} + \cdots + 1 = 0$.

Problem 1.21. For any set A of integers, let $s(A)$ denote the sum of elements of A . Let $S = \{1, 2, \dots, 1999\}$. Find the number of elements in T_0 where

$$T_0 = \{T : T \subset S, s(T) \equiv 0 \pmod{7}\}.$$

Analysis. This means we need to count the number of subsets of S whose sums of elements are congruent to 0 modulo 7. This kind of problem can be solved via generating functions with the aid of roots of unity, as they can extract an arithmetic sequence easily.

Solution. When we form a subset of S , each element can be included or not included. If an element k is included, then its contribution to the sum is k . Otherwise, the

contribution is 0. Therefore, each element corresponds to a generating function $1+x^k$. Hence, the generating function corresponding to this problem is

$$f(x) = (1+x)(1+x^2)\cdots(1+x^{1999}). \quad (1)$$

If we let $f(x) = a_0 + a_1x + \cdots + a_dx^d$ (where $d = 1 + 2 + \cdots + 1999$), then each a_k counts the number of subsets of S whose sums of elements are k . This gives

$$T_0 = a_0 + a_7 + a_{14} + \cdots.$$

Let $\omega = e^{i\frac{2\pi}{7}}$ be a root of unity. We sum up

$$\begin{aligned} f(1) &= a_0 + a_1 + a_2 + \cdots + a_d, \\ f(\omega) &= a_0 + a_1\omega + a_2\omega^2 + \cdots + a_d\omega^d, \\ f(\omega^2) &= a_0 + a_1\omega^2 + a_2\omega^4 + \cdots + a_d\omega^{2d}, \\ &\vdots \\ f(\omega^6) &= a_0 + a_1\omega^6 + a_2\omega^{12} + \cdots + a_d\omega^{6d} \end{aligned}$$

using the properties of roots of unity. Note that a_1 is multiplied to the term

$$1 + \omega + \omega^2 + \cdots + \omega^6 = 0.$$

This holds for many other terms. Indeed, for each $k = 0, 1, 2, \dots, d$, the coefficient a_k is multiplied to the term

$$1 + \omega^k + \omega^{2k} + \cdots + \omega^{6k}.$$

If $k \not\equiv 0 \pmod{7}$, then $\omega^k \neq 1$ and so ω^k is a root of unity. As indicated, the expression $1 + (\omega^k) + (\omega^k)^2 + \cdots + (\omega^k)^6$ is equal to 0. If $k \equiv 0 \pmod{7}$, the $\omega^k = 1$ (as $\omega^7 = 1$) and so the expression is equal to 7. Therefore, we obtain

$$f(1) + f(\omega) + f(\omega^2) + \cdots + f(\omega^6) = 7(a_0 + a_7 + a_{14} + \cdots) = 7T_0.$$

It remains to find the particular values using the formula (1). Note that the exponents in the formula can be taken modulo 7 since x is a root of unity or $x = 1$. Thus, the main task is to compute

$$g(x) = (1+x)(1+x^2)\cdots(1+x^7).$$

As 7 is a prime number, when we substitute $x = \omega^k$ in $g(x)$ for $k = 1, 2, \dots, 6$, the exponents form a complete set of residues modulo 7. This shows

$$g(\omega^k) = g(\omega) = (1+\omega)(1+\omega^2)\cdots(1+\omega^7).$$

Note that $h(x) = (x+\omega)(x+\omega^2)\cdots(x+\omega^7)$ is the monic polynomial of degree 7 whose roots are $-\omega, -\omega^2, \dots, -\omega^7$. Hence, we have $h(x) = x^7 + 1$. It follows that $g(\omega^k) = h(1) = 1 + 1 = 2$, while it is obvious that $g(1) = 2^7$.

We have

$$f(\omega) = g(\omega)^{285} (1+\omega)(1+\omega^2)(1+\omega^3)(1+\omega^4) = 2^{285} (1+\omega^3)$$

since $(1 + \omega)(1 + \omega^2)(1 + \omega^4) = 1 + \omega + \omega^2 + \cdots + \omega^7 = \omega^7 = 1$. Similarly, we find that

$$f(\omega^k) = 2^{285}(1 + \omega^{3k})$$

for $k = 1, 2, \dots, 6$. For $k = 0$, we have $f(1) = 2^{1999}$. We obtain

$$T_0 = \frac{1}{7}(2^{1999} + 2^{285}((1 + \omega^3) + (1 + \omega^6) + \cdots + (1 + \omega^{18}))) = \frac{2^{1999} + 5 \times 2^{285}}{7}.$$

Remark. In a similar manner, we can obtain the number of subsets of S whose sums of elements are congruent to r modulo 7 for some $r = 1, 2, \dots, 6$. For example, for $r = 1$, we can consider the sum

$$f(1) + \omega^6 f(\omega) + \omega^{12} f(\omega^2) + \cdots + \omega^{36} f(\omega^6).$$

The trick is to make each coefficient a_{7k+1} multiplied to a 7ℓ th power of ω (which is 1).

Although we usually cannot substitute a number in a generating function, we can do so in this problem because the generating function here is a finite sum, which is simply a polynomial.

2 Extremal Combinatorics and Existence Problems

Extremal combinatorics refer to a kind of problem which seek for the optimal value of a quantity such that a certain thing can occur. Such a problem usually consists of two parts. Firstly, we need to show that the value cannot exceed the bound as claimed. Secondly, we have to show that such a bound is attainable. This second part is related to existence problems, which usually only requires a simple construction. Such a construction can often be inspired by small cases.

The following two theorems may be useful in obtaining some bounds.

Theorem 2.1. (Bonferroni inequalities) Let $S_1, S_2, \dots, S_n$ be any finite sets. Then

$$\left| \bigcup_{i=1}^n S_i \right| \leq \sum_{i=1}^n |S_{i_1}| - \sum_{i_1 < i_2} |S_{i_1} \cap S_{i_2}| + \sum_{i_1 < i_2 < i_3} |S_{i_1} \cap S_{i_2} \cap S_{i_3}| - \dots + \sum_{i_1 < \dots < i_k} |S_{i_1} \cap \dots \cap S_{i_k}|$$

for odd $k \leq n$, and

$$\left| \bigcup_{i=1}^n S_i \right| \geq \sum_{i=1}^n |S_{i_1}| - \sum_{i_1 < i_2} |S_{i_1} \cap S_{i_2}| + \sum_{i_1 < i_2 < i_3} |S_{i_1} \cap S_{i_2} \cap S_{i_3}| - \dots - \sum_{i_1 < \dots < i_k} |S_{i_1} \cap \dots \cap S_{i_k}|$$

for even $k \leq n$.

This is the inequality version of the **inclusion-exclusion principle**, and can be proved using the same method.

Proof. Consider any element a in $\bigcup_{i=1}^n S_i$. Suppose a belongs to exactly m of the sets $S_1, S_2, \dots, S_n$. We move everything to the left-hand side of the inequality. Then the contribution of a to the quantity is

$$1 - \binom{m}{1} + \binom{m}{2} - \binom{m}{3} + \dots + (-1)^k \binom{m}{k}.$$

Denote this by $f(m)$. By expanding the binomial coefficients, we find that f is a polynomial in m of degree k and leading coefficient $\frac{(-1)^k}{k!}$. Using the **binomial theorem**, we find that $f(m) = 0$ for $m = 1, 2, \dots, k$. Therefore, these are the only roots of f , and hence

$$f(m) = \frac{(-1)^k}{k!} (m-1)(m-2)\dots(m-k).$$

For odd k , we have $f(m) \leq 0$ for $m = 1, 2, \dots, n$, while for even k , we have $f(m) \geq 0$ for $m = 1, 2, \dots, n$. Also, it is clear that equality holds iff m is one of $1, 2, \dots, k$. This means every element in the union belongs to at most k sets. $\square$

In the following, we say that a sequence $x_1 \geq x_2 \geq \dots \geq x_n$ majorizes another sequence $y_1 \geq y_2 \geq \dots \geq y_n$ if they satisfy the following conditions:

$$\begin{aligned} x_1 &\geq y_1, \\ x_1 + x_2 &\geq y_1 + y_2, \\ &\dots \\ x_1 + x_2 + \dots + x_{n-1} &\geq y_1 + y_2 + \dots + y_{n-1}, \\ x_1 + x_2 + \dots + x_n &= y_1 + y_2 + \dots + y_n. \end{aligned}$$

We write $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$ in that case. Note that the sums of the two sequences must be equal. Now we can state the next theorem.

Theorem 2.2. (Majorization inequality/Karamata's inequality(蓋不等式)) For any real numbers $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ with $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$ and for any convex function f , we have

$$f(x_1) + f(x_2) + \dots + f(x_n) \geq f(y_1) + f(y_2) + \dots + f(y_n).$$

We are not going to prove this inequality. Also, we are only interested in the application of this inequality in combinatorics. Thus, we will not define a convex function here. Indeed, we will only apply this inequality to the binomial function $\binom{k}{2}$, which is well-known to be convex. In other words, if $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$, then we have

$$\binom{x_1}{2} + \binom{x_2}{2} + \dots + \binom{x_n}{2} \geq \binom{y_1}{2} + \binom{y_2}{2} + \dots + \binom{y_n}{2}.$$

2.1 Induction

Induction is definitely the most useful technique in combinatorics. We are all supposed to be familiar with this tool. But due to its importance, we will still go through a few problems to show its application.

Problem 2.1. Each cell of an $m \times n$ board is in one of the k colours so that all colours are used. A group of four cells is called a rectangle if they are the intersections of any two rows and two columns. Find the maximum possible value of k such that there exists a colouring of the board for which no rectangle contains four distinct colours.

Analysis. If $m = 1$, then we can take $k = n$. If $m = 2$, then a simple way to make all rectangles contain a repeated colour is to use the same colour for the second row. Similarly, if we have more rows, we can use a single colour to colour an additional row. This provides a lower bound on k , which is very likely to be the optimal case as there seems not to be any natural construction for the general case. Using induction is convenient since the conditions for smaller cases form a subset of the conditions for larger cases.

Solution. We claim that the maximum possible value of k is $m + n - 1$.

We first provide a possible construction for $k = m + n - 1$. Suppose the colours are $C_1, C_2, \dots, C_{m+n-1}$. For the first row, we use the colours $C_1, C_2, \dots, C_n$ to colour the n cells respectively. Then for the i th row where $i = 2, 3, \dots, m$, we use the colour C_{n+i-1} to colour the whole row. All the colours are used. For any rectangle, at least two of its cells must lie in the same row which is not the first row. Then these two cells have the same colour, and hence the rectangle cannot contain four distinct colours.

Next, we prove that the number of colours cannot exceed $m + n - 1$ by induction. When $m = 1$ or $n = 1$, there are only $m + n - 1$ cells and hence this is trivially satisfied. Assume this bound holds for all cases (m, n) with $m \leq m_0$, $n \leq n_0$ and $(m, n) \neq (m_0, n_0)$. We consider the case (m_0, n_0) for some $m_0, n_0 \geq 2$.

Suppose on the contrary that at least $m_0 + n_0$ colours are used. WLOG assume $n_0 \geq m_0$. If we delete any column, then we obtain an $m_0 \times (n_0 - 1)$ sub-board which satisfies the same condition. Therefore, we can apply the inductive hypothesis to show that at most $m_0 + n_0 - 2$ colours are used in this sub-board. This implies the deleted column contains at least 2 colours which are not used in any other columns. We say that a cell is special if the colour used in that cell is not used in any other columns. This means there are at least 2 special cells in each column.

As there are at least $2n_0 \geq 2m_0$ special cells, there must be 2 special cells lying in the same row. WLOG assume cells $(1, 1)$, $(1, 2)$ are special (here, cell (x, y) is the cell in row x and column y). We can also assume the other special cell in column 1 is $(2, 1)$. From the constraint, the other special cell in column 2 cannot be $(2, 2)$. Thus, we may assume it is $(3, 2)$.

Consider the rectangle with cells $(1, 1)$, $(1, 2)$, $(2, 1)$, $(2, 2)$. The colour of $(2, 2)$ has to be the same as the colour of one of $(1, 1)$, $(1, 2)$, $(2, 1)$. But since $(1, 1)$ and $(2, 1)$ are special cells in column 1, the colour of $(2, 2)$ must be the same as that of $(1, 2)$. Similarly, by considering the rectangle with cells $(1, 1)$, $(1, 2)$, $(3, 1)$, $(3, 2)$, we conclude that the colour of $(3, 1)$ is the same as that of $(1, 1)$. But then $(2, 1)$, $(2, 2)$, $(3, 1)$, $(3, 2)$ will be a rectangle having four distinct colours. This is a contradiction.

$(1, 1)$	$(1, 2)$
$(2, 1)$	$(2, 2)$
$(3, 1)$	$(3, 2)$

Therefore, at most $m_0 + n_0 - 1$ colours can be used for the case (m_0, n_0) . By induction, we must have $k \leq m + n - 1$. Thus, the maximum possible value of k is $m + n - 1$.

Problem 2.2. (IMO shortlist 2010 C2) On some planet, there are 2^N countries ($N \geq 4$). Each country has a flag N units wide and one unit high composed of N fields of size 1×1 , each field being either yellow or blue. No two countries have the same flag.

We say that a set of N flags is diverse if these flags can be arranged into an $N \times N$ square so that all N fields on its main diagonal will have the same colour. Determine the smallest positive integer M such that among any M distinct flags, there exist N flags forming a diverse set.

Analysis. To guess the answer, it is rather hard to look for small cases. Indeed, the smallest case $N = 4$ is not easy to solve. Although the same problem makes sense for $N = 1, 2, 3$, the condition that $N \geq 4$ actually provides a hint that the answers for $N = 1, 2, 3$ are different from the general case $N \geq 4$. Therefore, solving the small cases $N = 1, 2, 3$ is not useful to find the general answer. However, these may be useful for us to establish the base case $N = 4$.

Other than looking for small cases, we guess the answer by finding a natural lower bound. This means we want to find a set of M distinct flags so that there is no subset which is diverse. A natural construction is to fix the colour of the last 2 fields of the flags.

Solution. We claim that the smallest M is $2^{N-2} + 1$.

We first show that there exists a set of 2^{N-2} flags which does not contain any diverse subset. Consider all flags whose second last field is yellow and the last field is blue. There are 2^{N-2} different flags of this kind. No matter which N of these flags are chosen and no matter how they are arranged into an $N \times N$ square, the $(N-1, N-1)$ field must be yellow while the (N, N) field must be blue. Thus, the main diagonal cannot consist of a single colour. This construction shows that M is at least $2^{N-2} + 1$.

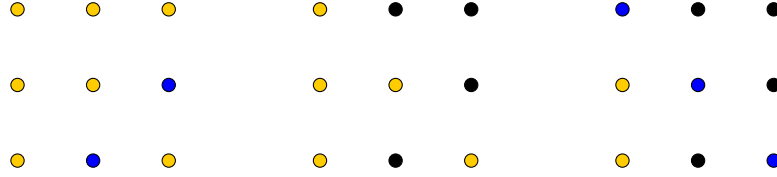
Next, we prove that any set of $2^{N-2} + 1$ flags contains a diverse subset. This can be proved by induction. The base case $N = 4$ (and $M = 2^2 + 1 = 5$) can be handled by some sort of exhaustive arguments. We shall begin with the cases $N = 2$ and $N = 3$. In the following, we use Y to denote yellow, B to denote blue, and X to denote an undetermined colour.

- For $N = 2$, we claim that any 3 flags contain a diverse subset. Up to symmetry, one of the flags must be YY and one of the flags must be YB . Then YB, YY form a diverse set in this order.



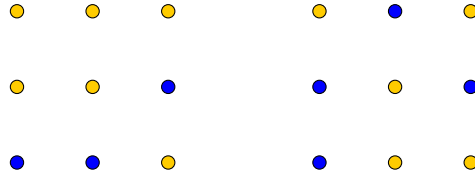
- For $N = 3$, we claim that any 4 flags contain a diverse subset. If all flags are YXX , then the flags must be YYY, YYB, YBY, YBB . Then $YYY, YYB,$

YBY form a diverse set. If 3 of the flags are YXX and the remaining is BXX , we apply the case $N = 2$ to the last 2 fields of the 3 flags of the form YXX . Then there are 2 flags which are YYX, YXY or YBX, YXB . In any case, we can find a diverse subset (YXX, YYX, YXY for the former and BXX, YBX, YXB for the latter). The same discussion works if the colours are interchanged, or if the order of the fields are changed.

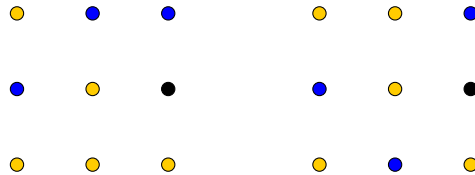


Therefore, it remains to work on the case when 2 flags are YXX and 2 flags are BXX . Only the following cases are possible up to symmetry.

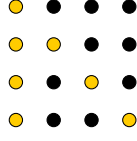
- The flags are YYY, YYB, BBY, BBB . Then YYY, YYB, BBY is a diverse set.
- The flags are YBY, YBB, BYY, BYB . Then YBY, BYB, BYY is a diverse set.



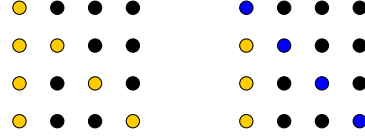
- Two of the flags are YYY and YBB . Then there must be a flag of the form BYX . In that case, YBB, BYX, YYY is a diverse set.
- Two of the flags are YYB and YBY . Then there must be a flag of the form BYX . In that case, YYB, BYX, YBY is a diverse set.



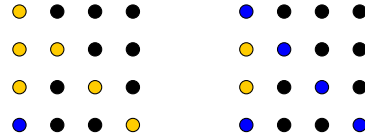
- For $N = 4$, consider any set of 5 flags. If all flags are $YXXX$, note that there must be a flag of the form $YXXY$, or otherwise there are at most 4 distinct flags. For the other 4 flags, apply the case $N = 3$ to the first 3 fields. Since the first field must be yellow, the main diagonal in the 3×3 square must be yellow. This means there are flags of the form $YXXX, YYXX, YXYX$. Then $YXXX, YYXX, YXYX, YXXY$ is a diverse set.



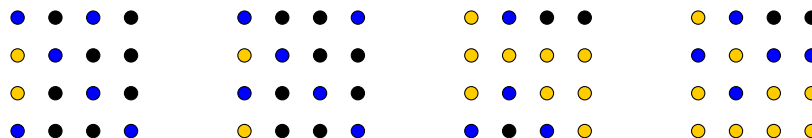
If 4 of the flags are $YXXX$ and the remaining is $BXXX$, we apply the case $N = 3$ to the last 3 fields of the flags of the form $YXXX$. Then there are flags of the form $YYXX$, $YXYX$, $YXXY$ or $YBXX$, $YXBX$, $YXXB$. In any case, we can find a diverse set ($YXXX$, $YYXX$, $YXYX$, $YXXY$ for the former and $BXXX$, $YBXX$, $YXBX$, $YXXB$ for the latter).



If 3 of the flags are $YXXX$ and the other 2 are $BXXX$, we apply the case $N = 3$ to the last 3 fields of the flags $YXXX$, $YXXX$, $BXXX$, $BXXX$. If the main diagonal of that 3×3 square is yellow, then we can put the remaining $YXXX$ flag to the first row to form a 4×4 square with a yellow main diagonal. If the main diagonal of that 3×3 square is blue and one of the flags $BXXX$ is not used, then we can put it to the first row to form a 4×4 square with a blue main diagonal.



Up to symmetry, it remains to consider the case when the flags are $YXXX$, $YXXX$, $YBXX$, $BXBX$, $BXXB$. If one of the first two flags is $YXBX$, then $BXBX$, $YBXX$, $YXBX$, $BXXB$ form a diverse set. The same holds if one of the first two flags is $YXXB$ due to symmetry. Therefore, we may assume the flags are $YYYY$, $YBYY$, $YBXX$, $BXBX$, $BXXB$. If the fourth flag is $BXBY$, then $YBXX$, $YYYY$, $YBYY$, $BXBY$ is a diverse set. We may assume the fourth flag is $BXBB$, and similarly for the fifth flag. Thus, the flags are $YYYY$, $YBYY$, $YBXX$, $BYBB$, $BBBB$, in which case $YBXX$, $BYBB$, $YBYY$, $YYYY$ form a diverse set. These solve the base case $N = 4$.



We now proceed to the inductive step. This is indeed similar to the arguments for the case $N = 4$ using $N = 3$, but it is much easier since the most troublesome scenario will not occur. Consider any $2^{N-2} + 1$ flags for some $N \geq 5$. At least half of the flags have the same colour in the first field. WLOG assume $2^{N-3} + 1$ flags have the form $YXX \cdots X$.

We apply the case $N - 1$ to the last $N - 1$ fields of the flags of the form $YXX \cdots X$. Then we either have $YYXX \cdots X$, $YXYXX \cdots X$, $\dots$, $YXX \cdots XY$, or $YBXX \cdots X$, $YXBXX \cdots X$, $\dots$, $YXX \cdots XB$. For the former case, since $2^{N-3} + 1 > N - 1$ for $N \geq 5$ by induction (this is the place we need $N \geq 4$ in the problem), there exists an unused flag $YXX \cdots X$. Then $YXX \cdots X$, $YYXX \cdots X$, $YXYXX \cdots X$, $\dots$, $YXX \cdots XY$ is a diverse set. For the latter case, we can argue in the same way if there exists a flag of the form $BXX \cdots X$.

It remains to consider the case when all flags have the form $YXX \cdots X$. Note that there must be a flag of the form $YXX \cdots XY$, or otherwise there are at most 2^{N-2} distinct flags. Next, we choose any $2^{N-3} + 1$ of the remaining flags and apply the case $N - 1$ to the first $N - 1$ fields. Since the first field must be yellow, the main diagonal in the $(N - 1) \times (N - 1)$ square must be yellow. This means there are flags of the form $YXX \cdots X$, $YYXX \cdots X$, $\dots$, $YXX \cdots XYX$. Then $YXX \cdots X$, $YYXX \cdots X$, $\dots$, $YXX \cdots XYX$, $YXX \cdots XY$ is a diverse set. This establishes the case N , and hence the bound holds by induction.

We conclude that the smallest M is $2^{N-2} + 1$ for $N \geq 4$.

Problem 2.3. (IMO shortlist 2009 C7/Problem 6) Let $a_1, a_2, \dots, a_n$ be distinct positive integers and let M be a set of $n - 1$ positive integers not containing

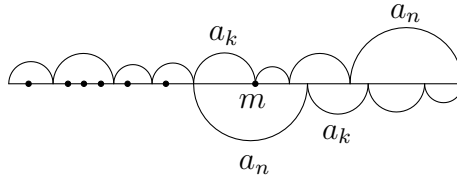
$$s = a_1 + a_2 + \cdots + a_n.$$

A grasshopper is to jump along the real axis, starting at the point 0 and making n jumps to the right with lengths $a_1, a_2, \dots, a_n$ in some order. Prove that the order can be chosen in such a way that the grasshopper never lands on any point in M .

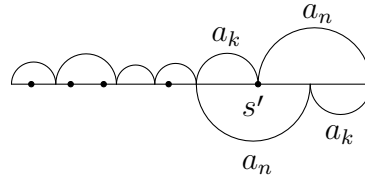
Analysis. It is natural to use induction to solve this problem. If we can use one step to jump over a point in M , then the claim follows from the inductive hypothesis easily. Therefore, it is natural to consider the largest jump first. We have to modify the jumping to handle the exceptional cases.

Solution. Suppose $a_1 < a_2 < \cdots < a_n$. For $n = 1$, the statement is trivially satisfied since $|M| = 0$. Assume the statement holds for all smaller n . Firstly, we tend to use a_n as the last step.

- If $s' = a_1 + a_2 \cdots + a_{n-1} \notin M$ and there exists an $m \in M$ such that $s' < m < s$, then we can remove the step a_n and the points in M which lie between s' and s . Then we are reduced to the case $n - 1$ since the size of the new set M' is at most $n - 2$. If $|M'| < n - 2$, we can add some points to M' arbitrarily to make it have size $n - 2$. Therefore, we can apply the inductive hypothesis to choose an order of $a_1, \dots, a_{n-1}$ in such a way that the grasshopper can avoid the points in M' and hence the points in M .
- If $s' \notin M$ but there is no point in M lying between s' and s , we remove the step a_n and the largest point $m \in M$ and apply the inductive hypothesis. This gives a certain order of $a_1, \dots, a_{n-1}$ for which all the points in M other than m are avoided. If the path does not pass through m , then we are done. Otherwise, suppose the jump with step a_k lands on m for some $k = 1, 2, \dots, n - 1$. Then we use a_n to replace a_k in that jump. As $a_n > a_k$, such a jump must jump over point m . Due to the maximality of m , the grasshopper will not land on any point in M afterwards. Thus, the remaining jumps can be done arbitrarily and we are done.



- If $s' \in M$ and s' is the largest point in M , we can argue in the same way as the second case. Indeed, we remove the step a_n and the point s' and apply the inductive hypothesis. This gives a path which is close to what we need, except that the second last jump with step a_k lands on s' . We can use a_n to replace a_k in that jump so that the point s' is jumped over. Next, we can use a_k as the last jump. This avoids every point in M .



- If $s' \in M$ and there exists another point $m \in M$ lying strictly between s' and s , we try to modify the jumping. We would like to use a_n as the second last step and some other a_j as the last step. Then the last two steps will jump over two points s' and m and hence we can apply the inductive hypothesis. This means it suffices to find an index j such that both $s - a_j$ and $s - a_j - a_n$ do not

lie in M . Suppose on the contrary that at least one of them lies in M for each $j = 1, 2, \dots, n-1$. Note that

$$\begin{aligned} s - a_1 &> s - a_2 > \dots > s - a_{n-1} > s - a_n \\ &> s - a_1 - a_n > s - a_2 - a_n > \dots > s - a_{n-1} - a_n. \end{aligned}$$

Therefore, the n points in M (one corresponding to each $j = 1, 2, \dots, n-1$ and the point $s' = s - a_n$) are all distinct. This contradicts $|M| = n-1$. Therefore, such an index j must exist. Removing the steps a_n and a_j , and removing the points s' and m , we are in a position of the case $n-2$ and hence the inductive hypothesis can be applied. This case is thus solved.

By induction, we can always choose a path which avoids every point in M . This completes the proof.

2.2 Pigeonhole Principle

Theorem 2.3. (Pigeonhole principle(抽屜原理/鴿巢原理)) Suppose there are n elements $a_1, a_2, \dots, a_n$ each of which belongs to exactly one of the sets $A_1, A_2, \dots, A_m$. Then there exist $\left\lceil \frac{n}{m} \right\rceil$ elements among the a_i 's which belong to the same set.

The pigeonhole principle is rather a common sense. Another common proving technique, i.e. proof by contradiction, can essentially provide the same result.

Problem 2.4. For any positive integer n , find the smallest m such that among any m integers, there must be some of them which add up to give a multiple of n .

Analysis. It suffices to consider modulo n . If all chosen integers are of the form $kn+1$, then we need at least n of them to give a sum of a multiple of n . This gives a lower bound on m , which seems to be the optimal value as well. The remaining step is to construct pigeonholes for our need. The construction is indeed a standard one for this kind of problem (which involves choosing an arbitrary number of elements from a given set).

Solution. Consider the $n-1$ integers $1, n+1, \dots, n(n-2)+1$. Each of them is congruent to 1 modulo n . Therefore, the sum of any of these integers must congruent to k modulo n for some $k = 1, 2, \dots, n-1$, which cannot be 0. Therefore, m is at least n .

For any n integers $a_1, a_2, \dots, a_n$, consider $s_1, s_2, \dots, s_n$ defined by

$$s_j = a_1 + \dots + a_j.$$

If one of the s_j 's is a multiple of n , then we are done by choosing $a_1, \dots, a_j$. Otherwise, these n integers must fall in the other $n-1$ residue classes modulo n . By the

pigeonhole principle, there exist indices $i < j$ such that $s_i \equiv s_j \pmod{n}$. Then

$$a_{i+1} + \cdots + a_j = s_j - s_i \equiv 0 \pmod{n}$$

and hence we can choose $a_{i+1}, \dots, a_j$. This shows n is the optimal value.

Problem 2.5. (Sperner's theorem(斯佩納定理)) Let $\mathcal{F}$ be a family of subsets of $\{1, 2, \dots, n\}$ such that there do not exist two distinct elements $S, T \in \mathcal{F}$ satisfying $S \subset T$. Find the greatest possible value of $|\mathcal{F}|$.

Analysis. Recall that a family of sets means a set whose elements are sets. For example, $\{\{3\}, \{1, 2\}, \{2, 3\}\}$ is a family of subsets of $\{1, 2, 3\}$. But it does not satisfy the condition as $\{3\} \subset \{2, 3\}$.

To guess the answer, one may find a sufficient condition for two sets S, T not being subsets of each other. Indeed, if $|S| = |T|$ and they are distinct subsets, then we cannot have $S \subset T$ or $T \subset S$. Therefore, we can choose all the k -element subsets of $\{1, 2, \dots, n\}$. The number $\binom{n}{k}$ is maximized when $k = \left\lfloor \frac{n}{2} \right\rfloor$. This is indeed the answer.

Solution. For each subset $S = \{a_1, \dots, a_k\}$ of $\{1, 2, \dots, n\}$, we associate it to some sequences $b_1, \dots, b_n$ which are permutations of $1, 2, \dots, n$ for which $\{b_1, \dots, b_k\} = S$. Therefore, each subset S with k elements is associated to exactly $k!(n-k)!$ different sequences.

The key observation is as follows. For any two sets $S, T \in \mathcal{F}$, the sequences that they associate are all distinct. Otherwise, suppose both of them associate to $b_1, \dots, b_n$. Then we may assume there exist $k \leq \ell$ such that $\{b_1, \dots, b_k\} = S$ and $\{b_1, \dots, b_\ell\} = T$. We cannot have $k = \ell$ since S and T are distinct. Then this implies $S \subset T$, which contradicts the constraint on $\mathcal{F}$.

Now, $\sum_{S \in \mathcal{F}} (|S|)!(n-|S|)!$ counts the number of sequences to which the subsets in $\mathcal{F}$ associate. If this number is larger than $n!$, then there exists a sequence to which two distinct subsets in $\mathcal{F}$ associate by the pigeonhole principle. This contradicts to our observation. Therefore, we obtain

$$\sum_{S \in \mathcal{F}} (|S|)!(n-|S|)! \leq n!.$$

This can be rewritten as

$$\sum_{S \in \mathcal{F}} \binom{n}{|S|}^{-1} \leq 1.$$

As $\binom{n}{|S|} \leq \binom{n}{\left\lfloor \frac{n}{2} \right\rfloor}$, this yields

$$\binom{n}{\left\lfloor \frac{n}{2} \right\rfloor} \geq \sum_{S \in \mathcal{F}} 1 = |\mathcal{F}|.$$

This gives an upper bound on $|\mathcal{F}|$. The upper bound can be attained when, for example, $\mathcal{F}$ contains all $\left\lfloor \frac{n}{2} \right\rfloor$ -element subsets of $\{1, 2, \dots, n\}$. Clearly, there do not exist two distinct S, T among these subsets such that $S \subset T$.

Problem 2.6. (CHKMO 2006 Problem 1) Let M be a subset of $\{1, 2, \dots, 2006\}$ with the following property: For any three elements x, y and z ($x < y < z$) of M , $x + y$ does not divide z . Determine the largest possible size of M .

Analysis. This is another common type of problem which can be solved using the pigeonhole principle (or any equivalent arguments). Usually, the optimal case can be attained by choosing all the large elements.

Solution. When $M = \{1003, 1004, \dots, 2006\}$, the sum of any two elements in M is at least 2007, which is larger than the largest element in M . Therefore, $x + y \nmid z$ for any $x, y, z \in M$. This gives a possible case for $|M| = 1004$.

Consider any subset M satisfying the constraint. Suppose the largest element of M is a . Then M contains at most one element in each of the pairs

$$(1, a-1), (2, a-2), \dots, \left(\left\lfloor \frac{a}{2} \right\rfloor, \left\lceil \frac{a}{2} \right\rceil \right)$$

(note that the last pair may contain the same number, which does not affect the validity of our claim). Therefore, M contains at most $\frac{a}{2}$ elements smaller than a . Since a is the largest element, M cannot contain elements larger than a . Therefore,

$$|M| \leq \frac{a}{2} + 1 \leq 1003 + 1 = 1004.$$

Hence, the largest possible size of M is 1004.

Problem 2.7. (CHKMO 2012 Problem 2) Some of the lattice points (x, y) , with $1 \leq x \leq 101$ and $1 \leq y \leq 101$ are marked so that no 4 marked points form the vertices of an isosceles trapezoid with bases parallel to the x -axis or the y -axis (a rectangle is counted as an isosceles trapezoid). Determine the maximum number of marked points. (A lattice point is a point with integral coordinates.)

Analysis. In establishing the upper bound, we only focus on those isosceles trapezoids with bases parallel to the x -axis. Each such isosceles trapezoid has vertices (x_1, y_1) , (x_2, y_1) , (x_3, y_2) and (x_4, y_2) with $x_1 + x_2 = x_3 + x_4$. This suggests us considering the sum of x -coordinates for a pair of points in the same horizontal line.

The construction for the optimal case is slightly harder than usual. Hints can be found from the small cases.

Solution. Consider pairs of marked points with the same y -coordinates. If there exist two pairs of marked points with the same sum of x -coordinates and different y -coordinates, then the 4 points in these pairs are the vertices of an isosceles trapezoid.

Suppose there are $a_1, a_2, \dots, a_{101}$ marked points in the 101 horizontal lines respectively (this means there are a_1 marked points with y -coordinate equal to 1, etc.). For the horizontal line $y = 1$, suppose the marked points are $(x_1, 1), (x_2, 1), \dots, (x_{a_1}, 1)$ where $x_1 < x_2 < \dots < x_{a_1}$. Since

$$x_1 + x_2 < x_1 + x_3 < \dots < x_1 + x_{a_1} < x_2 + x_{a_1} < \dots < x_{a_1-1} + x_{a_1},$$

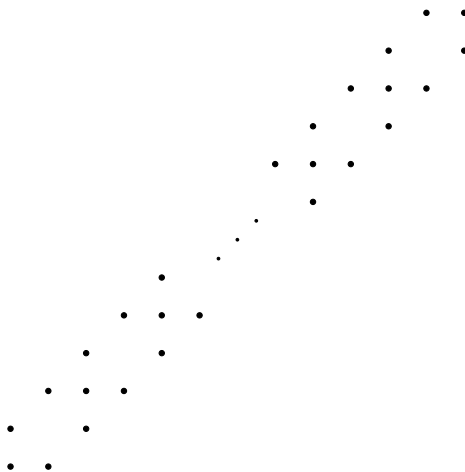
there are at least $2a_1 - 3$ distinct sums formed from the x -coordinates. This only holds for $a_1 \geq 2$. For $a_1 = 0$ or $a_1 = 1$, clearly, no sum can be formed. Therefore, there are at least

$$f(a_1) + f(a_2) + \dots + f(a_{101}) \tag{1}$$

sums where $f(n) = \max\{2n - 3, 0\}$. Each of these sums lies between $1 + 2 = 3$ and $100 + 101 = 201$. There are 199 possibilities. If (1) exceeds 199, then two of the sums are equal by the pigeonhole principle. Since the equal sums must correspond to distinct horizontal lines from the construction, we can find an isosceles trapezoid as suggested. Therefore, we may assume (1) is at most 199. Then

$$199 \geq f(a_1) + f(a_2) + \dots + f(a_{101}) \geq (2a_1 - 3) + (2a_2 - 3) + \dots + (2a_{101} - 3).$$

This implies $a_1 + a_2 + \dots + a_{101} \leq 251$. In other words, the number of marked points is at most 251. This can be attained, for example, from the following construction.



It is not hard to see that there is no isosceles trapezoid (a rigorous proof is to show that the sums of x -coordinates of pairs of points in the same horizontal line are all different, and the same holds for y -coordinates by symmetry). There are

$$3 + 5 \times 49 + 3 = 251$$

points in total.

Problem 2.8. (CHKMO 2011 Problem 4) Let $A_1A_2\cdots A_n$ be a cyclic polygon ($n \geq 3$). Find the maximum number of distinct acute-angled triangles whose vertices are chosen from $A_1, A_2, \dots, A_n$.

Analysis. We shall count the number of non-acute triangles, which means a triangle with an angle $\geq 90^\circ$. An angle $\angle ABC$ is at least 90° iff points A, B, C lie on the same side of a diameter of the circle. This suggests a simple way to count the total number in terms of the binomial coefficients. It remains to use algebraic method to find an upper bound.

Solution. We first count the number N_1 of non-acute triangles with A_1 as one vertex which is $< 90^\circ$. Let ℓ be the diameter of the circle which passes through A_1 . Then for points B, C on the circle, $\triangle A_1BC$ is not acute with $\angle BA_1C < 90^\circ$ iff B and C lie on the same side of ℓ (possibly on ℓ). Suppose there are a points on one side of ℓ and b points on the other side of ℓ other than A_1 . Note that the two sides share at most one common point which is the other intersection of ℓ and the circle. Therefore, we must have $a + b = n - 1$ or $a + b = n$. We find that

$$N_1 = \binom{a}{2} + \binom{b}{2}.$$

We have to minimize N_1 . Note that $b \geq n - 1 - a$ so that $\binom{b}{2} \geq \binom{n-1-a}{2}$. WLOG assume $a \geq n - 1 - a$. If n is odd, then $(a, n - 1 - a) > \left(\frac{n-1}{2}, \frac{n-1}{2}\right)$. Apply the **majorization inequality**. We obtain

$$N_1 \geq \binom{a}{2} + \binom{n-1-a}{2} \geq \binom{\frac{n-1}{2}}{2} + \binom{\frac{n-1}{2}}{2} = \frac{(n-1)(n-3)}{4}.$$

The same bound holds if we replace A_1 by any other vertex A_j . We sum up all the lower bounds. Note that each non-acute triangle is counted twice in this sum (one corresponding to each acute angle). Therefore, the number of non-acute triangles is at least

$$n \cdot \frac{(n-1)(n-3)}{4} \cdot \frac{1}{2} = \frac{n(n-1)(n-3)}{8}.$$

Thus, the number of acute triangles is at most

$$\binom{n}{3} - \frac{n(n-1)(n-3)}{8} = \frac{n(n-1)(n+1)}{24}.$$

This can be attained when $A_1A_2\cdots A_n$ is a regular polygon (since in that case all inequalities used become equalities).

Similarly, if n is even, then $(a, n - 1 - a) > \left(\frac{n}{2}, \frac{n-2}{2}\right)$. Apply the **majorization inequality**. We obtain

$$N_1 \geq \binom{a}{2} + \binom{n-1-a}{2} \geq \binom{\frac{n}{2}}{2} + \binom{\frac{n-2}{2}}{2} = \frac{(n-2)^2}{4}.$$

Thus, the number of acute triangles is at most

$$\binom{n}{3} - n \cdot \frac{(n-2)^2}{4} \cdot \frac{1}{2} = \frac{n(n-2)(n+2)}{24}.$$

The regular polygon case does not yield equality. Indeed, we first consider a regular polygon $B_1B_2\cdots B_n$. Then we rotate half of the points, say $B_1, B_2, \dots, B_{\frac{n}{2}}$, anti-clockwise about the centre of the circle with a sufficiently small angle θ to points $C_1, C_2, \dots, C_{\frac{n}{2}}$. Then $C_1C_2\cdots C_{\frac{n}{2}}B_{\frac{n}{2}+1}B_{\frac{n}{2}+2}\cdots B_n$ is one possible case which gives the upper bound $\frac{n(n-2)(n+2)}{24}$.

2.3 Colouring

Colouring refers to the method of partitioning a certain set into several subsets and use the distinct properties of these subsets to proceed, usually using the pigeonhole principle or double counting to deduce useful results. It is a useful tool to solve many tiling problems.

Problem 2.9. Let a be a positive integer. Find all $m \times n$ rectangles which can be tiled by $1 \times a$ (and $a \times 1$) rectangles completely.

Analysis. For a tiling, we mean every cell is covered by exactly one rectangle. Therefore, an immediate necessary condition for this to hold is that the area of the $m \times n$ rectangle is a multiple of the area of the small rectangle. However, this is not sufficient. Indeed, it is only clear that a tiling is possible if one of m, n is divisible by a .

Solution. If $a \mid m$ or $a \mid n$, then it is obvious that we can use $1 \times a$ rectangles to tile an $m \times n$ rectangle. Now, assume $a \nmid m$ and $a \nmid n$.

We use (i, j) to denote the cell in row i and column j , for $0 \leq i \leq m-1$ and $0 \leq j \leq n-1$. We use a colours $C_0, C_1, \dots, C_{a-1}$ to colour the cells of an $m \times n$ rectangle as follows. For the cell (i, j) , we set $i + j \equiv r \pmod{a}$ where $0 \leq r \leq a-1$. Then we use C_r to colour the cell. No matter horizontal or vertical rectangles are used, each $1 \times a$ rectangle must cover a cells with pairwise distinct colours. Therefore, if a tiling is possible, then the number of cells in each colour should be the same. We claim that this is not the case.

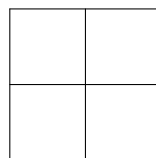
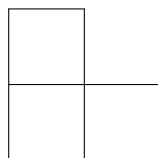
Let $m = q_1a + r_1$ and $n = q_2a + r_2$ where $1 \leq r_1, r_2 \leq a-1$. WLOG assume $r_1 \leq r_2$. We consider the number of cells in colour C_{r_1-2} and C_{r_1-1} respectively (where the indices are taken modulo a).

A pair (i, j) with $0 \leq i \leq m-2$, $0 \leq j \leq n-1$ and $i + j \equiv r_1 - 2 \pmod{a}$ corresponds to a pair $(i+1, j)$ with $1 \leq i+1 \leq m-1$, $0 \leq j \leq n-1$ and $(i+1) + j \equiv r_1 - 1 \pmod{a}$.

Removing these corresponding pairs, it suffices to count the number of solutions to the former equation with row number $m - 1$ and count the number of solutions to the latter equation with row number 0. Indeed, the number of pairs $(m - 1, j)$ with $0 \leq j \leq n - 1$ and $(m - 1) + j \equiv r_1 - 2 \pmod{a}$ is q_2 since $m \equiv r_1 \pmod{a}$ and $n - 1 = q_2 a + (r_2 - 1)$. Similarly, the number of pairs $(0, j)$ with $0 \leq j \leq n - 1$ and $0 + j \equiv r_1 - 1 \pmod{a}$ is $q_2 + 1$. Thus, the number of cells in colour C_{r_1-2} is different from the number of cells in colour C_{r_1-1} . This is a contradiction.

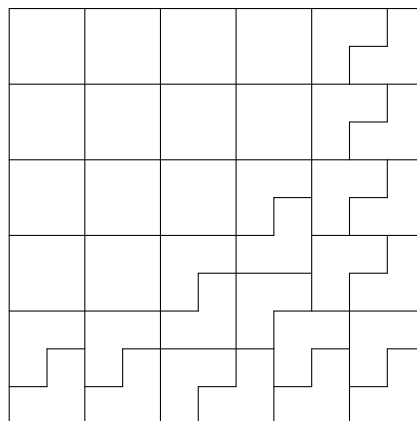
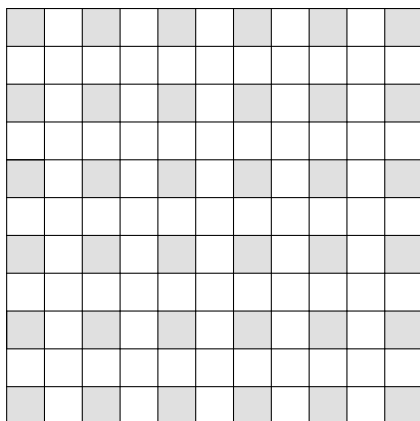
This shows a tiling is not possible if $a \nmid m$ and $a \nmid n$. This completes the proof.

Problem 2.10. (HKIMO TST 2015 Test 2 Problem 4) A 11×11 grid is to be covered completely without overlapping by some 2×2 squares and L -shapes each composed of three unit cells. Determine the smallest number of L -shapes used. (Each shape must cover some grids entirely and cannot be placed outside the 11×11 grid. The L -shapes may be reflected or rotated when placed on the grid.)



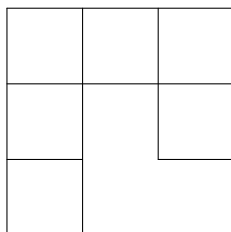
Analysis. Suppose x squares and y L -shapes are used. A simple counting gives $4x + 3y = 121$. This gives $y = 3, 7, 11, \dots$ and correspondingly, $x = 28, 25, 22, \dots$. To minimize y , we have to maximize x . However, it seems that the grid cannot contain too many squares, or otherwise it cannot be tiled completely. This suggests using colouring to find another constraint which has to be satisfied by x and y .

Solution. Suppose x squares and y L -shapes are used. Then the total number of cells is $4x + 3y$, which should be equal to $11^2 = 121$.



We colour the cells as shown. Then every square and every L -shape covers at most one blackened cell. As there are 36 blackened cells, we have $x + y \geq 36$. Therefore, $y = 4(x + y) - (4x + 3y) \geq 4(36) - 121 = 23$. The figure on the right gives one example using 23 L -shapes.

Problem 2.11. (IMO shortlist 2004 C7/Problem 3) Define a ‘hook’ to be a figure made up of six unit squares as shown below in the picture, or any of the figures obtained by applying rotations and reflections to this figure.

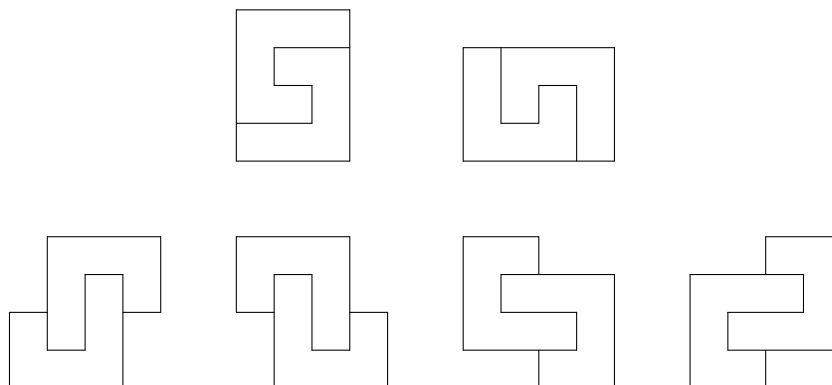


Determine all $m \times n$ rectangles that can be covered without gaps and without overlaps with hooks such that

- the rectangle is covered without gaps and without overlaps
- no part of a hook covers area outside the rectangle.

Analysis. Firstly, it is obvious that the hooks must come in pairs to form one of two possible shapes. Therefore, by counting the number of cells, we immediately know that mn is divisible by 12. However, this is not a sufficient condition for such a tiling to exist. Indeed, we shall use various colouring to obtain more information.

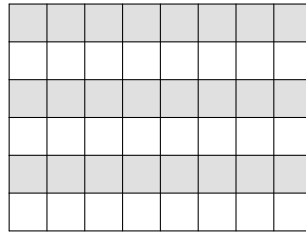
Solution. Note that the centre of a hook must be occupied by another hook. Thus, the hooks come in pairs in one of the following ways.



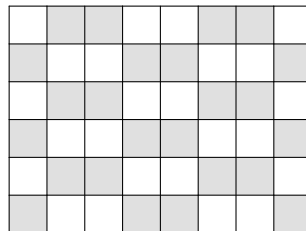
Thus, we may simply consider tiling using these shapes. We use R_1 and R_2 respectively to denote the shapes in the first row, and use S_1, S_2, S_3, S_4 to denote the shapes in the second row.

Since each shape consists of 12 cells, we must have $12 \mid mn$ if the tiling is possible. We first consider the hardest case $2 \mid m$ and $6 \mid n$. We would like to show that one of m, n is divisible by 4 so that this case is indeed contained in the other cases that we consider later.

We colour all the cells in the odd rows as shown below. Under this colouring, each R_1, S_1, S_2, S_3, S_4 occupies exactly 6 coloured cells, while each R_2 occupies 4 or 8 coloured cells. As m is even, the number of coloured cells is equal to the number of uncoloured cells. Hence, the number of R_2 's occupying 4 coloured cells is the same as the number of R_2 's occupying 8 coloured cells. This shows the number of R_2 's is even. If we colour the odd columns instead of rows, then the same argument shows that the number of R_1 's is even.



We further consider a colouring as shown below. This means for the odd rows, only cells in the k th column with $k \equiv 2, 3 \pmod{4}$ are coloured, while for the even rows, only cells in the k th column with $k \equiv 0, 1 \pmod{4}$ are coloured. Then each of R_1, R_2, S_1, S_2 occupies 6 coloured cells. For S_3 and S_4 , those which are at an even distance from the leftmost edge (and hence the rightmost edge) each occupies 6 coloured cells, while each of the remaining occupies 4 or 8 coloured cells. Therefore, there is an even number of shape of the second kind. Similarly, if we translate this colouring to the left by 1 column (meaning that a cell (ℓ, k) is coloured if ℓ is odd and $k \equiv 1, 2 \pmod{4}$, or ℓ is even and $k \equiv 0, 3 \pmod{4}$), then the number of S_3 's and S_4 's which are at even distances from the leftmost edge is even. By combining these results, we find that the total number of S_3 's and S_4 's is even. By symmetry, the total number of S_1 's and S_2 's is also even.



Now, we have found that in the case $2 \mid m$ and $6 \mid n$, the number of shapes is even. Therefore, the number of cells is a multiple of 24. This shows at least one of m, n is divisible by 4. This will be covered in the following cases.

If $4 \mid m$ and $3 \mid n$, then we can use only R_1 to tile the rectangle in the natural way. Up to symmetry, it remains to consider the case where $12 \mid m$ and n is not divisible by 3 or 4. Since any $m \times 3$ or $m \times 4$ rectangle can be tiled, a tiling exists if n can be written in the form $3a + 4b$ for some nonnegative integers a, b . It is known that every n which is at least $(3 - 1)(4 - 1) = 6$ can be written in this form (indeed, for this particular case, it suffices to check that all residue classes $1, 2, 5, 7, 10, 11 \pmod{12}$ greater than 5 have such representations). Therefore, the only cases left are $n = 1, 2, 5$.

Since every shape occupies at least 3 columns, the cases $n = 1, 2$ cannot be tiled. For $n = 5$, no matter which shape we use to cover the $(1, 1)$ cell, that shape must occupy the $(3, 3)$ cell. Due to symmetry, the shape covering the $(1, 5)$ cell must also occupy the $(3, 3)$ cell. Thus, these two shapes overlap but they cannot be the same shape. This shows no tiling is possible.

We conclude that an $m \times n$ rectangle can be tiled if it belongs to at least one of the following cases:

- $4 \mid m$ and $3 \mid n$; or
- $3 \mid m$ and $4 \mid n$; or
- $12 \mid m$ and $n \geq 7$; or
- $m \geq 7$ and $12 \mid n$.

2.4 Extremal Principle

Sometimes we may consider an extremal situation. This means we may choose a situation for which a certain quantity is maximized or minimized. This usually allows us to get more information than simply working on an arbitrary situation.

Problem 2.12. Let n be a positive integer. Suppose $3n$ cells of a $2n \times 2n$ table are blackened. Prove that one can choose n rows and n columns such that all blackened cells lie in a chosen row or a chosen column.

Analysis. Since we want the chosen rows and columns to contain many blackened cells, we may simply consider a maximal case and claim that it satisfies the condition. On average, every 1.5 of the blackened cells should lie in a chosen row or a chosen column. Thus, it suffices to have half of the chosen rows and columns each of which contains 2 blackened cells.

Solution. Firstly, we choose n rows which contain the greatest number of blackened cells. Such a choice may not be unique but it does not matter. Secondly, we choose n columns which contain the greatest number of blackened cells which do not lie in any of the chosen rows. We claim that this choice satisfies the condition.

Suppose on the contrary that there is a blackened cell not lying in the chosen rows and columns. Then there are at least $n + 1$ blackened cells not lying in any of the chosen rows (and those cells should lie in different columns). Since only n rows are not chosen, 2 of these blackened cells must belong to the same row by the pigeonhole principle. By the maximality of the first step, every chosen row must contain at least 2 blackened cells. This shows at most $3n - 2n = n$ blackened cells do not lie in the chosen rows, which is a contradiction. Therefore, our choice does meet the condition.

Problem 2.13. There is a real number in each cell of an $m \times n$ table. In each step, we can change the signs of all the numbers in a row or in a column. Prove that it is possible to make the sum of every row and every column nonnegative after finitely many steps.

Analysis. Since the sum of every row and every column has to be nonnegative, the sum of all the numbers should be large. We may pick the case corresponding to the largest sum. To do so, we first need to ensure that there are only finitely many possibilities for the sum.

Solution. If we change the signs of a certain row or column twice, then the effects are cancelled out. Therefore, we may assume the sign of each row and each column is changed at most once. Hence, there are only finitely many possible configurations by applying the steps. Among all configurations which can be attained, we choose one for which the sum of all numbers is the largest. We claim that the condition is satisfied under this configuration.

Suppose on the contrary (and WLOG) that the sum $-S$ of the first row is negative. If we change the signs of numbers in the first row, then this sum is changed to S , which is greater than $-S$. Thus, the total sum is increased. This contradicts the maximal choice. Therefore, the sum of any row and any column is nonnegative and we are done.

Problem 2.14. (APMO 2008 Problem 2) Students in a class form groups each of which contains exactly three members such that any two distinct groups have at most one member in common. Prove that, when the class size is 46, there is a set of 10 students in which no group is properly contained.

Analysis. Since we want to find a subset of the class of size 10 which does not contain any group, we may choose such a subset with the largest size. If the size exceeds 10, we may simply remove the extra students. Therefore, the key point is to derive a contradiction by assuming the size is smaller than 10.

Solution. Consider any set of students which do not contain a group. We choose such a set S with the largest size. It suffices to show that the size is at least 10. Suppose on the contrary that $|S| \leq 9$. By the maximality, we cannot add any student to this set. Therefore, for any student not in S , he must form a group with two other students in S . We count the number N of groups with students a, b, c such that $a, b \in S$ and $c \notin S$.

On the one hand, any $c \notin S$ corresponds to at least one such group as mentioned. Therefore, $N \geq 46 - |S|$. On the other hand, every pair of $a, b \in S$ corresponds to at most one such group since two distinct groups have at most one common member. This gives $N \leq \binom{|S|}{2}$. Combining the inequalities, we obtain $|S|(|S| + 1) \geq 92$. This contradicts the assumption

$$|S|(|S| + 1) \leq 9 \times 10 = 90.$$

Therefore, we must have $|S| \geq 10$. Thus, we can choose any subset of S with size 10 to meet the condition.

Remark. A proof using the extremal principle can usually be replaced by a proof using induction. For example, in this problem, one can extend the set S by adding more and more students until the size becomes 10 using basically the same argument.

Problem 2.15. (CHKMO 2015 Problem 4) Let $n \geq 2$ be an integer. There are n distinct circles on the plane such that any two circles have two distinct intersections and no three circles have a common intersection. Initially there is a coin on each of the intersection points of the circles. Starting from X , players X and Y alternately take away a coin, with the restriction that one cannot take away a coin lying on the same circle as the last coin just taken away by the opponent in the previous step. The one who cannot do so will lose. In particular, one loses when there is no coin left. For what values of n does Y has a winning strategy?

Analysis. Since the number of coins is even, Y is the one who take the last coin if the restriction is ignored. Therefore, we believe that Y should be the winner. It suffices to divide the intersection points into pairs such that the two intersection points in the same pair do not lie on the same circle (and thus we need $n \geq 4$ to apply this argument). One can give a construction directly. However, we shall use the extremal principle as an alternative solution. This shows such a pairing exists without providing the explicit construction.

Solution. We use (i, j) to denote a coin on the intersection of circles i and j . Thus, the two coins on the intersection of circles i and j are (i, j) and (j, i) for $i \neq j$.

For $n = 2$ or $n = 3$, X wins by taking away $(1, 2)$ as Y cannot take the other coins. For $n = 4$, WLOG assume X takes away $(1, 2)$ in the first step. Then Y should take away $(3, 4)$. Afterwards, X can only take away $(2, 1)$, and then Y can only take away $(4, 3)$. At this stage, X cannot take away any coin and hence Y wins.

Now, we claim that Y has a winning strategy for $n \geq 5$. It suffices to pair up the $n(n-1)$ intersection points such that the two points in the same pair do not lie on the same circle. If this can be done, whenever X takes away any coin on a point, Y then takes away the coin on the point which is in the same pair as that. This shows Y can always take away a coin. The game will end when X cannot take a coin, which must happen since there is a finite number of coins. Then Y will win the game.

Consider all pairings of the $n(n-1)$ intersection points which do not contain a pair $\{(i,j), (j,i)\}$. Such pairings exist for $n \geq 3$. For each pairing, we count the number of bad pairs, meaning pairs of intersection points lying on the same circle, and denote the number by N . Among all possible pairings, we choose one for which N is minimized. So we have to prove $N = 0$.

Suppose on the contrary that $N \geq 1$. WLOG assume there is a pair $\{(a,b), (a,c)\}$ where $b \neq c$. We say that an intersection point (i,j) is special if $i = a$ or $j = a$ or $\{i,j\} = \{b,c\}$. We claim that there is another pair containing two intersection points which are not special. Indeed, there are $\frac{n(n-1)}{2}$ pairs in total, and there are only $(n-1) + (n-1) + 2 = 2n$ special points. Since $\frac{n(n-1)}{2} \geq 2n$ for $n \geq 5$ and there exists a pair containing two special points as assumed, this claim is true. There are two possibilities for this non-special pair.

- Suppose there is a pair $\{(w,x), (y,z)\}$ of non-special points where w, x, y, z are pairwise distinct. WLOG we may assume w, x are distinct from b , and y, z are distinct from c (since (b,c) and (c,b) are special). We swap the points (a,c) and (w,x) . We have the new pairs $\{(a,b), (w,x)\}$ and $\{(a,c), (y,z)\}$. From our assumption, both new pairs are not bad. Therefore, N is decreased by 1, which contradicts the minimality of our choice.
- Suppose there is a bad pair $\{(w,x), (w,y)\}$ of non-special points where w, x, y are pairwise distinct. WLOG we may assume w, x are distinct from b . (If $w = b$, then we may assume w, x are distinct from c .) We swap the points (a,c) and (w,x) . We have the new pairs $\{(a,b), (w,x)\}$ and $\{(a,c), (w,y)\}$. The first pair is not bad from the assumption. Therefore, N is decreased by at least 1. Again, this is a contradiction.

Therefore, we must have $N = 0$. Thus, Y can apply the strategy to win the game for $n \geq 5$. To conclude, Y wins iff $n \geq 4$.

3 Game and Operation

For operation problems, we mean combinatorial problems which define a certain rules of operations and ask whether something can be done by applying the operations repeatedly. Induction is certainly a helpful tool in this topic. Other than that, a number of problems require one to think of some algorithms to apply the operations. Another way of thinking is to look for patterns or invariances. We will emphasize on these two techniques in this section.

For game problems, they share some similarities with operation problems. The difference is that this kind of problem usually involve two players and ask if one side has a winning strategy. To confirm this, when we provide an algorithm for the winning side, we need to consider all possibilities that the opponent plays because we cannot control what the opponent does.

3.1 Algorithm

When we construct an algorithm so as to meet our goal, we may first consider some small cases and look for patterns. Sometimes cases corresponding to different parities may give different results. If some patterns can be found, then usually the way to solve the small cases can be generalized to all cases by induction. If there seems to be no hints from the small cases, then we cannot avoid working on the general case. But then we can still simplify the problem by transforming the general case to certain special cases for consideration. The exact way to do so will be explained when we go through the problems, since this depends very much on the rules of the operations.

Problem 3.1. (CMO 1994 Problem 2) There are n ($n \geq 4$) plates containing a total of at least 4 sweets. For each operation, we take away a sweet from each of the two chosen plates and put the sweets into another plate. Is it possible to move all sweets to a single plate after finitely many operations?

Analysis. By trying some small cases, one should find that the answer is yes. Since the operation is independent of the total number of sweets, we can apply induction directly. This allows us to work on a single configuration other than all initial configurations.

Solution. We use $(a_1, a_2, \dots, a_n)$ to denote a configuration for which the plates contain $a_1, a_2, \dots, a_n$ sweets respectively. We prove that it is always possible to move all sweets to a single plate by induction on the total number of sweets, which we denote by m .

For the base case $m = 4$, the only possible configurations are $(4, 0, 0, 0)$, $(3, 1, 0, 0)$, $(2, 2, 0, 0)$, $(2, 1, 1, 0)$ and $(1, 1, 1, 1)$. There may be other empty plates but we can

ignore them. From the chain of operations

$$(1, 1, 1, 1) \longrightarrow (3, 1, 0, 0) \longrightarrow (2, 0, 2, 0) \longrightarrow (1, 2, 1, 0) \longrightarrow (0, 4, 0, 0),$$

we find that all the configurations can be turned into the desired one. Thus, the claim is true when $m = 4$.

Consider the case $m \geq 5$ and assume all smaller cases have been established. Then we can first ignore one sweet and apply the inductive hypothesis. This allows us to move $m-1$ sweets to a single plate. If this happens to be the plate which contains the extra sweet, then we are done. Otherwise, we are left to work on the only configuration $(m-1, 1, 0, 0)$. Again, all other empty plates can be ignored. We simply need to find an algorithm to put the extra sweet to the first plate. One example is given as follows:

$$\begin{aligned} (m-1, 1, 0, 0) &\longrightarrow (m-2, 0, 2, 0) \longrightarrow (m-3, 2, 1, 0) \longrightarrow (m-4, 2, 0, 2) \\ &\longrightarrow (m-2, 1, 0, 1) \longrightarrow (m, 0, 0, 0). \end{aligned}$$

(Indeed, the key step is the third step since the previous steps are uniquely determined. The goal of our choice is to make the two plates have the same number of sweets so that they can be cleared using the same number of operations.)

This establishes the case m . By induction, we can always attain the desired configuration.

Problem 3.2. (CMO 2010 Problem 5) We carry out some operations on the cards placed on the points $A_1, A_2, \dots, A_n$ and the point O ($n \geq 3$). An operation refers to one of the following:

- (1) If the number of cards on a certain point A_i is at least 3, then we can take away 3 of them and put one on each of the points A_{i-1}, A_{i+1} and O (here $A_0 = A_n, A_{n+1} = A_1$); or
- (2) If the number of cards on the point O is at least n , then we can take away n of them and put one on each of the points $A_1, A_2, \dots, A_n$.

Prove that if the total number of cards placed on these $n+1$ points is at least n^2+3n+1 , then we can always apply some operations so that the number of cards on each point is not less than $n+1$.

Analysis. We would like to handle as few cases as possible. Due to the special point O , we may first apply only operations (1) to move most of the cards to point O . If the number of cards on O is large enough, then we can apply operations (2) to get what we want. This leaves us only a case when the number of cards on O lies in a certain range. The remaining task is to make the total number of cards on the A_i 's as small as possible.

Solution. We apply operation (1) repeatedly in an arbitrary way until this cannot be done. This must come to an end since the number of cards on O is increased by 1 in each operation. Then each point A_i contains at most 2 cards, so that O contains at least

$$n^2 + 3n + 1 - 2n = n^2 + n + 1$$

cards. We focus on this state and let $a_1, a_2, \dots, a_n, b$ be the number of cards on $A_1, A_2, \dots, A_n, O$ respectively. The above says $a_i \leq 2$ and $b \geq n^2 + n + 1$.

If $b \geq n^2 + 2n + 1$, then we can apply $n + 1$ operations (2) so that each point has at least $n + 1$ cards. Therefore, we may now assume $n^2 + n + 1 \leq b \leq n^2 + 2n$.

At this stage, we can only apply operation (2). We apply operation (2) once so that each a_i satisfies $1 \leq a_i \leq 3$ and b satisfies $n^2 + 1 \leq b \leq n^2 + n$. Note that we would like to make $S = a_1 + a_2 + \dots + a_n$ small so that b exceeds its upper bound to get what we want. Therefore, we should reduce the number of points A_i with 3 cards. Thus, we apply operation (1) on those points. Since the positions of these points are interdependent, we should do so step by step. There are a few cases to consider. We claim that we can always decrease the value of S while keeping the condition $1 \leq a_i \leq 3$ (which ensures that we can argue repeatedly).

- If each a_i is at least 2, then since $b \geq n^2 + 3n + 1 - 3n = n^2 + 1$, we can apply $n - 1$ operations (2) so that each a_i becomes at least $n + 1$ and b is at least $n + 1$. Thus we are done.
- If $a_2 = a_n = 3$, $2 \leq a_3, a_4, \dots, a_{n-1} \leq 3$ and $a_1 = 1$, we apply operation (1) to each $A_2, A_3, \dots, A_n$ in that order. For $k = 2, 3, \dots, n - 1$, after the operation to A_k , the value of a_{k+1} is increased by 1 and hence it is at least 3. Therefore, we can apply the operations as stated. After these operations, we have $a_1 = 3$, $a_2 = a_n = 1$ and $1 \leq a_3, a_4, \dots, a_{n-1} \leq 2$. The condition $1 \leq a_i \leq 3$ is kept and S is decreased (since S is decreased by 1 during each operation (1)).
- If $a_2 = a_k = 3$, $2 \leq a_3, a_4, \dots, a_{k-1} \leq 3$ and $1 \leq a_1, a_{k+1} \leq 2$ for some $3 \leq k \leq n - 1$, we apply operation (1) to each $A_2, A_3, \dots, A_k$ in that order. Then we have $a_2 = a_k = 1$, $1 \leq a_3, a_4, \dots, a_{k-1} \leq 2$ and $2 \leq a_1, a_{k+1} \leq 3$. The condition $1 \leq a_i \leq 3$ is kept and S is decreased.

Following any of these cases repeatedly, we must come to an end since S cannot decrease indefinitely. This means the configuration does not contain any of the specified cases up to symmetry. Indeed, the last two cases mean that there are two distinct points A_2, A_k each containing 3 cards so that any point between them (in one of the directions) contains either 2 or 3 cards. (The restrictions $1 \leq a_1, a_{k+1} \leq 2$ can be ignored since we can replace k by $k + 1$ if $a_{k+1} = 3$, etc.)

Suppose among $a_1, a_2, \dots, a_n$, there are x, y, z of them equal to 1, 2, 3 respectively. If $z > x$, then either $z = 1$ or there are two distinct $a_2, a_k = 3$ with $a_3, \dots, a_{k-1} \geq 2$ by the pigeonhole principle. In the former case, we have $x = 0$ so that we are done

as discussed. In the latter case, we can decrease the value of S and repeat the arguments. Therefore, it suffices to consider $z \leq x$. Indeed, at this stage the value of S is small enough for our purpose. We find that

$$b \geq n^2 + 3n + 1 - x - 2y - 3z \geq n^2 + 3n + 1 - 2(x + y + z) = n^2 + n + 1.$$

As explained, we simply need to apply n operations (2) so that each point contains at least $n + 1$ cards. This completes the proof.

Problem 3.3. (IMO shortlist 2009 C8) For any integer $n \geq 2$, we compute the integer $h(n)$ by applying the following procedure to its decimal representation. Let r be the rightmost digit of n .

- (1) If $r = 0$, then the decimal representation of $h(n)$ results from the decimal representation of n by removing this rightmost digit 0.
- (2) If $1 \leq r \leq 9$ we split the decimal representation of n into a maximal right part R that solely consists of digits not less than r and into a left part L that either is empty or ends with a digit strictly smaller than r . Then the decimal representation of $h(n)$ consists of the decimal representation of L , followed by two copies of the decimal representation of $R - 1$. For instance, for the number $n = 17,151,345,543$, we will have $L = 17,151$, $R = 345,543$ and $h(n) = 17,151,345,542,345,542$.

Prove that, starting with an arbitrary integer $n \geq 2$, iterated application of h produces the integer 1 after finitely many steps.

Analysis. Induction seems to be a useful tool to solve this problem. However, if one tries to work on the small cases, one will realize that even the case $n = 3$ is nontrivial and the case $n = 4$ is almost nonsense. Indeed, one should note that the process is tedious if there exist some large digits. Conversely, if n only consists of digits 0 and 1, then it is relatively simpler. Therefore, we may try to use a different inductive approach in contrast to the usual ordering.

Another useful observation is that if there is a digit 0, then every digit on the left of this 0 remains unchanged before this 0 is deleted. This allows us to break an integer into several parts each of which does not contain the digit 0. The same applies to larger digits using induction.

Solution. We say that a positive integer n is good if the problem statement holds. Firstly, suppose n has the form $\overline{X0Y}$ where X and Y are strings of digits. We note that the string X remains unchanged before the 0 is deleted using operation (1). If we can prove that both X and Y are good, then we can first apply the operations repeatedly until Y becomes 1. From the above observation, the integer becomes $\overline{X00\cdots01}$. After one more operation, we obtain $\overline{X00\cdots0}$. By deleting the zeros, we are left with the case X , which is good by assumption. Therefore, the integer n is also good. From this, it suffices to show that every n without the digit 0 is good.

Next, we consider $n = \overline{X1}$ where X does not contain the digit 0. After one more operation, the integer becomes $\overline{X0X0}$. Breaking the integer into two parts, it suffices to show X is good. In other words, we can actually delete the unit digit 1 from $\overline{X1}$ to obtain X . This can be used to replace the original operation (1) (as the unit digit will never be 0 under our consideration).

In the same way, suppose n has the form $\overline{X1Y}$ where X and Y do not contain 0. Suppose both X and Y are good. We apply the operations repeatedly using our new operation (1). Then the digit 0 never appears and hence X remains unchanged throughout the operations. The same holds until Y is transformed to an integer consists of only the digit 1, so that the integer becomes $\overline{X11\cdots 1}$. By deleting the 1's and using the fact that X is good, we conclude that n is good. In other words, we may now break the integer into several parts using the digits 1. Then we only need to show that each such part is good.

We can use the same argument repeatedly, or argue neatly using induction. At the end, it suffices to show that all integers containing only the digit 9 are good, which is simple. Indeed, consider $n = \underbrace{99\cdots 9}_{m \text{ times}}$ where $m \geq 1$. Applying the operation once,

we obtain $\underbrace{99\cdots 9}_{(m-1)} 8 \underbrace{99\cdots 9}_{(m-1)} 8$. We can delete the unit digit 8 and break this integer

into two parts using the middle 8. It suffices to show that $\underbrace{99\cdots 9}_{(m-1) \text{ times}}$ is good. By the

method of descent or by induction, it is rather obvious that the integer can eventually be transformed to 9. After one more operation, this becomes 88. The unit digit is deleted and leaves a single 8. Inductively, the integer will be transformed to 7, 6, $\dots$, 1 step by step. This shows any n is good as desired.

Next, we shall work on some game problems.

Problem 3.4. Let $n \geq 3$ be an integer. Initially, a number 1 and a number 2 are written on the blackboard. Players X and Y alternately write down a number on the blackboard according to the following rules:

- (1) If a k has been written on the blackboard, then one may write down a $k + 1$; or
- (2) If a k and an m have been written on the blackboard, then one may write down a km .

Note that the same integer may appear on the blackboard more than once. In operation (2), k and m must be distinct numbers on the blackboard but they may have the same value. For example, player X may write down an extra 2 in the first step, and then player Y may write down 4 in the next step as $4 = 2 \times 2$.

The game will end when one of the following is satisfied:

- (i) There are n numbers having the same remainder when divided by n ; or

(ii) There are n numbers having pairwise distinct remainders when divided by n .

The one who writes down the last number will win. Find all n for which player X has a winning strategy.

Analysis. If one tries to work on the small cases, one would find that player X has a winning strategy for $n = 3, 4, 5, 6, 7$. This may lead to a wrong feeling that player X always has a winning strategy, which is indeed not the case. This shows we should not easily believe in any pattern observed from small cases.

To find out who is the potential winner, we may simplify the rules of the game. For example, if the players can write down any integer on the blackboard without any constraint, then the players only need to prevent the number of distinct residue classes to exceed $n - 2$ and prevent the number of integers in each residue class to exceed $n - 2$. In other words, the players only have $(n - 2)^2$ chances in total. This shows the potential winner should depend on the parity of n . The rules of the game only affect the results of small cases. The potential winner only needs to guarantee that the residue class 1 can be written down.

Solution. In view of the rules, we shall consider everything modulo n .

For $n = 3$, player X wins immediately by writing down 3.

For $n = 4$, the one who writes down 3 or 4 will lose because the opponent will complete all residue classes. Thus, both players will only write down 2 (because 1 cannot be written down before 3 and 4 appear). By considering the parity, player X is the winner.

For odd $n \geq 5$, the number $(n - 2)^2 - 2$ is odd so that player X is the potential winner. The strategy is to write down $3, 4, \dots, \frac{n+1}{2}$ one by one regardless of what player Y has written down. Meanwhile, player Y has written down $\frac{n-5}{2}$ numbers.

So there can be at most $\frac{n-3}{2} \leq n-2$ numbers of the same residue class, and at most $2 + \frac{n-3}{2} + \frac{n-5}{2} = n-2$ numbers of different residue classes. Thus, player Y cannot win at this stage. One can write down the residue class 1 as $2 \times \frac{n+1}{2} = n+1$. Therefore, any residue class which has appeared can be written down.

To prevent losing the game, both players will avoid to have $n - 1$ distinct residue classes and avoid to have $n - 1$ numbers in the same residue classes. Hence, there are only $(n - 2)^2$ chances for them to write the numbers (note that one can always add a new residue class due to operation (1)). Considering the parity, player X is the winner.

For $n = 6$, player X can still win the game. He can first write down another 2. If player Y does not add a new residue class, then the two players can only write down 2 alternately, and player X will win. If player Y writes down 3 or 4, then player

X writes down 4 or 3 correspondingly. Afterwards, both players will avoid writing down 5 or 6, and hence they can only add more 2, 3, 4's. There are only 13 chances in this case (including the initial number 1). Thus, player X will win.

For even $n \geq 8$, we claim that player Y is the potential winner. No matter player X writes down 2 or 3 initially, player Y can always write down 4. Afterwards, player Y only needs to write down $5, 6, \dots, \frac{n}{2}$ and n one by one regardless of what player X has written down. Meanwhile, player X has written down $\frac{n-4}{2}$ numbers. So there can be at most $\frac{n-2}{2} \leq n-2$ numbers of the same residue class, and at most $2 + \frac{n-4}{2} + \frac{n-4}{2} = n-2$ numbers of different residue classes. Thus, player X cannot win at this stage. One can write down the residue class 1 by operation (1). Therefore, any residue class which has appeared can be written down. As in the odd case, player Y will win.

Therefore, player X wins iff n is odd or $n = 4, 6$.

Problem 3.5. Initially, two positive integers $m, n \geq 2$ are written down on the blackboard. Players X and Y alternately choose one number $k \geq 2$ on the blackboard and replace it by $k-1$ or $\left\lfloor \frac{k}{2} \right\rfloor$. A player wins if both numbers on the blackboard become 1. Find all pairs (m, n) such that player X has a winning strategy.

Analysis. If we consider the same problem with one initial number instead of two, then we can easily deduce the small cases for which player X can win. Indeed, player X wins iff n belongs to the set $S = \{2, 3, 5, 7, 8, 9, 11, 12, 13, 15, 17, \dots\}$. One observes that those n 's for which player X cannot win are $4, 6, 10, 14, 16, \dots$, which are twice the numbers in S . Such an observation can easily be proved via induction. The remaining task is to find a close form to express the answer. Note that the case of two numbers should be closely related to the case of one number. Thus, the set S should play an important role.

Solution. In view of the discussion, we let T be the set containing all positive integers of the form 2^{2k} or $2^{2k-1}(2m+1)$ for some $k, m \geq 1$ (which is essentially the set $\{4, 6, 10, 14, 16, \dots\}$) and let S be the complement of T in $\{2, 3, 4, \dots\}$. Equivalently, S contains all positive integers of the form 2^{2k-1} or $2^{2k-2}(2m+1)$ for some $k, m \geq 1$.

We consider the game with one initial number n instead of two. We claim that player X has a winning strategy iff $n \in S$. This can be proved via induction. When $n = 2$, we have $2 \in S$ and player X clearly wins.

For the inductive step, we first consider even $n \geq 4$. If $n \in S$, then $\frac{n}{2} \in T$ by definition. Player X can replace n by $\frac{n}{2}$ in the first step. Then the game is reduced to the case

$\frac{n}{2}$ where player X is the second player. From $\frac{n}{2} \in T$ and the inductive hypothesis, player X has a winning strategy. If $n \in T$, then $n - 1 \in S$ and $\frac{n}{2} \in S$. Thus, no matter what player X has done in the first step, the game is reduced to a case for which the first player, which is now player Y , has a winning strategy.

Similarly, suppose $n \geq 3$ is odd. Then we must have $n \in S$. By definition, either $n - 1$ or $\frac{n - 1}{2}$ belongs to T . Player X can replace n by that element, and hence he has a winning strategy. Our claim follows by induction.

We return to the original problem. If both m, n belong to T , player Y has a winning strategy. Indeed, this can be viewed as two separate games with a single positive integer. Whenever player X makes a move in the first game, player Y will do the same by following the strategy described, and similarly if player X makes a move in the second game. One of these games will end when player Y replaces the number by 1 eventually. The same holds for the remaining game and player Y is the one who wins.

Similarly, for $m \in S$ and $n \in T$ or the symmetric case, player X has a winning strategy. This is because he can replace the number m by a number in T , so that he can win from the previous case.

The hardest case is $m, n \in S$. A similar reasoning does not work. If one works on some small cases, then there are different occasions where player X and player Y can win respectively. Thus, we should further partition this into subcases. Indeed, if one of the numbers becomes an integer in T , then it is reduced to the previous case and hence the next player will win. Therefore, both players should keep both integers in set S . We note the following fact: for any $k \in S$, exactly one of $k - 1$ and $\left\lfloor \frac{k}{2} \right\rfloor$ belongs to S . Thus, the operations are uniquely determined if both players keep the integers in S .

We partition S into two sets $S_1 = \{2, 3, 8, 11, 13, 15, \dots\}$ and $S_2 = \{5, 7, 9, 12, 17, \dots\}$ depending on the parity of the number of steps required for the integer to become 1. For example, since $8 \rightarrow 7 \rightarrow 3 \rightarrow 1$, both 8 and 3 belong to the same set while 7 does not. If m, n both belong to S_1 or both belong to S_2 , then player Y has a winning strategy. If m, n belong to different sets (S_1 and S_2), then player X has a winning strategy. This can be easily proved by induction as above, since the situation is uniquely determined. This completes the proof.

Remark. It is more desirable to define S_1 and S_2 mathematically. We shall give a brief description without proof. Recall that any integer in S is of the form 2^{2s-1} or $2^{2s-2} \cdot (2t + 1)$ where $s, t \geq 1$.

- Every integer of the form 2^{2k-1} where $k \geq 1$ belongs to S_1 .
- An integer of the form $2^k - 1$ where $k \geq 2$ belongs to S_1 iff k is even.

- For any other integer in set S , let N be the number of 1's in its binary representation. Then the integer belongs to S_1 iff N is odd.

3.2 Invariance

Invariance refers to a quantity or a property which remains unchanged when operations are carried out. Some simple invariances can be easily observed, like total number of coins, the parity of a value, etc. To look for nontrivial invariances, one may link up both the operations and the results we need.

Problem 3.6. The first $2n$ positive integers lie in a row in an arbitrary order. Each step we can swap two adjacent numbers, or choose three adjacent numbers a, b, c in this order and replace them by c, a, b (but not b, c, a). Find the smallest number of steps required so that we can always rearrange the positive integers in ascending order regardless of the initial order.

Analysis. The ‘worst’ situation is probably the case when the integers are in descending order. With some trial, one would guess the answer is n^2 . It is not hard to use induction to prove that n^2 steps are always sufficient. However, induction could not be used in a straightforward way to prove the lower bound. This is because when there are more numbers, there are more flexibilities for us to choose the operations. Thus, the inductive hypothesis cannot be applied (which probably requires us to fix and ignore 2 numbers). In fact, in this solution we will find a suitable invariance to obtain the lower bound.

Solution. The answer is n^2 . We first prove by induction that we can always rearrange the numbers in ascending order in n^2 steps. This is trivial when $n = 1$. Suppose this holds for some $n = k$. When $n = k + 1$, there are $2k + 2$ numbers. Let the positions of 1 and 2 be x and y respectively.

If $x \geq 3$, we can choose 3 adjacent numbers $a, b, 1$ and replace them by $1, a, b$ in one step. Thus, in each step the position of 1 can be moved by 2 steps to the left, until $x = 1$ or $x = 2$. In the latter case, we can swap 1 and the number on its left. So 1 can be put to the expected position in at most $\left\lceil \frac{2k + 2 - 1}{2} \right\rceil = k + 1$ steps. Similarly, after that 2 can be put to the expected position in at most $\left\lceil \frac{2k + 2 - 2}{2} \right\rceil = k$ steps. After fixing 1 and 2, we can apply the inductive hypothesis to rearrange the remaining $2k$ numbers in k^2 steps. So at most

$$(k + 1) + k + k^2 = (k + 1)^2$$

steps are required to rearrange the numbers in ascending order. By induction, we get the upper bound n^2 for the number of steps required.

Next, consider the initial ordering $2n, 2n-1, \dots, 1$. We claim that at least n^2 steps are needed in this case. Under each operation, a number either moves to the right by 1 step or moves to the left by 1 or 2 steps.

Consider any number k . It needs to move to the left by $2n+1-2k$ steps (note that this also holds when $k > n$, with the convention that moving -1 step to the left means moving 1 step to the right). However, it is not hard to see that most numbers have to move to the right in some operations throughout the whole process. In fact, for any other number $m < k$, the pair (m, k) is in wrong order initially in the sense that m lies on the right of k . Since m has to lie on the left of k at the end, there must be an occasion for which their order is changed. This only happens in an operation for which m moves to the left while k moves to the right (probably together with another number). Also, each time there is at most one such m . Therefore, k has to move to the right by at least $k-1$ steps throughout the process. So there are at least

$$\left\lceil \frac{(2n+1-2k) + (k-1)}{2} \right\rceil = \left\lceil n - \frac{k}{2} \right\rceil$$

operations for which k is the number moving to the left. Again, since there is exactly one number moving to the left in each operation, there should be at least

$$\sum_{k=1}^{2n} \left\lceil n - \frac{k}{2} \right\rceil = n + (n-1) + (n-1) + (n-2) + (n-2) + \dots + 1 + 1 + 0 = n^2$$

operations in total.

Therefore, the minimum number of steps is n^2 .

Problem 3.7. Initially, there are three positive integers with sum S written on the blackboard. An operation consists of the following two steps in order. Suppose the positive integers on the blackboard are a, b, c .

- Firstly, we replace two of the integers a and b with difference at least 2 by $\left\lfloor \frac{a+b}{2} \right\rfloor$ and $\left\lceil \frac{a+b}{2} \right\rceil$ respectively.
- Secondly, we choose one of the integers $\left\lfloor \frac{a+b}{2} \right\rfloor$, $\left\lceil \frac{a+b}{2} \right\rceil$ or c (we denote the chosen integer by d) and replace this integer d by $S-d$.

For example, if the initial numbers are 1, 3, 7 with $S = 11$, then we can first replace these numbers by 4, 3, 4 since $\frac{1+7}{2} = 4$, and then replace these numbers by 4, 3, 7 since $7 = 11 - 4$. Note that only a single operation has been carried out. Also, the value of S is fixed throughout the process (although the sum of numbers may change). Prove that the process must come to an end no matter how the operations are carried out.

Analysis. The process will end when any two of the numbers differ by at most 1. It is not simple to find an invariance since the sum of the numbers may vary greatly, although it is not hard to prove that the numbers must be smaller than S throughout the process. We would like to find a function which keeps on decreasing. In view of our goal, such a function should be related to the differences of the numbers. Any such function must be decreasing in the first step. However, it does not make sense when a number d is replaced by $S - d$, unless S is involved in the differences in a certain way. This suggests the following choice.

Solution. Suppose the three numbers are a, b, c . We define

$$f = \left| a - \frac{S}{2} \right| + \left| b - \frac{S}{2} \right| + \left| c - \frac{S}{2} \right|.$$

We claim that f is monotonic decreasing throughout the process. Suppose we carry out the first step on a and b where $a \geq b + 2$.

- If $\frac{S}{2} \geq a > b$, then $\frac{S}{2} \geq \left\lfloor \frac{a+b}{2} \right\rfloor, \left\lceil \frac{a+b}{2} \right\rceil$. Note that the sum of these two numbers remains unchanged in the first step. Therefore,

$$f = S - a - b + \left| c - \frac{S}{2} \right|$$

remains unchanged.

- Similarly, if $a > b \geq \frac{S}{2}$, then f remains unchanged.
- Suppose $a > \frac{S}{2} > b$. Since $\left\lfloor \frac{a+b}{2} \right\rfloor, \left\lceil \frac{a+b}{2} \right\rceil$ differ by at most 1, they are both at least $\frac{S}{2}$ or both at most $\frac{S}{2}$. Therefore, the value of f is changed to

$$|S - (a + b)| + \left| c - \frac{S}{2} \right|.$$

In this case, one can show that

$$|S - (a + b)| < a - b = \left| a - \frac{S}{2} \right| + \left| b - \frac{S}{2} \right|.$$

Thus, f is strictly decreasing.

Next, in the second step, suppose we replace a number a by $S - a$. Then f remains unchanged since

$$\left| a - \frac{S}{2} \right| = \left| (S - a) - \frac{S}{2} \right|.$$

This shows f is monotonic decreasing. Since f can only take nonnegative rational values with denominator 2, it cannot decrease indefinitely. Thus, either f is so small

that $|a-b|, |b-c|, |c-a| \leq 1$, in which case we are done, or f becomes a constant. From our discussion, this can only happen if we only apply the first step to two numbers which are both $\geq \frac{S}{2}$ or both $\leq \frac{S}{2}$. We shall show that this is impossible, meaning that the two numbers on the same side of $\frac{S}{2}$ must become closer and closer to each other.

To do so, we further define

$$g = |d_a - d_b| + |d_b - d_c| + |d_c - d_a|$$

where $d_x = \left|x - \frac{S}{2}\right|$ for $x = a, b, c$. This describes how close the differences of the numbers to $\frac{S}{2}$ are. WLOG assume we apply the first step to two numbers a, b with $a > b \geq \frac{S}{2}$. Then $|d_a - d_b| = |a - b|$ must decrease. Also,

$$|d_b - d_c| + |d_c - d_a| = \left|b - \frac{S}{2} - d_c\right| + \left|a - \frac{S}{2} - d_c\right|$$

is monotonic decreasing as before (here we may assume $\frac{S}{2} + d_c$ is a constant and argue in the same way). In the second step, the value of g remains unchanged. Therefore, if we always choose two numbers on the same side of $\frac{S}{2}$, then g is strictly decreasing, which is impossible. Therefore, the process must come to an end as desired.

Problem 3.8. (APMO 2009 Problem 1) Consider the following operation on positive real numbers written on a blackboard: Choose a number r written on the blackboard, erase that number, and then write a pair of positive real numbers a and b satisfying the condition $2r^2 = ab$ on the board.

Assume that you start out with just one positive real number r on the blackboard, and apply this operation $k^2 - 1$ times to end up with k^2 positive real numbers, not necessarily distinct. Show that there exists a number on the board which does not exceed kr .

Analysis. The strategy is to find a function f such that $f(r)$ and $f(a) + f(b)$ have some relations under the constraint $2r^2 = ab$. As it seems that the optimal operation is to take $a = b = \sqrt{2}r$, we would like to have $f(r) = 2f(\sqrt{2}r)$. The variable r can be ignored if f is homogeneous. This suggests the choice of f in the solution.

Solution. Let $f(x) = \frac{1}{x^2}$. Whenever $2r^2 = ab$, we have

$$f(r) = \frac{1}{r^2} = \frac{2}{ab} \leq \frac{1}{a^2} + \frac{1}{b^2} = f(a) + f(b).$$

Therefore, the sum of the function values is monotonic increasing. Suppose the k^2 numbers obtained at the end are $r_1 \leq r_2 \leq \dots \leq r_{k^2}$. Then

$$\frac{1}{r^2} \leq \sum_{i=1}^{k^2} \frac{1}{r_i^2} \leq \frac{k^2}{r_1^2}.$$

This gives $r_1 \leq kr$ as desired.

Problem 3.9. (IMO shortlist 2005 C5) There are n markers, each with one side white and the other side black. In the beginning, these n markers are aligned in a row so that their white sides are all up. In each step, if possible, we choose a marker whose white side is up (but not one of the outermost markers), remove it, and reverse the closest marker to the left of it and also reverse the closest marker to the right of it. Prove that, by a finite sequence of such steps, one can achieve a state with only two markers remaining if and only if $n - 1$ is not divisible by 3.

Analysis. To show that the final state can be attained, one simply needs to find an algorithm, probably with the use of induction. The harder part is to show that for the case $3 \mid n - 1$, the final state cannot be attained no matter how we operate. We have to find a function which is fixed throughout the process. If this can be done, then we know more about the configurations when the number of markers decreases.

The first simple idea may be assigning 1 for a white marker and 0 for a black marker and consider the sum of all values (in other words, we count the number of white markers). However, this is not an invariance, even if we only consider modulo 3 (which should be the case from the answer). Indeed, the sum modulo 3 is fixed only when we remove a white marker with both neighbours being white. In other cases, the number of white markers is decreased or increased by 1. It seems that only the parity is related, but not modulo 3.

We can try to modify the assignment of values by considering the operations. Suppose we remove the middle white marker in WWB (with obvious meaning of the labels). They become BW and all other markers are unaffected. The number of white markers is decreased by 1, but we would like the values assigned to the two white markers in WWB to have the same sum as that assigned to the white marker in BW . This suggests the relative position of the white and black markers should be taken into account. That is how we think of the solution given below.

Solution. For $n = 2$ and $n = 3$, it is obvious that the statement is true. Suppose we can achieve the final state (the state with only two markers remaining) for some n not congruent to 1 (mod 3). Consider the case $n + 3$. We use W to denote a marker with the white side facing upwards and B for black similarly. We apply the operations as follows. A marker with an underscore is the marker to be removed in the next step.

$$W\underline{W}WW\dots W \longrightarrow BB\underline{W}W\dots W \longrightarrow B\underline{W}BW\dots W \longrightarrow WW\dots W.$$

Since 3 steps are carried out, the number of markers is decreased by 3. Thus, we easily see that we can achieve the final state whenever $n \not\equiv 1 \pmod{3}$ by induction.

Suppose $n \equiv 1 \pmod{3}$. The case $n = 1$ is trivial. For the general case, we give a score to each marker for any given configuration. For each black marker, the score is 0. For each white marker, the score is 1 if there is an even number of black markers to its left, and the score is -1 if there is an odd number of black markers to its left. For example, the scores for the markers in $WWBW BWW$ are $1, 1, 0, -1, 0, 1, 1$ respectively. We claim that the sum $S \pmod{3}$ of all the scores assigned is an invariance.

- If WWW is changed to BB , then the scores of all markers not shown are unaffected. The scores for the white markers WWW are the same. Therefore, their contribution to $S \pmod{3}$ is 0. As black markers do not contribute to the sum, $S \pmod{3}$ remains unchanged.
- If WWB is changed to BW , then the scores of all markers not shown are unaffected. The scores for the markers involved change either from $1, 1, 0$ to $0, -1$ or from $-1, -1, 0$ to $0, 1$. In both cases, $S \pmod{3}$ remains unchanged.
- If BWW is changed to WB , then the scores of all markers not shown are unaffected. The scores for the markers involved change either from $0, 1, 1$ to $-1, 0$ or from $0, -1, -1$ to $1, 0$. In both cases, $S \pmod{3}$ remains unchanged.
- If BWB is changed to WW , then the scores of all markers not shown are unaffected. The scores for the markers involved change either from $0, 1, 0$ to $-1, -1$ or from $0, -1, 0$ to $1, 1$. In both cases, $S \pmod{3}$ remains unchanged.

This proves the claim. Initially, we have $S = n \equiv 1 \pmod{3}$. Assume it is possible to have two markers remaining. Then the sum of their scores is $\equiv 1 \pmod{3}$. This can happen only when the two markers are WB . However, we also note that the parity of the number of black markers is fixed in each operation. Therefore, it cannot happen that there is exactly one black marker. This completes the proof.

Problem 3.10. (IMO shortlist 2000 A5/Problem 3) k is a positive real. N is an integer greater than 1. N points are placed on a line, not all coincident. A move is carried out as follows. Pick any two points A and B which are not coincident. Suppose that A lies to the right of B . Replace B by another point B' to the right of A such that $AB' = kBA$. For what values of k can we move the points arbitrarily far to the right by repeated moves?

Analysis. When $N = 2$, it amounts to see when $1 + k + k^2 + \dots$ is divergent. The goal is met when $k \geq 1$. When $N = 3$, it is natural to consider the unique operation applied on the two points with the greatest distance. Suppose the initial points are A, B, C from left to right. Let $AB = a$ and $BC = b$. After one operation, the distances change

from a, b to $b, k(a+b)$. We want to find a certain function in terms of the distances a, b . If the value of the function does not decrease after an operation, then we can move the points arbitrarily far to the right. Although the negation needs not be true, it gives us a good hint to the answer.

Indeed, we consider $f = \alpha a + \beta b$ for some constants α, β . After one operation as specified, we have $f = \alpha b + \beta k(a+b) = \beta k a + (\alpha + \beta k)b$. If k is the critical value, then f should remain unchanged. This suggests we should have $\alpha = \beta k$ and $\beta = \alpha + \beta k$ by comparing coefficients. This gives $\beta = 2\alpha$. So we can simply choose $f = a + 2b$.

From this, we also believe that the answer for the case $N = 3$ is $k \geq \frac{1}{2}$. Indeed, the function $f = a + 2b = (a+b) + b$ has a special meaning in our situation. It simply means the sum of distances of the points from the rightmost point. This can easily be generalized to the case N .

Solution. Let the points be $a_1 \geq a_2 \geq \dots \geq a_N$ with $a_1 > a_N$. Define

$$f = (a_1 - a_2) + (a_1 - a_3) + \dots + (a_1 - a_N).$$

Suppose $k \geq \frac{1}{N-1}$. In each operation, we choose A to be the rightmost point and B to be the leftmost point. Then a_N is replaced by $a_1 + k(a_1 - a_N)$. However, the order of the points also changes. Therefore, the value of f becomes

$$\tilde{f} = \sum_{j=1}^{N-1} (a_1 + k(a_1 - a_N) - a_j) = k(N-1)(a_1 - a_N) + \sum_{j=2}^{N-1} (a_1 - a_j).$$

Since $k \geq \frac{1}{N-1}$ and $a_1 > a_N$, we have $\tilde{f} \geq f$. Since f is monotonic increasing and $a_1 - a_N \geq \frac{1}{N-1}f$, the distance between the two furthest points is bounded below by a positive number, say C_0 . In each step, the rightmost point is increased by at least $C_0 k$. Therefore, after finitely many steps, the rightmost point, and hence all the points, can be made sufficiently far to the right.

We now turn to the case $k < \frac{1}{N-1}$. Although the above function is decreasing in this case, it does not mean the points cannot be made sufficiently large (for example, $1 + \frac{1}{2} + \frac{1}{3} + \dots$ is divergent). Thus, the absolute size of the points should also be taken into account. This suggests we should consider

$$g = a_1 + C_1((a_1 - a_2) + (a_1 - a_3) + \dots + (a_1 - a_N))$$

for some positive constant C_1 . We claim that g is monotonic decreasing if we choose C_1 appropriately. Suppose the operation is applied on the points $A = a_u$ and $B = a_t$. This means a_t is replaced by $a_u + k(a_u - a_t)$. If $a_u + k(a_u - a_t) < a_1$, then it is obvious that g decreases. If $a_u + k(a_u - a_t) \geq a_1$, then the order of the numbers is changed.

Therefore, the value of g becomes

$$\tilde{g} = a_u + k(a_u - a_t) + C_1 \sum_{\substack{j=1 \\ j \neq t}}^N (a_u + k(a_u - a_t) - a_j).$$

Note that

$$\begin{aligned} & \tilde{g} \leq g \\ \Leftrightarrow & (a_u - a_1 + k(a_u - a_t)) + C_1 \left((N-1)(a_u - a_1 + k(a_u - a_t)) - a_1 + a_t \right) \leq 0 \\ \Leftrightarrow & (1 + C_1(N-1))(a_u - a_1 + k(a_u - a_t)) - C_1(a_1 - a_t) \leq 0. \end{aligned}$$

Note that $a_u - a_1 + k(a_u - a_t) \leq k(a_1 - a_t)$ since this is equivalent to $(a_1 - a_u)(k+1) \geq 0$.

Thus, if we choose C_1 such that $(1 + C_1(N-1))k = C_1$, i.e. $C_1 = \frac{k}{1 - (N-1)k}$, then

we obtain $\tilde{g} \leq g$. Note that C_1 is positive since $k < \frac{1}{N-1}$.

As g is monotonic decreasing, the value of g is bounded above. Since $a_1 < g$, the rightmost point a_1 is also bounded. This shows we cannot move the points to arbitrarily right. Thus, the answer is $k \geq \frac{1}{N-1}$.

4 Graph Theory

Many mathematical problems can be rephrased in terms of graph theory. It is sometimes easier for us to think in this direction. However, due to the many well-known facts discovered in this topic, the popularity of graph theory has declined in Olympiad. But it is still necessary for us to learn those terminologies and facts.

Firstly, we provide the usual terminologies in graph theory. We shall not use the formal definition.

A *graph*(圖) is usually denoted by $G = (V, E)$ where V is a set of *vertices*(頂點) and E is a set of *edges*(邊) consisting of pairs of vertices. We shall visualize a graph geometrically. Thus, vertices are denoted by points and edges are represented by segments joining two vertices. Since only the endpoints matter, we may use curved lines instead of segments.

In most cases, we shall only work on *simple graph*(簡單圖). This means every edge only connects two distinct vertices, and there is at most one edge between every pair of vertices. Sometimes we may give directions to the edges. Such a graph is called a *directed graph*(有向圖). In a simple directed graph, it is possible to have both the edge from A to B and the edge from B to A .

The *degree*(度) of a vertex A is the number of edges with A as one of the endpoints. For a simple graph with n vertices, the degree of each vertex can only be $0, 1, \dots, n-1$. For a directed graph, we may define in-degree (number of edges with A as the terminal vertex) and out-degree (number of edges with A as the initial vertex) analogously.

A *path*(路徑) of length s is a sequence of edges $e_1, e_2, \dots, e_s$ so that the terminal vertex of e_i coincides with the initial vertex of e_{i+1} for each $i = 1, 2, \dots, s-1$. We may also represent a path by the vertices it passes through in the obvious way. A *simple path*(簡單路徑) is a path for which no vertex is travelled more than once. Also, we define a *circuit* or *cycle*(圈) to be a path $e_1, e_2, \dots, e_s$ such that the terminal vertex of e_s also coincides with the initial vertex of e_1 . A *simple cycle*(簡單圈) is a cycle for which no vertex is travelled more than once except the initial and terminal vertex. For a directed graph, the same definitions apply but the direction of the edges needs to be considered as well.

There are a few special kinds of graph. A *complete graph*(完全圖) of n vertices is the graph where every pair of vertices is connected by an edge. We denote such a graph by K_n . A *bipartite graph*(二分圖) is a graph for which the vertices can be partitioned into two disjoint groups V_1, V_2 such that every edge must have one endpoint in V_1 and another endpoint in V_2 . A *connected graph*(連通圖) is a graph for which every pair of distinct vertices is connected by a simple path. For a directed graph, we say that it is strongly connected if the same holds when the direction is taken into account. Otherwise, it is only weakly connected if it is a connected graph when viewed as an undirected graph.

To begin, we go through a few well-known facts in graph theory.

Problem 4.1. (Handshaking theorem(握手定理)) Let $G = (V, E)$ be a simple graph. Let $d_1, d_2, \dots, d_n$ be the degrees of the n vertices respectively. Prove that

$$d_1 + d_2 + \dots + d_n = 2|E|.$$

Analysis. This is an immediate consequence of the definition.

Solution. For any edge in E , it contributes 2 to the sum of degrees as it has 2 endpoints. The result follows immediately.

Remark. This gives a simple relationship between the degrees and the number of edges.

Problem 4.2. Prove that a simple graph is bipartite iff it does not contain any cycle of odd length.

Analysis. Although this result is less trivial, the proof is simply based on the definitions.

Solution. Firstly, suppose G is a bipartite graph. Let V_1, V_2 be the partition of the vertices. Consider any cycle consisting of vertices $A_1, A_2, \dots, A_k$ in that order so that $A_i A_{i+1}$ is an edge of G for $i = 1, 2, \dots, k-1$, as well as $A_k A_1$. WLOG assume $A_1 \in V_1$. Since G is bipartite, we have $A_2 \in V_2$. Inductively, $A_j \in V_1$ iff j is odd, and $A_j \in V_2$ iff j is even. Note that the last vertex A_k should belong to V_2 , as $A_k A_1$ is an edge. Therefore, the length k of the cycle must be even.

Conversely, suppose a graph does not contain any odd cycle. WLOG assume the graph is connected. Otherwise, we can simply consider each connected piece of the graph to work out the partition. Firstly, we put a vertex A in V_1 . For any other vertex B , we consider any path from A to B . We put B in V_1 if the path has an even length, and put it in V_2 if not. This is well-defined. Indeed, if there exist paths $A, C_1, C_2, \dots, C_k, B$ and $A, D_1, D_2, \dots, D_\ell, B$ such that k and ℓ have different parities, then the cycle $A, C_1, C_2, \dots, C_k, B, D_\ell, D_{\ell-1}, \dots, D_1$ has an odd length. This contradicts the assumption.

We have succeeded in partitioning the set of vertices into V_1, V_2 . Suppose there exists an edge connecting some vertices C, D in the same set. Consider a path from A to C , followed by the edge from C to D , and followed by another path from D to A . Since the two paths have the same parity, this gives a cycle of odd length, which is a contradiction. Therefore, V_1, V_2 satisfy the bipartite condition.

Problem 4.3. (Turán's theorem(圖蘭定理)) Let $n \geq r$ be positive integers. Suppose $G = (V, E)$ is a simple graph which does not contain any K_{r+1} subgraph. Prove that $|E|$ is maximized when the vertices in V can be partitioned into r sets $V_1, V_2, \dots, V_r$ for which the following holds:

- for each pair i, j , $|V_i|$ and $|V_j|$ differ by at most 1; and
- two vertices are joined by an edge iff they belong to different sets.

Analysis. We consider any such graph with the greatest number of edges. The key step is to show that the vertices can be partitioned into r sets so that two vertices in the same set are not connected. It is easier to think if we consider the complement of the graph (meaning that all edges are removed and we draw an edge between two vertices iff they were not connected).

Solution. Let $G = (V, E)$ be any simple graph without K_{r+1} and with the maximum $|E|$. Consider the complement $\overline{G} = (V, \overline{E})$ which is the graph with the same vertex set and with the edge set $\overline{E}$ being the complement of E in $K_{|V|}$. Hence, $\overline{G}$ does not contain $r+1$ vertices which are pairwise not joined by an edge, and $|\overline{E}|$ is minimized. We claim that $\overline{G}$ is a union of at most r complete graphs. This will be done via induction on r . We shall call a set of vertices discrete if the vertices in this set are pairwise not joined by an edge.

The base case $r = 1$ is trivial since the graph must be a complete graph. For $r > 1$, if $\overline{G}$ is not connected, then it can be partitioned into $G_1, \dots, G_k$ each of which is a connected graph. For each G_i , let V_i be a discrete set with the greatest number of vertices in G_i . Then the union of the V_i 's is also discrete with size $|V_1| + \dots + |V_k|$. Conversely, any discrete set in $\overline{G}$ is a union of discrete sets in $G_1, \dots, G_k$. Hence, it suffices to investigate each connected component. The component G_i has a discrete set of size at most $|V_i|$. Since $k \geq 2$, $|V_i|$ is smaller than r . Applying the inductive hypothesis, G_i must be a union of complete graphs. Since it is connected, G_i is a complete graph. This shows $\overline{G}$ is a union of complete graphs as claimed.

If $\overline{G}$ is connected, we pick a vertex A with the smallest degree. Let $B_1, B_2, \dots, B_m$ be the neighbours of A . We first claim that $A, B_1, \dots, B_m$ is a complete subgraph. Thus, we may assume $m \geq 2$ and B_1, B_2 are not joined by an edge. We remove all the edges joined to either B_1 or B_2 , and then we join all the edges $AB_1, AB_2, B_1B_2, B_1B_i, B_2B_i$ for $3 \leq i \leq m$. In other words, A, B_1, B_2 are pairwise joined, and they have exactly the same neighbours other than themselves. In the new graph G' formed, it does not contain any discrete set of size $r+1$. Indeed, if such a set S exists, it should not exist in $\overline{G}$. Therefore, S must contain B_1 or B_2 . Note that S contains at most one of A, B_1, B_2 (otherwise it is not discrete). WLOG assume $B_1 \in S$. If we replace B_1 by A , the set is still discrete as B_1 and A have the same neighbours. However, such a set exists in $\overline{G}$ as it does not involve B_1 or B_2 . This is a contradiction.

Therefore, the new graph does not contain any discrete set of size $r+1$. But we shall show that the number of edges is decreased. Indeed, let x_1, x_2 be the number of edges B_1B_i, B_2B_i (note that we must have $i \geq 3$) in the original graph $\overline{G}$, and let y_1, y_2 be the number of edges B_1C_i, B_2D_j for some other vertices C_i, D_j (which could be A) other than the B_i 's. Then we have removed $x_1 + x_2 + y_1 + y_2$ edges and added $2m - 1$ edges. Recall that A has the smallest degree. Therefore,

$$(x_1 + y_1) + (x_2 + y_2) \geq m + m > 2m - 1.$$

This shows there is a graph with a smaller number of edges which does not contain any discrete set of size $r + 1$. This contradicts the choice of G and $\overline{G}$. Therefore, any two of $A, B_1, \dots, B_m$ are joined by an edge. If this is not the whole graph $\overline{G}$, then WLOG B_1 is joined to another point C not shown due to the connectedness. We can simply remove the edge B_1C without affecting the key property. This is due to the same reason as before. Indeed, if a discrete set contains B_1, C and some other vertices, then the set cannot contain $A, B_2, \dots, B_m$, and the same holds if we replace B_1 by A . Thus, the size of this discrete set cannot exceed r . In any case, we have derived a contradiction.

Now, we know that $\overline{G}$ is a union of complete graphs of sizes $t_1, t_2, \dots, t_\ell$ respectively. We claim that any two of the t_i 's differ by at most 1. Suppose on the contrary that $t_1 - t_2 \geq 2$. Note that

$$(t_1, t_2) > \left(\left\lceil \frac{t_1 + t_2}{2} \right\rceil, \left\lfloor \frac{t_1 + t_2}{2} \right\rfloor \right).$$

A complete graph of size w has $\binom{w}{2}$ edges. By the **majorization inequality**, we have

$$\binom{t_1}{2} + \binom{t_2}{2} \geq \binom{\lceil \frac{t_1+t_2}{2} \rceil}{2} + \binom{\lfloor \frac{t_1+t_2}{2} \rfloor}{2}.$$

Equality holds when $t_1 = \left\lceil \frac{t_1 + t_2}{2} \right\rceil$, which is not the case. This contradicts the assumption that the number of edges is minimized.

Therefore, the complement of $\overline{G}$, which is G , has exactly the form as described in this problem.

Remark. The graph in this problem is called the *Turán graph*(圖蘭圖).

Problem 4.4. (Hall's theorem) Let $G = (V, E)$ be a bipartite graph with the vertices partitioned into V_1 and V_2 . For any subset W of V_1 , let d_W be the number of vertices in V_2 which are neighbours of at least one vertex in W . Then the following are equivalent.

- Let $V_1 = \{A_1, A_2, \dots, A_m\}$. There exists a set of vertices $\{B_1, B_2, \dots, B_m\}$ in V_2 such that $A_i B_i$ is an edge for each $i = 1, 2, \dots, m$.
- For each subset W of V_1 , we have $d_W \geq |W|$.

Analysis. The second statement is clearly a necessary condition for the first statement to hold. This theorem mainly tells us that it is also a sufficient condition. Here is an example to illustrate our goal. If $V_1 = \{A_1, A_2\}$, $V_2 = \{B_1, B_2, B_3\}$ and the only edges are $A_1 B_2, A_2 B_2, A_2 B_3$, then we can pair up the vertices like (A_1, B_2) and (A_2, B_3) so that each A_i belongs to a pair and the partners of the A_i 's are distinct.

The idea of the proof is simply induction and algorithm.

Solution. Firstly, suppose the first statement holds. Suppose each vertex A_i in V_1 is joined to a vertex B_i in V_2 . For any subset W of V_1 , if A_i belongs to W , then B_i contributes 1 to d_W by definition. Therefore, d_W is at least the number of B_i 's, which is the number of A_i 's, i.e. $|W|$.

Conversely, suppose the second statement holds. We show that such a pairing exists by induction on m , the number of vertices in V_1 . For the base case $m = 1$, we simply take $W = \{A_1\}$ so that A_1 must be joined to a vertex B_1 in V_2 . Now, assume $m \geq 2$.

Let $V_1 = \{A_1, A_2, \dots, A_m\}$. Consider the subset $\{A_1, A_2, \dots, A_{m-1}\}$. By the inductive hypothesis, we can find $B_1, B_2, \dots, B_{m-1} \in V_2$ such that $A_i B_i$ is joined by an edge for $i = 1, 2, \dots, m-1$. It remains to find a partner for A_m following the steps given below.

- If A_m is joined to some other vertex $B_m \in V_2$, then we are done. Otherwise, WLOG we may assume A_m is only joined to $B_1, \dots, B_{i_1}$ for some $1 \leq i_1 \leq m-1$ (again i_1 must be at least 1).
- Consider $W = \{A_1, \dots, A_{i_1}, A_m\}$. Since $|W| = i_1 + 1$, from the assumption, the number of vertices joined to at least one of the vertices in W is at least $i_1 + 1$. So there should be some vertices other than $B_1, \dots, B_{i_1}$. If there exists such a vertex $B_m \in V_2$ not involved yet, say $A_{i_1} B_m$ is an edge, then we simply change the pair (A_{i_1}, B_{i_1}) to (A_{i_1}, B_m) and (A_m, B_{i_1}) . This gives a desired pairing. If not, then WLOG the only vertices joined to at least one of $A_1, \dots, A_{i_1}, A_m$ are $B_1, \dots, B_{i_2}$ for some $i_1 < i_2 \leq m-1$.
- Consider $W = \{A_1, \dots, A_{i_2}, A_m\}$. Arguing in the same way, there are two possibilities. One possibility is that there exists a vertex $B_m \in V_2$ such that $A_{i_2} B_m$ is an edge, and WLOG $A_{i_1} B_{i_2}$ is an edge. Then we change the pairs (A_{i_1}, B_{i_1}) and (A_{i_2}, B_{i_2}) to (A_{i_1}, B_{i_2}) , (A_{i_2}, B_m) and (A_m, B_{i_1}) . The other possibility is that the only vertices joined to at least one of $A_1, \dots, A_{i_2}, A_m$ are $B_1, \dots, B_{i_3}$ for some $i_2 < i_3 \leq m-1$.

Following the process repeatedly, since the indices $i_1, i_2, \dots$ are strictly increasing, it must come to an end. This means we must be able to find a vertex B_m not involved in previous steps such that we can change the pairs to form a new pairing including A_m . This completes the proof by induction.

Problem 4.5. (APMO 2003 Problem 5) Given two positive integers m and n , find the smallest positive integer k such that among any k people, either there are $2m$ of them who form m pairs of mutually acquainted people or there are $2n$ of them forming n pairs of mutually unacquainted people.

Analysis. WLOG assume $m \geq n$. We first start with some small cases. When $n = 1$, the condition is satisfied whenever there are 2 unacquainted people. So we may assume every pair of people are acquainted. In that case, we clearly have $k = 2m$.

When $n = 2$, in order to avoid having 2 pairs of mutually unacquainted people, it is not hard to see the optimal case is that there exists a person who is unacquainted to any other people, and these are the only pairs of unacquainted people. In that case, we should have $k = 2m + 1$. We can indeed confirm this by some exhaustive arguments.

From these small cases, we may observe that k is increased by 1 when n is increased by 1. This allows us to guess the answer and we can try to prove this using induction. Since there are two parameters, choosing the appropriate quantity to carry out induction is important so as to simplify the arguments.

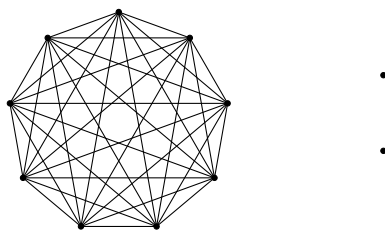
Solution. WLOG assume $m \geq n$. We claim that the smallest k is $2m + n - 1$. In the following, we identify the people as vertices of a complete graph. Two vertices are joined by a red edge if the two people they represent are acquainted, and they are joined by a blue edge otherwise.

Firstly, we claim that $k \geq 2m + n - 1$. It suffices to give a counterexample when there are exactly $2m + n - 2$ vertices. Inspired by the construction in the special case $n = 2$, we give the following construction. Let the vertices be $V_1, V_2, \dots, V_{2m+n-2}$. We colour the edges between any two of $V_1, V_2, \dots, V_{2m-1}$ red, and all other edges blue (the figure below corresponds to the case $m = 5$ and $n = 3$, where the edges not shown are blue). Since only the $2m - 1$ vertices $V_1, V_2, \dots, V_{2m-1}$ are joined by some red edges, it is clear that we cannot find m disjoint pairs of red edges.

Suppose there are n disjoint pairs of blue edges. As there are

$$(2m + n - 2) - (2m - 1) = n - 1$$

vertices not belonging to the red K_{2m-1} , the n pairs of blue edges must be formed by some vertices including at least $2n - (n - 1) = n + 1$ of the vertices $V_1, V_2, \dots, V_{2m-1}$. By the pigeonhole principle, 2 of them must form a blue edge, which is a contradiction. So this is indeed a counterexample.



Secondly, we prove that the condition is satisfied when $k = 2m + n - 1$. When $n = 1$, it is obvious that any $k = 2m$ vertices must contain a red K_{2m} or a blue K_2 , since a blue K_2 merely means a blue edge. We can easily extract m pairs of red edges from the former case if there is no blue edge. This serves as the base case of the induction on n for a fixed difference $m - n$.

Assume the claim holds for some pair (m, n) . For the inductive step, we work on the case $(m + 1, n + 1)$. In other words, we need to show that for any colouring of

$2m + n + 2$ vertices, there must be $m + 1$ disjoint pairs of red edges or $n + 1$ disjoint pairs of blue edges. If there exist 3 vertices A, B, C such that AB is red and AC is blue, we remove these 3 points and apply the inductive hypothesis to the remaining $2m + n - 1$ vertices. This either gives m disjoint pairs of red edges or n disjoint pairs of blue edges not involving A, B, C . By including the edge AB or the edge AC , we easily get what we want.

So it suffices to consider the case when any 3 vertices form a monochromatic triangle. If any 2 edges among A, B, C are red, then by considering A, B, D for any other vertex D , the edges AD and BD must be red. By considering A, D, E for any other vertex E , the edge DE must be red. Therefore, all edges in the graph are red. Similarly, if any 2 edges among A, B, C are blue, then all edges in the graph are blue. Both cases are trivial as $2m + n - 1 \geq 2m \geq 2n$. The inductive step is complete.

By induction, the condition is satisfied when $k = 2m + n - 1$. So the minimum k is $\max\{m, n\} + m + n - 1$ in general.

Problem 4.6. In a tournament, each player has played against every other player once and there is a unique winner in each match. We say that player A beats player B if there exist distinct players $A = C_0, C_1, \dots, C_n = B$ with $n \geq 1$ such that C_i wins over C_{i+1} for $i = 0, 1, \dots, n - 1$. Suppose for any two players A and B , we have A beats B and B beats A . Prove that all the players can be arranged in a circle such that each of them wins over the player to his left.

Analysis. We can use vertices to denote players and directed edges to denote the results of matches. In terms of graph theory, it means that if we assign direction to the edges of a complete graph such that the directed graph is strongly connected, then it must contain a cycle which passes through every vertex once.

Using induction on the number of players is not that simple because the condition is completely different when more players are added. This is because if there exists a path from A to B , it does not guarantee that such a path still exists when one of the vertices of the graph is removed. Thus, a better way is to use extremal principle to deal with this problem.

Solution. We shall use the formulation of graph theory. We first show that there exists a simple cycle. Take arbitrary vertices A and B . From the condition, there exists a path from A to B and a path from B to A . They combine to give a cycle. If the same vertex appears twice apart from the initial and the terminal vertex, then we simply remove all vertices in between to shorten the cycle. At the end, we obtain a simple cycle.

We pick a simple cycle with the greatest number of vertices. Suppose the cycle is $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_k \rightarrow A_1$. If this cycle contains all vertices, then we are done. Consider any vertex B not belonging to this cycle. Note that we cannot have the two edges $A_i \rightarrow B$ and $B \rightarrow A_{i+1}$ (with $A_{k+1} = A_1$ as usual), since otherwise we can

form a larger cycle

$$A_1 \rightarrow \cdots \rightarrow A_i \rightarrow B \rightarrow A_{i+1} \rightarrow \cdots \rightarrow A_k \rightarrow A_1.$$

Hence, if $A_i \rightarrow B$ for some i , then inductively we show that $A_j \rightarrow B$ for all j . Similarly, if $B \rightarrow A_i$ for some i , then $B \rightarrow A_j$ for all j .

Since there exists a path from any vertex not in the cycle to A_1 , there must be a vertex B such that $B \rightarrow A_i$ for all i . Next, we take a simple path from A_1 to B , say $A_1 \rightarrow C_1 \rightarrow \cdots \rightarrow C_m \rightarrow B$. We may assume $C_1, \dots, C_m$ are different from the A_i 's as we can simply remove all vertices in between two A_i 's (this amounts to replace A_1 by another vertex A_i , which does not affect the argument). But then we have found a larger cycle

$$A_1 \rightarrow C_1 \rightarrow \cdots \rightarrow C_m \rightarrow B \rightarrow A_2 \rightarrow \cdots \rightarrow A_k \rightarrow A_1.$$

This is a contradiction. Therefore, this cycle must contain all vertices.

Problem 4.7. (CMO 1987 Problem 3) In a sport tournament, every two players play against once. There is a winner for each match. After the tournament, we select some outstanding players. A player A is said to be outstanding if for any other player B , either A wins over B or there exists a player C such that C wins over B and A wins over C .

Suppose there is only one outstanding player according to the above rule. Prove that this player wins over all other players.

Analysis. Since the outstanding player should win over all other players, it is natural to consider the player who wins the greatest number of matches. We shall show that this player is the outstanding player, and he cannot lose to other players.

Solution. Let A be a player who wins the greatest number of matches. Suppose A wins over $B_1, B_2, \dots, B_k$. For any other player C (if exists), since C wins at most k matches and C wins over A , player C cannot win over all of $B_1, B_2, \dots, B_k$. WLOG assume B_k wins over C . Then we have the chain $A \rightarrow B_k \rightarrow C$. As this holds for any player C , this shows A is outstanding by definition. Thus, A is the only outstanding player.

Suppose on the contrary that A does not win every match. We remove $A, B_1, \dots, B_k$ and consider the remaining subgraph. Using exactly the same argument as above, we can find a player C which is outstanding in this subgraph. Also, C must win over A , and hence we have the chains $C \rightarrow A \rightarrow B_i$. This shows C is outstanding by definition. This contradicts the given fact that there is only one outstanding player. Therefore, A should win over all other players.

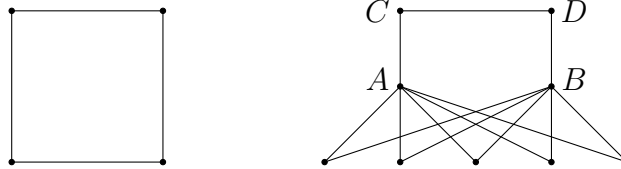
Remark. In graph theory, it is always useful to consider a vertex with the smallest or greatest degree.

Problem 4.8. There are $n \geq 4$ railway stations in a city. For any two different stations, either there is a rail joining them directly or not. Suppose for any pair of stations, they are either joined directly by a rail or there exists another station to which both of them are joined. Also, there is no central station in the city, i.e. a station which is joined to all other stations directly. Find the minimum number of rails that are running in the city.

Analysis. Since the distance between any two vertices is either 1 or 2, there should be some vertices with large degrees. The condition prohibits the existence of a vertex with degree $n - 1$. So it is natural to consider some cases when there exist some vertices with degrees $n - 2$ or $n - 3$. By working out a few possibilities and comparing these, it is not hard to find an optimal case to run the smallest number of rails.

To establish the lower bound, we may choose a vertex with the smallest degree, classify the vertices into some groups and apply the condition to show that the number of edges cannot be too small. The **majorization inequality** is another useful tool to solve this kind of graph theory problem.

Solution. We use terminologies in graph theory. For $n = 4$, a 4-cycle satisfies the conditions. If there are only 3 edges in total, then we may assume the degree of one of the vertices A is 1 by the pigeonhole principle (note that the degree cannot be 0). Let B be the vertex adjacent to A . Then the degree of B has to be 3 in order that the distance between A and any other vertex is at most 2. This contradicts the given condition. So the minimum number of rails needed is 4 when $n = 4$.



For $n \geq 5$, we claim that the minimum value is $2n - 5$. Consider the following case. Suppose C is adjacent to A and D , while B is adjacent to D . Each of the other $n - 4$ vertices is adjacent to both A and B . There are

$$3 + 2(n - 4) = 2n - 5$$

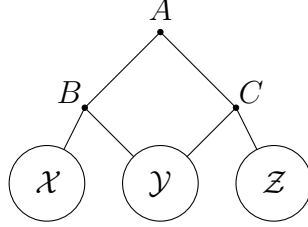
edges in total. One can check that all conditions are satisfied. So $2n - 5$ edges suffice.

Let A be a vertex with the smallest degree. Clearly, its degree is positive. If the degree of A is 1, say B is adjacent to A , then B should be adjacent to all other vertices in order that the distance between A and any other vertex is at most 2. This is a contradiction.

If the degree of A is 2, say B and C are adjacent to A , we define the following.

- Group $\mathcal{X}$: the group of remaining vertices adjacent to B but not C .

- Group $\mathcal{Y}$: the group of remaining vertices adjacent to both B and C .
- Group $\mathcal{Z}$: the group of remaining vertices adjacent to C but not B .



We first consider the general case where $\mathcal{X}$ and $\mathcal{Z}$ are nonempty. For any $X \in \mathcal{X}$ and $Z \in \mathcal{Z}$, either XZ is an edge or there exists W distinct from A, B, C which is adjacent to both X and Z . Let $\mathcal{S}$ be the set of all vertices in $\mathcal{X}, \mathcal{Z}$ and all W involved (some W 's may belong to $\mathcal{Y}$). Note that $\mathcal{S}$ is a connected subgraph since there exists a path from each $X \in \mathcal{X}$ to each $Z \in \mathcal{Z}$, and each W is adjacent to some $X \in \mathcal{X}$. We make use of the following well-known fact in graph theory.

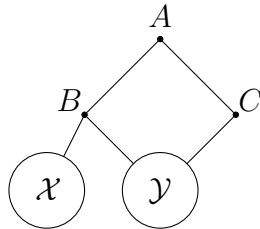
Claim. Any connected graph of $k \geq 2$ vertices consists of at least $k - 1$ edges.

Proof. We prove this by induction on k . The base case $k = 2$ is trivial. Assume this holds for the case $k - 1$. For the case k , suppose on the contrary that there are at most $k - 2$ edges. Then the sum of degrees is at most $2(k - 2)$. By the pigeonhole principle, there is a vertex V with degree $\leq \frac{2(k - 2)}{k} < 2$. As the graph is connected, its degree must be 1. Then we remove V together with the edge with V as an endpoint. The remaining graph is a connected graph of $k - 1$ vertices but it has at most $k - 3$ edges. This is a contradiction. So the claim holds by induction. $\square$

From this claim, we find that the number of edges in $\mathcal{S}$ is at least $|\mathcal{S}| - 1 \geq |\mathcal{X}| + |\mathcal{Z}| - 1$. Together with the $|\mathcal{X}| + |\mathcal{Y}| + 1$ edges with B as an endpoint and $|\mathcal{Y}| + |\mathcal{Z}| + 1$ edges with C as an endpoint, we find that

$$\begin{aligned} |E| &\geq (|\mathcal{X}| + |\mathcal{Z}| - 1) + (|\mathcal{X}| + |\mathcal{Y}| + 1) + (|\mathcal{Y}| + |\mathcal{Z}| + 1) \\ &= 2(|\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}|) + 1 = 2(n - 3) + 1 = 2n - 5. \end{aligned}$$

Hence, we have established the lower bound if $\mathcal{X}$ and $\mathcal{Z}$ are nonempty.



Next, WLOG assume $\mathcal{Z}$ is empty. Then B and C are not adjacent or otherwise B is adjacent to all vertices. For any $X \in \mathcal{X}$, there exists $Y \in \mathcal{Y}$ which is adjacent to both X and C . This means there is an edge joining X and a vertex in $\mathcal{Y}$ for each $X \in \mathcal{X}$. Thus, there are at least $|\mathcal{X}|$ such edges in total. Therefore,

$$|E| \geq |\mathcal{X}| + (|\mathcal{X}| + |\mathcal{Y}| + 1) + (|\mathcal{Y}| + 1) = 2(|\mathcal{X}| + |\mathcal{Y}|) + 2 = 2(n - 3) + 2 = 2n - 4.$$

We may now assume the degree of each vertex is at least 3. We count the number of pairs of vertices. On the one hand, there are exactly $\binom{n}{2}$ pairs. On the other hand, each pair of vertices A, B either corresponds to an edge AB or to a pair of edges AC, CB for some vertex C . For each vertex C with degree d , there are $\binom{d}{2}$ pairs of edges with C as a common endpoint. This yields

$$\binom{n}{2} \leq |E| + \sum_{j=1}^n \binom{d_j}{2}$$

where $d_1 \geq d_2 \geq \dots \geq d_n$ are the degrees of the vertices. (Note that this is an inequality instead of an identity since some pairs of vertices A, B may correspond to more than one pair of edges.) As each d_j is at least 3 and the sum of degrees is $2|E|$, the sequence $(d_1, d_2, \dots, d_n)$ is majorized by $(2|E| - 3(n - 1), 3, 3, \dots, 3)$. By the majorization inequality, we have

$$\sum_{j=1}^n \binom{d_j}{2} \leq \binom{2|E| - 3(n - 1)}{2} + (n - 1) \binom{3}{2} = \frac{(2|E| - 3n + 3)(2|E| - 3n + 2)}{2} + 3(n - 1).$$

It follows that

$$n(n - 1) \leq 2|E| + (2|E| - 3n + 3)(2|E| - 3n + 2) + 6(n - 1).$$

Suppose on the contrary that $|E| \leq 2n - 6$. Then we get

$$n(n - 1) \leq 2(2n - 6) + (n - 9)(n - 10) + 6(n - 1).$$

This implies $8n \leq 72$, and hence $n \leq 9$. However, we also have

$$2n - 6 \geq |E| = \frac{1}{2} \sum_{j=1}^n d_j \geq \frac{3n}{2}$$

since the degree of each vertex is at least 3. This yields $n \geq 12$, which contradicts $n \leq 9$.

In any case, the number of edges is at least $2n - 5$ when $n \geq 5$. So this is the minimum number of rails needed.

Problem 4.9. Two players X and Y play the chasing game as follows: X draws a simple connected graph with n vertices (with $n \geq 3$) and places two predators and one prey on different vertices. The game then begins. For each round, Y can control the two predators and let one or each of them move to an adjacent vertex. Then X can allow the prey to stay or move to an adjacent vertex. The game ends when one of the predators catch the prey (by going to the same vertex as the prey) and Y will win the game. Find the smallest n for which X has a strategy to prevent Y from winning the game.

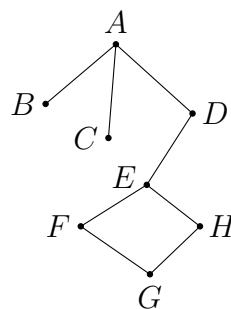
Analysis. Suppose the prey stays at a vertex A . Since the two predators can block two adjacent vertices of A , it is better to have the degree of A being 3. It will be good if the degree of every vertex is 3. Also, the neighbours of A should not have any other common neighbour. Otherwise, a predator can be placed at that common neighbour to guard the neighbours of A . In view of this, the number of vertices is at least $1 + 3 + 2 \times 3 = 10$. This suggests a possible way to draw the graph.

Solution. We claim that the smallest n is 10.

We first prove that $n \geq 10$. If each vertex has degree at most 2, then the graph can only be a single path or a cycle. In both cases, Y can win by fixing one predator and moving the other predator towards the prey.

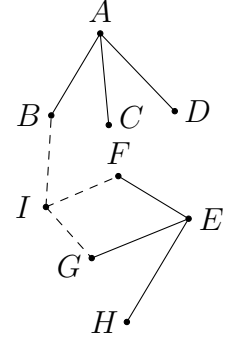
Suppose Y fixes one predator at a vertex with the largest degree (which is ≥ 3 from above). Then the prey cannot go through this vertex and all its adjacent vertices. So we may remove these vertices from the graph and the prey can only stay in one of the connected components in the remaining graph. If the connected component that the prey stays has at most 3 vertices, then one of these vertices must be joined to all other vertices. Y can then let the other predator move to this vertex so that the prey has nowhere to go. Therefore, we may assume one of the connected components has at least 4 vertices. Note that this implies $n \geq 8$.

If $n = 8$, we may further assume each vertex has degree at most 3, or otherwise Y can use the above strategy. Pick a vertex A with degree 3 which is connected to B, C, D . The remaining vertices E, F, G, H should form a connected graph after removing all edges joining to A, B, C, D . Also, none of E, F, G, H can be adjacent to all other three vertices, or otherwise Y can still use the same strategy. Then there are only two possible cases, either E, F, G, H form a path, in which case a predator can stay at A and the other predator can catch the prey by going towards it along the path, or E, F, G, H form a cycle as shown on the right.



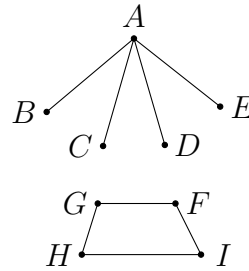
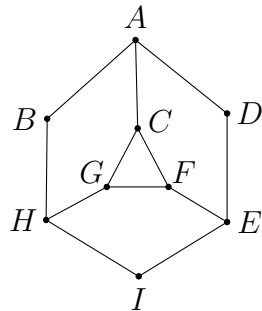
Since the graph is connected, WLOG we may assume E is adjacent to D . Note that the degree of E is at most 3 so it cannot be connected to other vertices. Now, the two predators can stay at A and G respectively so that the prey must stay at E . Then the predator at A can move to D and the prey has nowhere to go. Therefore, player Y has a strategy when $n = 8$.

If $n = 9$, we again pick A to be a vertex with the largest degree. Again, we may assume there exists a connected component with at least 4 vertices after removing A together with its neighbours. If A has degree 3, suppose A is adjacent to B, C, D . For the other vertices E, F, G, H, I , if I is not connected to the graph formed by E, F, G, H , then we can use the same argument in the case $n = 8$ to deduce a strategy for Y . Hence, we may assume E, F, G, H, I are connected. In the subgraph formed by E, F, G, H, I , suppose one of the vertices has degree 3, say, E is adjacent to F, G, H . When the predators stay at A and E , the prey can only stay at I .



If I is connected to only one of F, G, H , say, IF is an edge, then Y can win by moving the predator at E to F . Similarly, if I is connected to at most one of B, C, D , then Y can also win by moving the predator at A to this vertex (or to a neighbour of I if I is not adjacent to B, C, D). Note that one of these two cases must occur, or otherwise I has total degree at least 4, which is larger than that of A .

Next, consider the case when each vertex of the subgraph composed of E, F, G, H, I has degree at most 2. Since it is connected, either it is a path or a cycle. For the former case, Y may let one predator stay at A and let the other one chase the prey along the path. For the latter case, suppose $EFGHI$ is a cycle as shown. Since the whole graph is connected, WLOG assume D, E are adjacent. Then E has degree 3. By the same argument, if we remove E together with its neighbours instead of A , then the remaining vertices A, B, C, G, H should also form a cycle. WLOG, we may assume BH and CG are edges. Similarly, if we take away H with its neighbours B, G, I , the other 5 vertices form a cycle. So CF is an edge. Now, suppose the predators stay at A and I . Then the prey can only stay at F or G . Note that F and G has degree 3 so there cannot be any other edge with one of them as endpoint. Then Y will win by moving the predator at A to C .

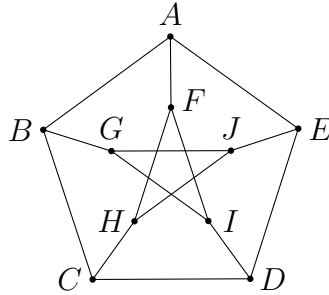


The only case left for $n = 9$ is that A has degree 4. Suppose A is adjacent to B, C, D, E . As in the case $n = 8$, the remaining vertices F, G, H, I should form a cycle $FGHI$. Also, in order that the strategy in the case $n = 8$ cannot be applied, each of F, G, H, I should connect to at least 2 (and hence 2 by counting degrees) of B, C, D, E . There are a few possible cases. If a pair of opposite vertices in the

cycle share a common vertex among B, C, D, E (say, BG, BI, CG are edges), then the predators can stay at A and I . G is the only vertex available for the prey. But it has nowhere to go when the predator at A moves to C . If a pair of adjacent vertices in the cycle share the same neighbours among B, C, D, E (say, CF, DF, CG, DG are edges, so that H is connected to B, E), then the predators can force the prey to go to G by staying at A and I . Afterwards, the predator at A can go to B , while the one at I can go to F simultaneously. Y can win.

If all these cases do not happen, then we may assume the edges among these vertices are $BG, CG, CF, DF, DI, EI, EH, BH$. When the predators stay at A and I , the prey will stay at G . In the next step, they can move to B and F respectively so that the prey will be caught. Hence, Y has a strategy in all cases.

Lastly, we claim that X can prevent Y from winning for $n = 10$. Suppose X draws the graph as shown below. All vertices are symmetric to each other so that we may assume the prey stays at A . To catch the prey, the predators need to guard the vertices B, E, F simultaneously. For B , one predator should stay at one of B, C, G . For E , one predator should stay at one of D, E, J . For F , one predator should stay at one of F, H, I . Hence, it is obvious that the two predators cannot guard all 3 vertices at the same time. Hence, the prey can always escape.

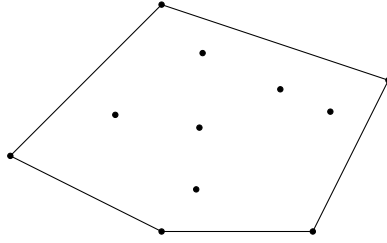


Remark. The graph in the case $n = 10$ is called the *Petersen graph*(佩特森圖). It often appears as an example in graph theory since it has several special properties.

5 Combinatorial Geometry

Combinatorial geometry mainly refers to combinatorial problems with geometric settings. Those problems do not require deep knowledge in geometry, but rather emphasize more on combinatorial thinking. Many techniques introduced, such as induction, double counting and colouring, are useful in this kind of problem.

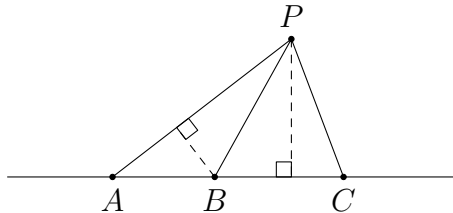
There is one major fact that is frequently used for point set problems. If there is a finite set S containing points on the plane which do not lie on the same line, then there exists a unique convex polygon containing all these points with the smallest area. Such a polygon is called the *convex hull* (凸包) of the point set S . Clearly, the vertices of the convex hull must belong to S .



Problem 5.1. (Sylvester-Gallai theorem(西爾維斯特-加萊定理)) Suppose there are $n \geq 3$ points on the plane which do not lie on the same straight line. Prove that there exists a line passing through exactly two of these points.

Analysis. This solution is a standard proof of this theorem. The key idea is to consider the distances from the points to the lines formed by joining some of the points. We choose a line ℓ and a point P not on ℓ such that the distance from P to ℓ is the smallest. If ℓ contains three distinct points A, B, C , then we derive a contradiction by finding a further smaller distance.

Solution. Consider any pair (ℓ, P) where ℓ is a line formed by two of the points and P is a point not lying on ℓ . Such a pair exists from the condition. Among all these pairs, we choose the one such that the distance from P to ℓ is the smallest. We claim that ℓ contains exactly two points.



Suppose on the contrary that there are three distinct points A, B, C on ℓ in that order. As $\angle PBA + \angle PBC = 180^\circ$, we may assume $\angle PBA \geq 90^\circ$. Consider $\triangle PAB$.

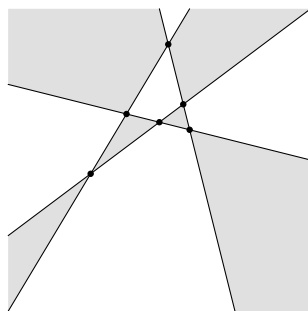
As $\angle PBA$ is the largest interior angle, PA is the longest side of this triangle. By considering the area of $\triangle PAB$, we find that the distance from B to PA is smaller than the distance from P to AB . This contradicts the minimal choice of the pair (ℓ, P) . Therefore, ℓ only contains two points as desired.

Remark. In combinatorial geometry, there are some geometric facts which often appear. For example, in this solution we have used the fact that larger interior angles of a triangle are opposite to longer sides. Another commonly used fact is the triangle inequality.

Problem 5.2. (EGMO 2017 Problem 3) There are 2017 lines in the plane such that no three of them go through the same point. Turbo the snail sits on a point on exactly one of the lines and starts sliding along the lines in the following fashion: she moves on a given line until she reaches an intersection of two lines. At the intersection, she follows her journey on the other line turning left or right, alternating her choice at each intersection point she reaches. She can only change direction at an intersection point. Can there exist a line segment through which she passes in both directions during her journey?

Analysis. With some trial and errors, one should know that the answer is negative. Since the directions chosen at each intersection point alternate, the parity should be the key thing to consider. In combinatorial geometry, one usual way to consider the parity is to use colouring.

Solution. The 2017 lines divide the plane into a finite number of regions, some of which are finite and some are not. We claim that we can colour these regions in black and white such that every two adjacent regions (which share a common edge) have distinct colours.



This claim can be proved by induction on the number of lines. When there is one line, it divides the plane into two halves. So we can colour one half black and the other half white. Suppose a colouring for n lines exists. We add one more line to the plane. This line divides the plane into two halves. We keep the colours on one side of the line and change the colours on the other side of the line. The adjacent regions which share a common edge formed by this line have distinct colours. Also, any two

regions which have been adjacent before this line is added remain to have distinct colours. By induction, a colouring exists for the case $n = 2017$.

Initially, we call the region to the right of the snail region I and the region to the left of the snail region II. WLOG, assume region I is black. When the snail reaches an intersection point, there are two possibilities. If the snail turns to the right, then the region to the right of her is still region I, which is black in colour. If the snail turns to the left, we call the region to the right of her region III. Since region III and region II are adjacent, region II and region I are adjacent, we know that region III has the same colour as region I, which is black. The same argument applies whenever the snail reaches an intersection point. Therefore, no matter how far she goes, the region to her right must be black in colour. So it is obvious that the direction travelled by the snail on every line segment is uniquely determined. The answer is negative.

Problem 5.3. (APMO 2011 Problem 2) Five points A_1, A_2, A_3, A_4, A_5 lie on a plane in such a way that no three among them lie on a same straight line. Determine the maximum possible value that the minimum value for the angles $\angle A_i A_j A_k$ can take where i, j, k are distinct integers between 1 and 5.

Analysis. This time we shall see the use of convex hull. We can pick the smallest angle formed by some consecutive vertices of the convex hull. Then all points lie inside this angle and we are able to find an angle which is small enough. The optimal case is probably the case of a regular pentagon.

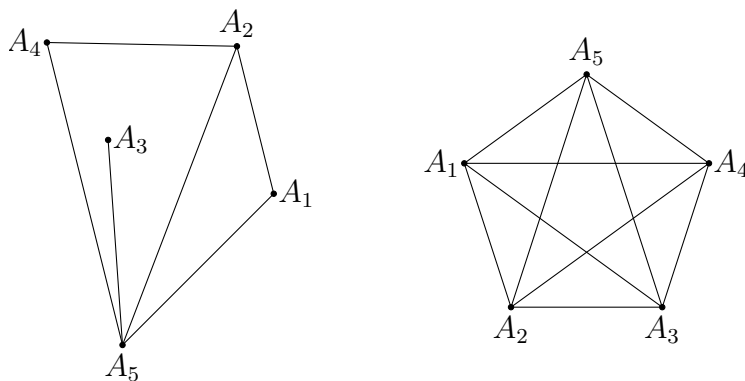
Solution. WLOG assume $\angle A_1 A_5 A_4$ is one of the smallest interior angle of the convex hull formed by the 5 points. Then A_2 and A_3 lie inside this angle. We may assume the lines $A_5 A_1, A_5 A_2, A_5 A_3, A_5 A_4$ lie in that order. By the pigeonhole principle, one of

$$\angle A_1 A_5 A_2, \angle A_2 A_5 A_3, \angle A_3 A_5 A_4$$

is at most $\frac{1}{3} \angle A_1 A_5 A_4$. If the convex hull has n sides, where $n = 3, 4, 5$, then

$$\angle A_1 A_5 A_4 \leq \frac{(n-2) \times 180^\circ}{n} \leq \frac{(5-2) \times 180^\circ}{5} = 108^\circ.$$

Thus, there exists $\angle A_i A_j A_k$ which is at most $\frac{1}{3} \times 108^\circ = 36^\circ$.



On the other hand, if $A_1A_2A_3A_4A_5$ is a regular pentagon, then each angle is equal to one of

$$\angle A_1A_5A_2, \angle A_2A_5A_3, \angle A_1A_5A_3, \angle A_1A_5A_4$$

due to symmetry. These are equal to $36^\circ, 36^\circ, 72^\circ, 108^\circ$ respectively. In this case the smallest angle is 36° .

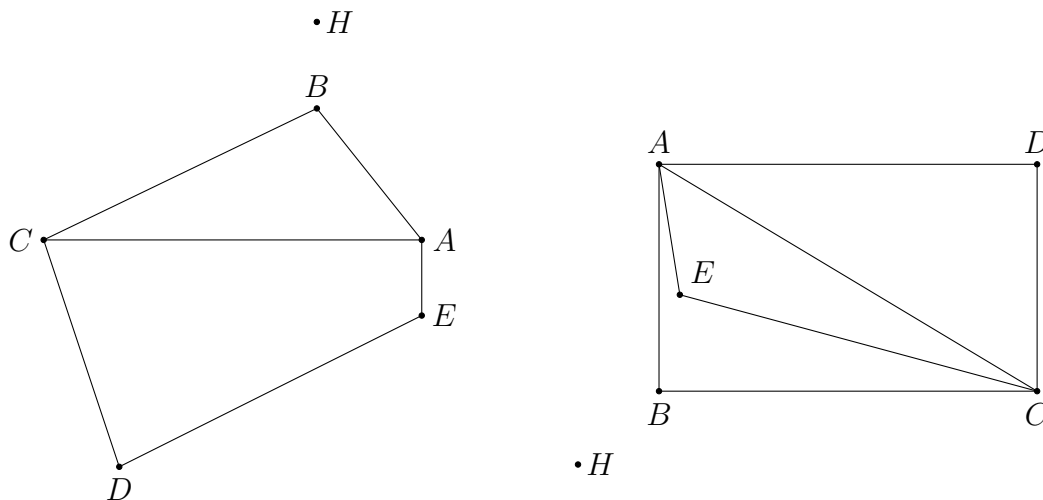
We conclude that the maximum possible value of the smallest angle is 36° .

Problem 5.4. Find all integers $n \geq 1$ and all sets S of n points on the plane such that no three points in S are collinear and the orthocentre of the triangle formed by any three distinct points in S also belongs to S .

Analysis. This problem has more geometric ingredient. The key property that one should know is that if H is the orthocentre of $\triangle ABC$, then C is the orthocentre of $\triangle ABH$, etc. Also, the position of the orthocentre should be important.

Solution. For $n = 1, 2$, S can be any set as the condition is trivially satisfied.

For $n \geq 3$, consider the convex hull of S . If the convex hull is a polygon of $m \geq 5$ vertices, then one of the interior angles $\angle ABC$ of this convex polygon is at least $\frac{m-2}{m} \cdot 180^\circ > 90^\circ$. Then the orthocentre H of $\triangle ABC$ lies outside $\triangle ABC$ and lies on the same side of AC as B . This implies H does not lie inside the convex hull, and hence $H \notin S$.



Therefore, the convex hull can only be a quadrilateral or a triangle. For the former case, it has to be a rectangle or otherwise we can use the same argument to an obtuse interior angle.

Suppose the convex hull of S is a rectangle. If $n = 4$, then S satisfies the condition since the orthocentre of a right-angled triangle is the vertex with the right angle. If

$n \geq 5$, let E be a point in S inside $ABCD$. WLOG, assume E lies inside $\triangle ABC$. Then $\angle CEA > \angle CBA = 90^\circ$. The orthocentre H of $\triangle ACE$ satisfies

$$\angle AHC = 180^\circ - \angle AEC < 90^\circ = \angle ABC.$$

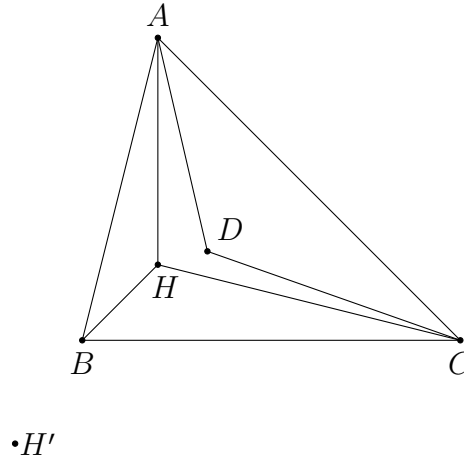
Hence, H lies outside the circumcircle of $\triangle ABC$. This implies H lies outside $ABCD$ so that it cannot belong to S .

Suppose the convex hull of S is $\triangle ABC$. Again, $\triangle ABC$ cannot be obtuse. If $\triangle ABC$ is right-angled and $n = 3$, then the condition is satisfied. If $\angle ABC = 90^\circ$ and $n \geq 4$, let D be a point in S inside $\triangle ABC$. Analogous to the argument of the rectangle, we deduce that the orthocentre of $\triangle ACD$ cannot belong to S .

If $\triangle ABC$ is acute, its orthocentre H must lie inside the triangle. If $n = 4$, then the condition is satisfied since the orthocentre of the triangle formed by any three of A, B, C, H is the remaining point. If $n \geq 5$, let D be a point in S . WLOG, assume D lies inside $\triangle ACH$. Let H' be the orthocentre of $\triangle ACD$. Then

$$\angle AH'C = 180^\circ - \angle ADC < 180^\circ - \angle AHC = \angle ABC.$$

Again, this implies H' lies outside $\triangle ABC$ and hence H' does not belong to S .



We conclude that n can only be 1, 2, 3, 4. For $n = 1, 2$, S can be anything. For $n = 3$, the three points in S form a right-angled triangle. For $n = 4$, the four points in S either form a rectangle or a triangle together with its orthocentre.

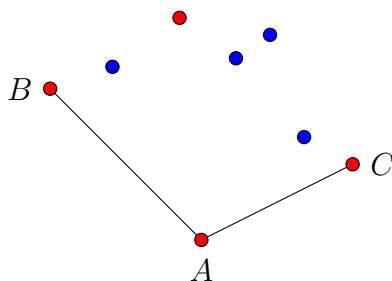
Problem 5.5. Let $n \geq 2$ be an integer. There are n red points and n blue points on the plane such that no three of them are collinear. Prove that there exists a line joining two of the points such that there is an equal number of red points and blue points (possibly none) on each side determined by the line.

Analysis. Clearly, the line must be formed by a red point and a blue point. In that case, it suffices to show that one side of the line contains an equal number of red

points and blue points. The trick of rotating a line with a fixed pivot and making use of the discreteness of some function with range $\mathbb{Z}$ is a useful tool in combinatorial geometry.

Solution. Consider the convex hull of the points. Let B, A, C be three consecutive vertices of the convex hull. WLOG assume A is red. If B is blue, then the line AB satisfies the condition since all the other $n - 1$ red points and $n - 1$ blue points lie on the same side of AB , and there is no point on the other side. Thus, we may assume B is red. Similarly, assume C is red.

Consider a line ℓ passing through A . Initially, ℓ coincides with the line AB . We rotate ℓ with A as the centre so that it sweeps across the convex hull until it coincides with the line AC . Let r and b be the number of red points and blue points respectively on the same side of ℓ as B . We consider the value $r - b$.



When ℓ meets the first point, we have $r - b = 1 - 0 = 1$ since only B lies on one side of ℓ . When ℓ reaches AC , we have $r - b = (n - 2) - n = -2$ since all points except A and C lie on one side of ℓ . Each time when ℓ sweeps across a point, the value of $r - b$ is increased or decreased by 1, depending on whether that point is red or blue. Hence, there must be some situations for which $r - b$ equals 0. We consider the last situation for which $r - b = 0$.

Suppose D is the next point met by ℓ . If D is blue, then there are r red points and $b = r$ blue points on one side of ℓ , and there are $n - 1 - r$ red points and $n - 1 - b = n - 1 - r$ blue points on the other side. So this is the line we need. If D is red, then $r - b$ becomes 1 immediately after ℓ meets D . Again from the discreteness, there must be some situations afterwards for which $r - b = 0$. This is a contradiction. Therefore, we can always get such a line.

Problem 5.6. (The University of Melbourne School Mathematics Competition 2017) A finite number of points in the plane are chosen, and each is coloured either red or blue. Suppose that for any line ℓ in the plane, the difference between the number of red points on ℓ and the number of blue points on ℓ is at most 1. Prove that the points all lie on a single line.

Analysis. This time we shall use the method of double counting. We count the number of pairs of points according to their colours. This sets up some equations

relating the total number of red/blue points and the number of red/blue points on the lines. Then we use algebraic ways to show that there must be a line with many points.

Solution. Let r and b be the number of red points and the number of blue points respectively. Let $n = r + b$ be the total number of points. It suffices to consider those lines which pass through at least two of the points. There are only finitely many of them. These lines can be partitioned into 3 groups depending on the differences of the number of points of distinct colours on them.

- For a line in group I, the number of red points and the number of blue points are the same. Suppose there are x_i red points and x_i blue points.
- For a line in group II, the number of red points is greater than the number of blue points. Suppose there are y_i red points and $y_i - 1$ blue points.
- For a line in group III, the number of red points is smaller than the number of blue points. Suppose there are $z_i - 1$ red points and z_i blue points.

The number of pairs of red points is $\binom{r}{2}$. This is equal to the sum of pairs of red points on all the lines. Hence, we obtain

$$\sum \binom{x_i}{2} + \sum \binom{y_i}{2} + \sum \binom{z_i - 1}{2} = \binom{r}{2}. \quad (1)$$

Each sum runs over all lines in the respective groups. The sum could be empty. Similarly, we have

$$\sum \binom{x_i}{2} + \sum \binom{y_i - 1}{2} + \sum \binom{z_i}{2} = \binom{b}{2}. \quad (2)$$

By counting the number of pairs of points with distinct colours, we obtain

$$\sum x_i^2 + \sum y_i(y_i - 1) + \sum (z_i - 1)z_i = rb. \quad (3)$$

We first rewrite (1) and (2) by expanding the binomial coefficients. This gives

$$\sum x_i(x_i - 1) + \sum y_i(y_i - 1) + \sum (z_i - 1)(z_i - 2) = r(r - 1) \quad (4)$$

and

$$\sum x_i(x_i - 1) + \sum (y_i - 1)(y_i - 2) + \sum z_i(z_i - 1) = b(b - 1). \quad (5)$$

We observe that the left-hand side of (3) is larger than those of (4) and (5), but the right-hand side is not in general. Thus, there should be some tight inequalities that have to be satisfied. We combine these equations by considering $2 \times (3) - (4) - (5)$. This gives

$$\sum 2x_i + \sum 2(y_i - 1) + \sum 2(z_i - 1) = r + b - (r - b)^2 \leq n. \quad (6)$$

Note that each of $2x_i$, $2(y_i - 1)$ and $2(z_i - 1)$ is approximately the number of points on a line. Hence, (6) tells us that it is almost certain that there can only be one line,

since otherwise some points contribute more than 1 to the left-hand side and thus the sum is greater than n . We can make this precise algebraically. Adding (4) and (5), we obtain

$$\sum 2x_i(x_i - 1) + \sum 2(y_i - 1)^2 + \sum 2(z_i - 1)^2 = r(r - 1) + b(b - 1) \geq \frac{n(n - 2)}{2} \quad (7)$$

as $r^2 + b^2 \geq \frac{(r + b)^2}{2} = \frac{n^2}{2}$. Suppose the line with the greatest number of points contains M points. Then $x_i \leq \frac{M}{2}$, $y_i \leq \frac{M + 1}{2}$ and $z_i \leq \frac{M + 1}{2}$. Applying these bounds to the left-hand side of (7), we get

$$\frac{M - 1}{2} (\sum 2x_i + \sum 2(y_i - 1) + \sum 2(z_i - 1)) \geq \frac{n(n - 2)}{2}.$$

Using (6), we have $M \geq n - 1$, meaning that there is a line containing almost all points.

If $M = n$, then we are done. If $M = n - 1$, suppose ℓ contains $n - 1$ points and we may assume there is a red point P not on ℓ . Since ℓ contains at least 2 points, it must contain a red point Q . Then the line PQ does not meet the condition. This is a contradiction. Hence, we are done.

Links

Theorems

- [Binomial theorem](#)
- [Bonferroni inequalities](#)
- [Generalized binomial theorem](#)
- [Hall's theorem](#)
- [Handshaking theorem](#)
- [Inclusion-exclusion principle](#)
- [Majorization inequality/Karamata's inequality](#)
- [Pigeonhole principle](#)
- [Sperner's theorem](#)
- [Sylvester-Gallai theorem](#)
- [Turán's theorem](#)

Propositions

- [1.1: Addition principle](#)
- [1.2: Multiplication principle](#)
- [1.3: Properties of permutation](#)
- [1.4: Properties of combination](#)
- [1.5: Recurrence relations](#)

Problems

- [1.1: Basic counting 1](#)
- [1.2: Basic counting 2](#)
- [1.3: Choosing arithmetic sequences](#)
- [1.4: HKIMO TST 2013 Test 2 Problem 2](#)
- [1.5: HKIMO TST 2014 Test 1 Problem 4 \(inclusion-exclusion principle\)](#)
- [1.6: An identity for Euler's totient function](#)
- [1.7: An identity for binomial coefficients](#)
- [1.8: CHKMO 2007 Problem 3](#)
- [1.9: IMO 2001 Problem 3](#)
- [1.10: Recurrence relation](#)

- 1.11: CHKMO 2003 Problem 2
- 1.12: IMO 2008 Problem 5
- 1.13: APMO 2017 Problem 3
- 1.14: Bijection
- 1.15: CMO 2001 Problem 5
- 1.16: IMO shortlist 2002 C3
- 1.17: IMO 2013 Problem 6
- 1.18: HKIMO TST 2014 Test 1 Problem 4 (generating function)
- 1.19: Generating function and Fibonacci sequence
- 1.20: Catalan numbers
- 1.21: Generating function and complex numbers
- 2.1: Extremal combinatorics and induction
- 2.2: IMO shortlist 2010 C2
- 2.3: IMO 2009 Problem 6
- 2.4: Pigeonhole principle
- 2.5: Sperner's theorem
- 2.6: CHKMO 2006 Problem 1
- 2.7: CHKMO 2012 Problem 2
- 2.8: CHKMO 2011 Problem 4
- 2.9: Tiling with rectangles
- 2.10: HKIMO TST 2015 Test 2 Problem 4
- 2.11: IMO 2004 Problem 3
- 2.12: Extremal principle 1
- 2.13: Extremal principle 2
- 2.14: APMO 2008 Problem 2
- 2.15: CHKMO 2015 Problem 4
- 3.1: CMO 1994 Problem 2
- 3.2: CMO 2010 Problem 5
- 3.3: IMO shortlist 2009 C8
- 3.4: Game 1
- 3.5: Game 2
- 3.6: Invariance 1

- 3.7: Invariance 2
- 3.8: APMO 2009 Problem 1
- 3.9: IMO shortlist 2005 C5
- 3.10: IMO 2000 Problem 3
- 4.1: Handshaking theorem
- 4.2: Bipartite graph
- 4.3: Turán's theorem
- 4.4: Hall's theorem
- 4.5: APMO 2003 Problem 5
- 4.6: Directed graph
- 4.7: CMO 1987 Problem 3
- 4.8: Application of the majorization inequality in graph theory
- 4.9: Chasing game
- 5.1: Sylvester-Gallai theorem
- 5.2: EGMO 2017 Problem 3
- 5.3: APMO 2011 Problem 2
- 5.4: Combinatorial geometry and orthocentre
- 5.5: Discreteness
- 5.6: The University of Melbourne School Mathematics Competition 2017

Terminologies

- addition principle
- binomial coefficient
- binomial function
- bipartite graph
- Catalan numbers
- circuit
- combination
- complete graph
- connected graph
- convex hull
- cycle

- degree
- dictionary order
- directed graph
- edge
- graph
- lexicographic order
- method of exhaustion
- multiplication principle
- Pascal's identity
- path
- permutation
- Petersen graph
- product rule
- simple cycle
- simple graph
- simple path
- sum rule
- Turán graph
- Vandermonde's identity
- vertex

References

- A Path to Combinatorics for Undergraduates: Counting Strategies
(by Titu Andreescu, Zuming Feng)

「遠看山有色，近聽水無聲。春去花還在，人來鳥不驚。」

《畫》